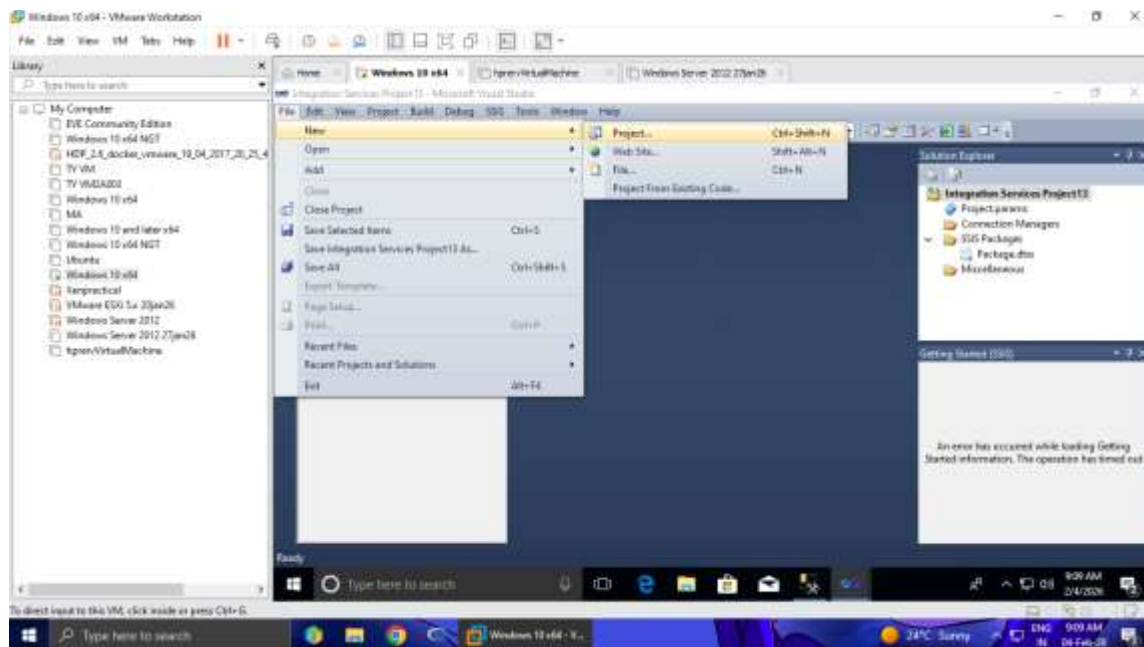


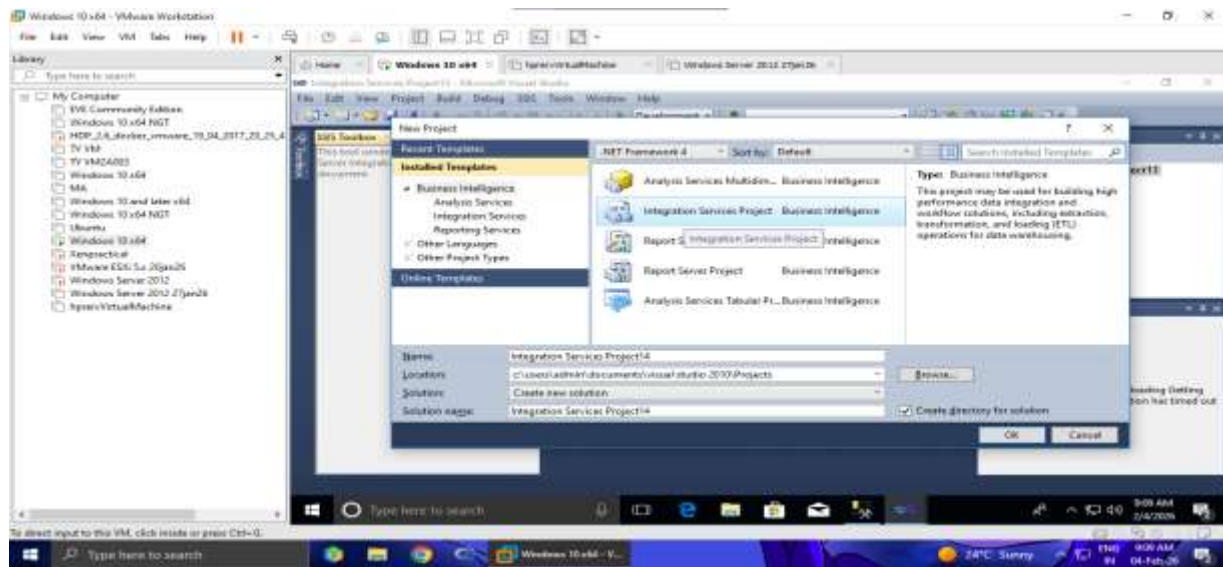
Practical 1

Import the legacy data from different sources such as (excel, sqlserver,oracle etc) and load in the target system.

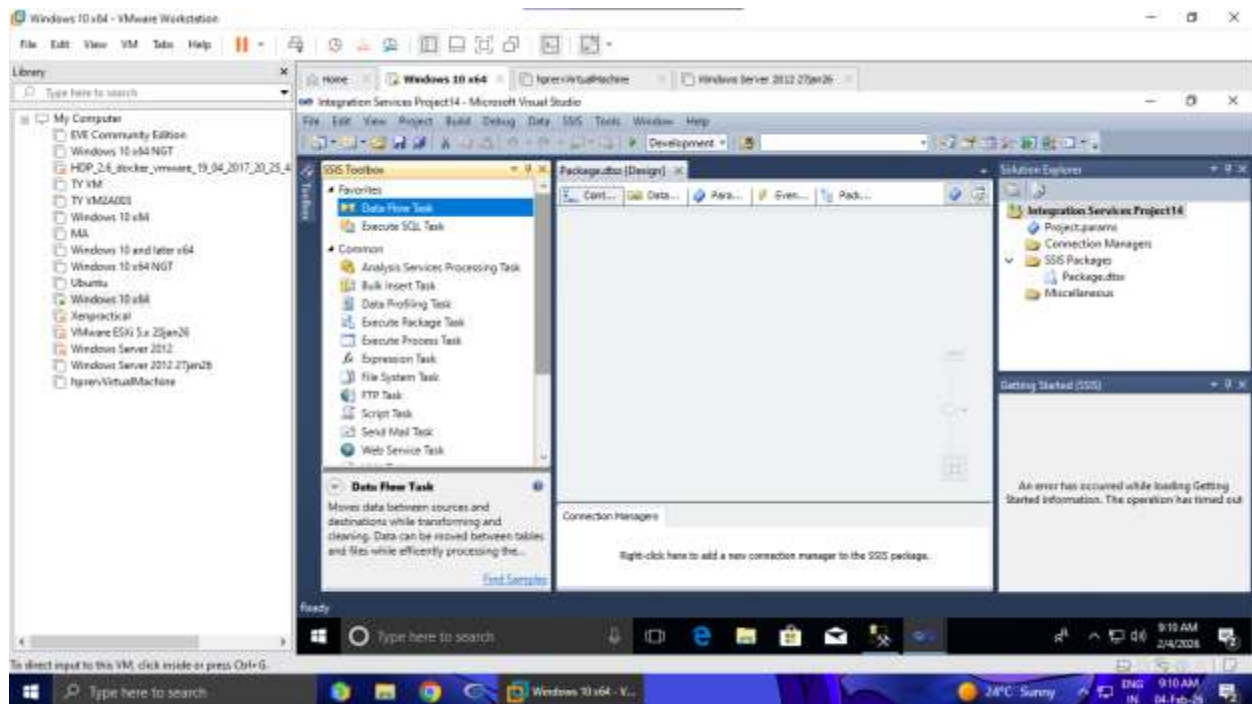
Step 1: In VMware open the windows 10x64 virtual machine> and search for sql server data tool> open > new > project.



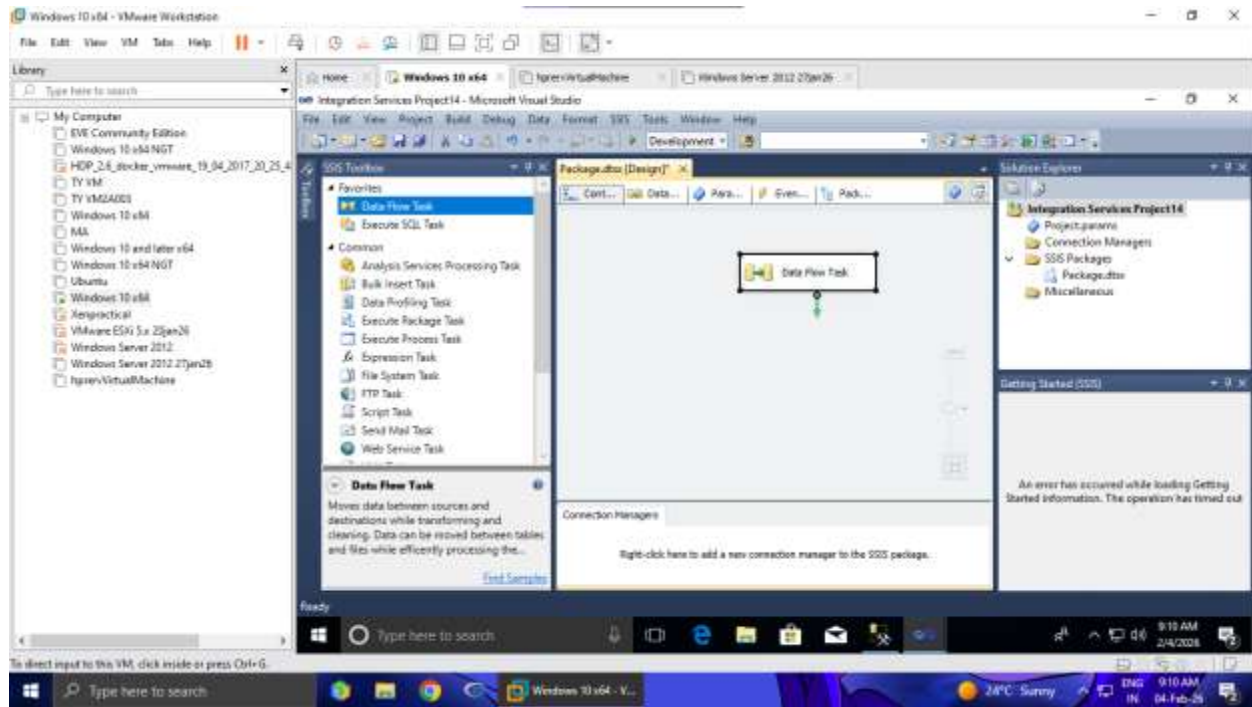
Step 2: choose “Integration service project” > Open.



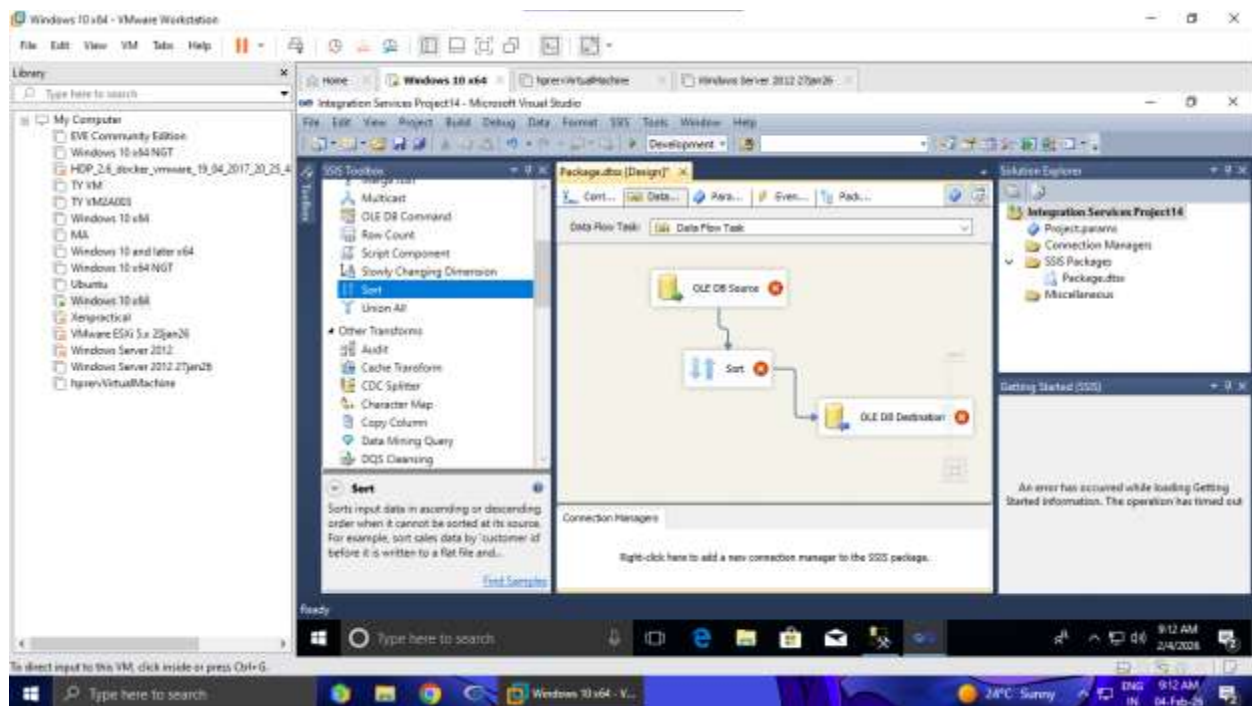
Step 3: Drag and drop the “data flow task”.



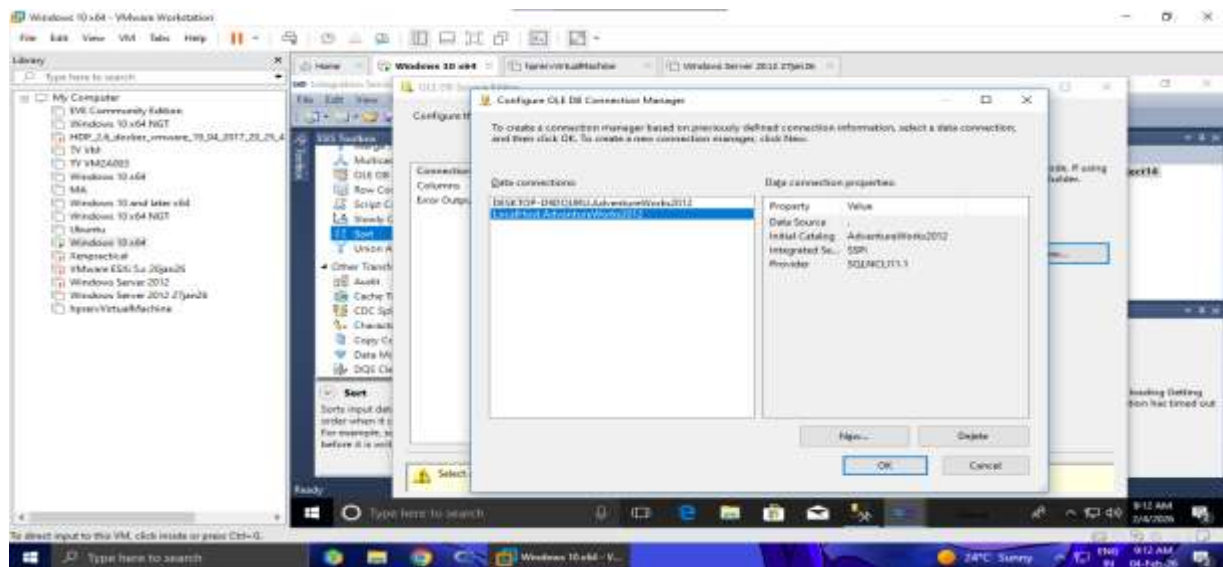
Double click on it.



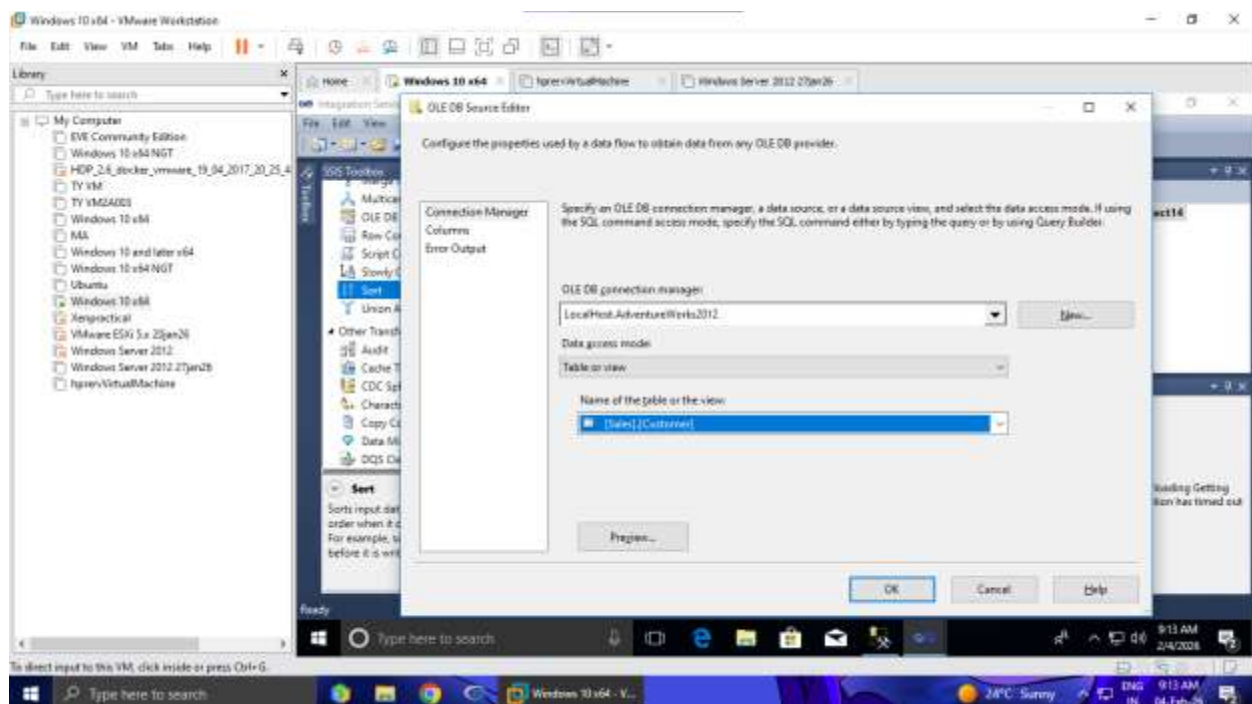
Step 4: Now from Source take “OLE DB Source” and from destination take “OLE DB destination” and Sort method and connect each of them with blue arrow.



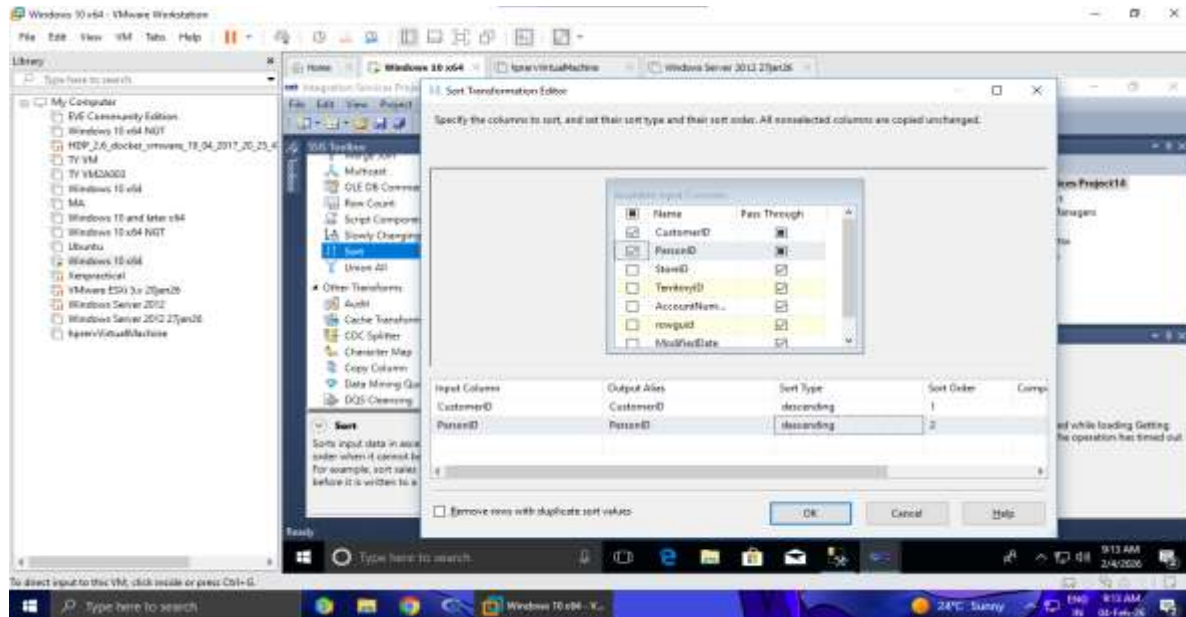
Step 5: Now go inside the OLE DB source and take the “AdventureWorks” Database > ok.



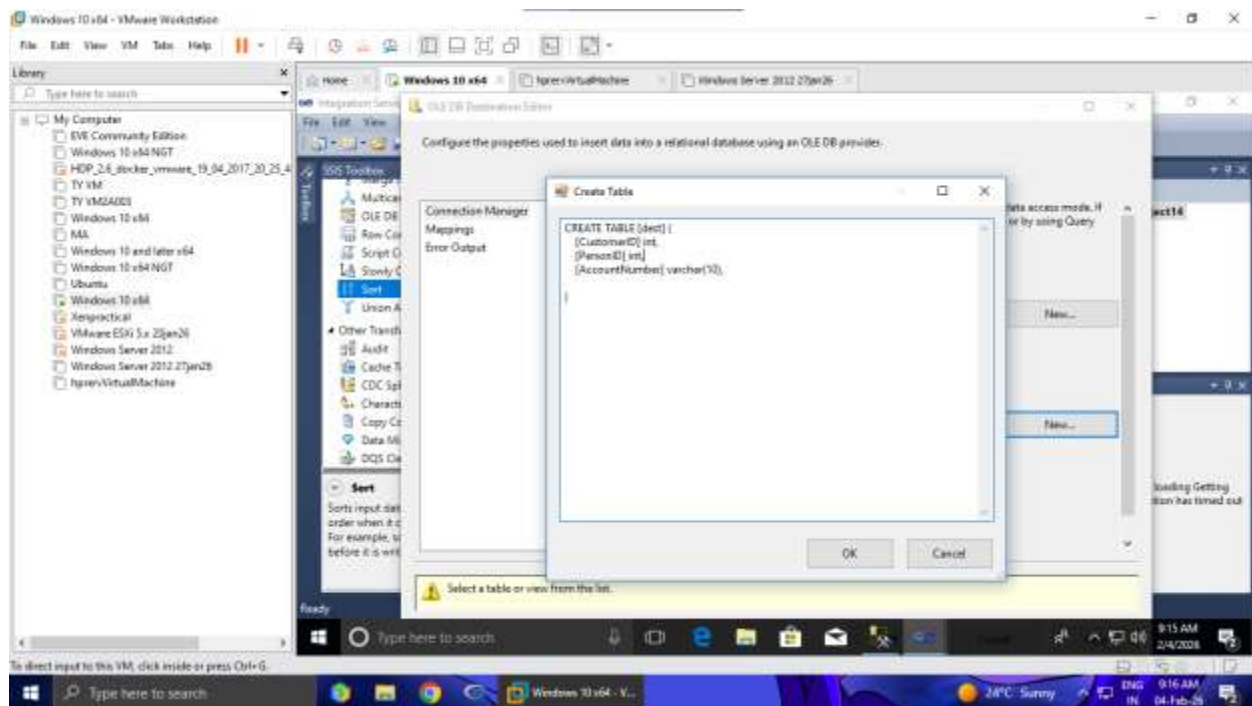
Step 6: Take proper “table name “ > sales.customers > ok.



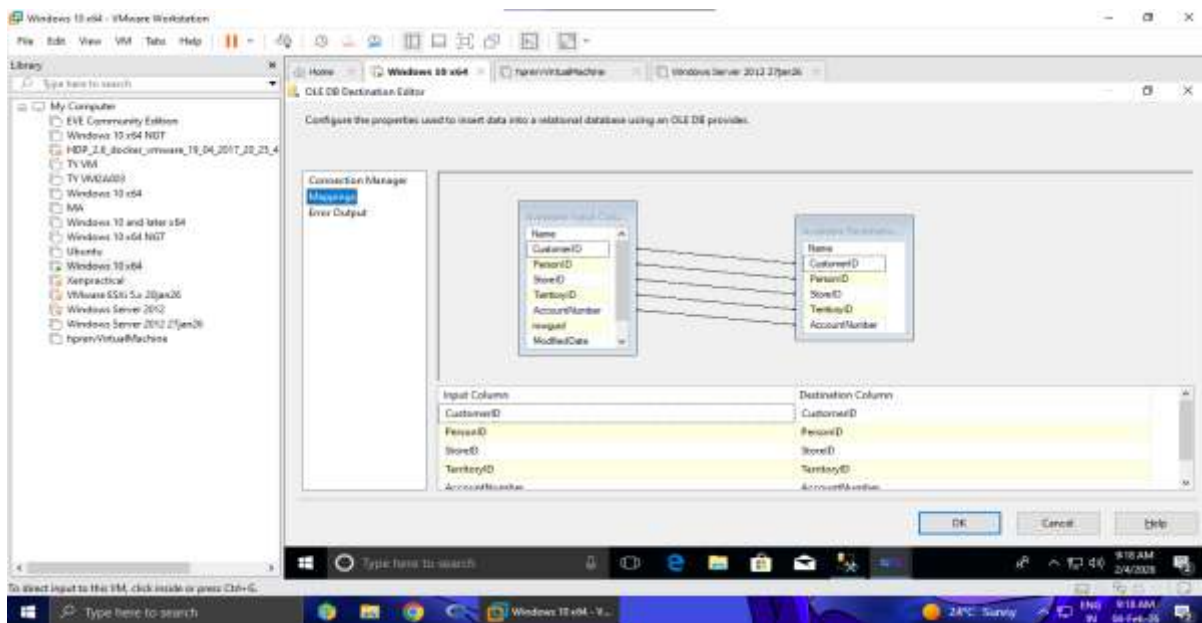
Step 7: Click on “Sort” (that we have took for transformation) > choose which column we have to sort > and also choose in which way we have to sort ascending or descending



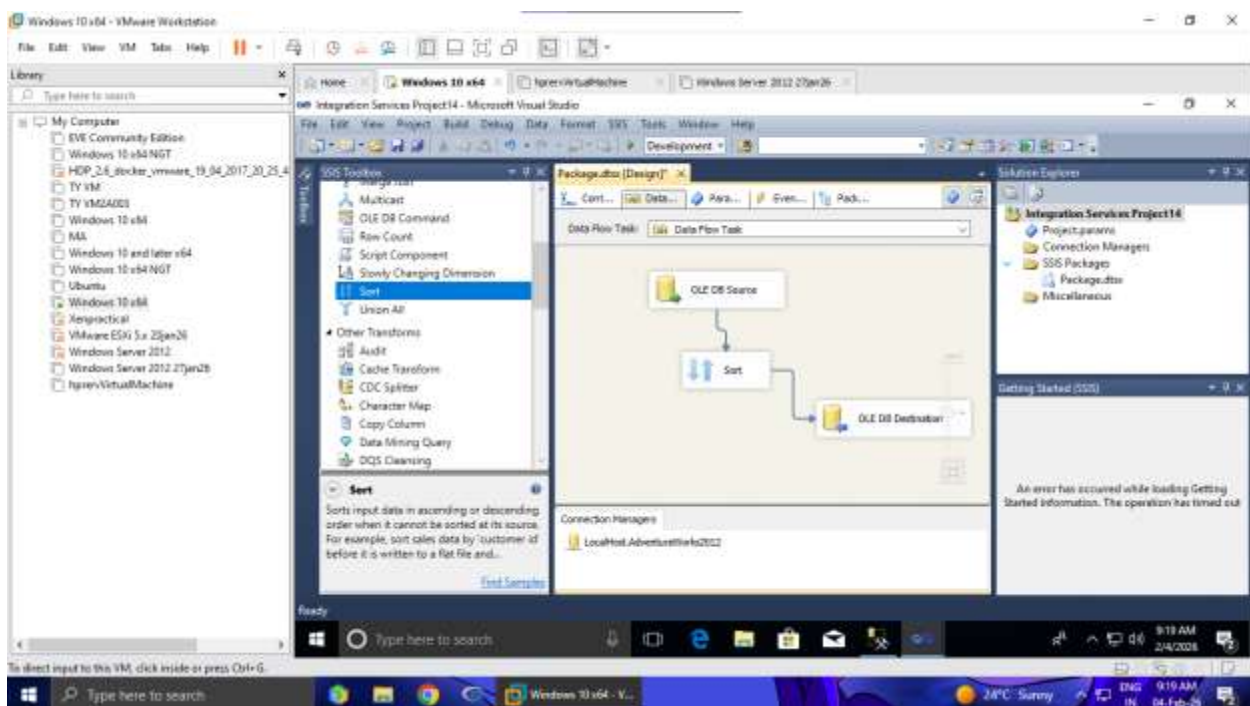
Step 8: After that go in OLE DB destination and create the table in which we have to store the sorted data > also choose the column which we want to display data.



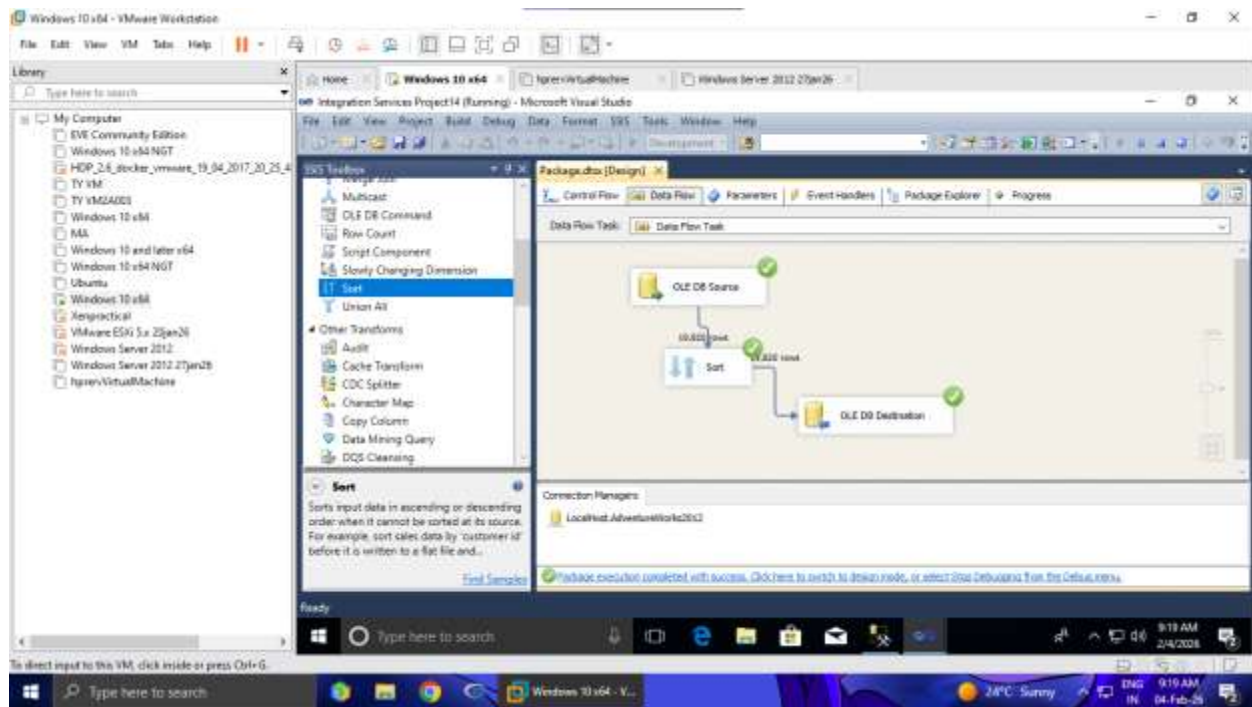
Step 9 : After selecting the appropriate destination table > go in mapping > and see whether all the column are mapped properly or not >ok



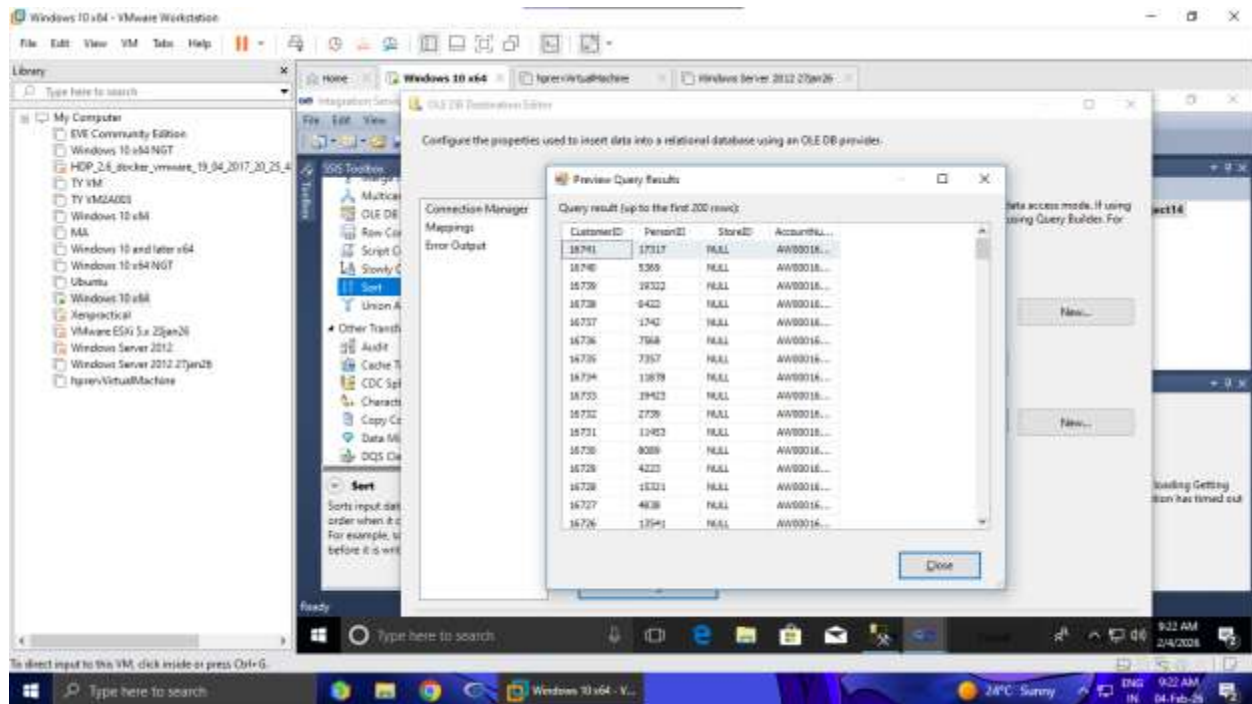
Step 10 : This is the final look after doing all setting



Step 11: Successfully got executed



Step 12: For seeing output > go in OLE DB destination > preview>(this is one way)



Step 13: otherwise > go in SQL > and type the following query > Output

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'AdventureWorks2012' database selected. The right pane shows the 'SQL Query' window with the following query:

```
use AdventureWorks2012;
select * from [OLE DB Destination];
```

The 'Results' pane shows the output of the query, which is a table with 10 rows and 6 columns: CustomerID, PersonID, StoreID, TerritoryID, AccountNumber, and rowguid. The data is as follows:

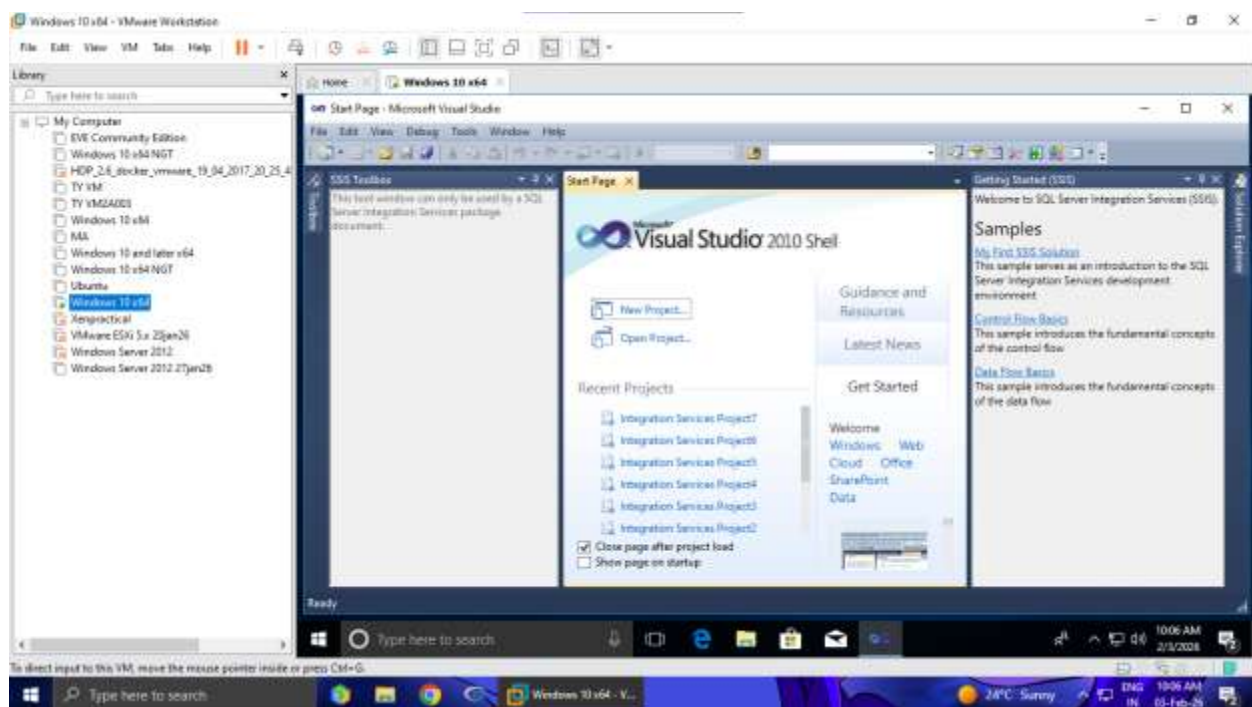
CustomerID	PersonID	StoreID	TerritoryID	AccountNumber	rowguid
12347	3036	NULL	9	AW00012347	5DFF98C5-2D37-48D6-A5D3-CDC5533894B
12348	15657	NULL	9	AW00012348	8FD4A549-6D34-4D1D-B121-DE28B84C3A3
12349	12420	NULL	9	AW00012349	20910E77-9154-48DA-A1F6-7E45D0E5FD42
12350	16994	NULL	9	AW00012350	8D3F88EA-84B5-48CC-8596-24C570F14266
12351	4827	NULL	9	AW00012351	AAA8653F-8AB3-48FA-A28E-5469C858B520
12352	6385	NULL	9	AW00012352	38036A71-069C-4C3B-A00C-A09FC8572ACF
12353	11079	NULL	9	AW00012353	813D3703-96A5-4697-8967-F945E187268B
12354	14883	NULL	9	AW00012354	DA3E56C3-8F38-4E13-8925-3626D388C486
12355	13796	NULL	9	AW00012355	7F3D0F8-436F-481E-89F1-F369D1087331
12356	20387	NULL	9	AW00012356	E20695F8-C8C5-4CE8-8421-708BFA4718E8

The status bar at the bottom indicates the query was successful: 'Query succeeded. [local] (11.0 SP1) | DESKTOP-08DQURP\Admin | AdventureWorks2012 | 00:00:00 | 18820 rows'.

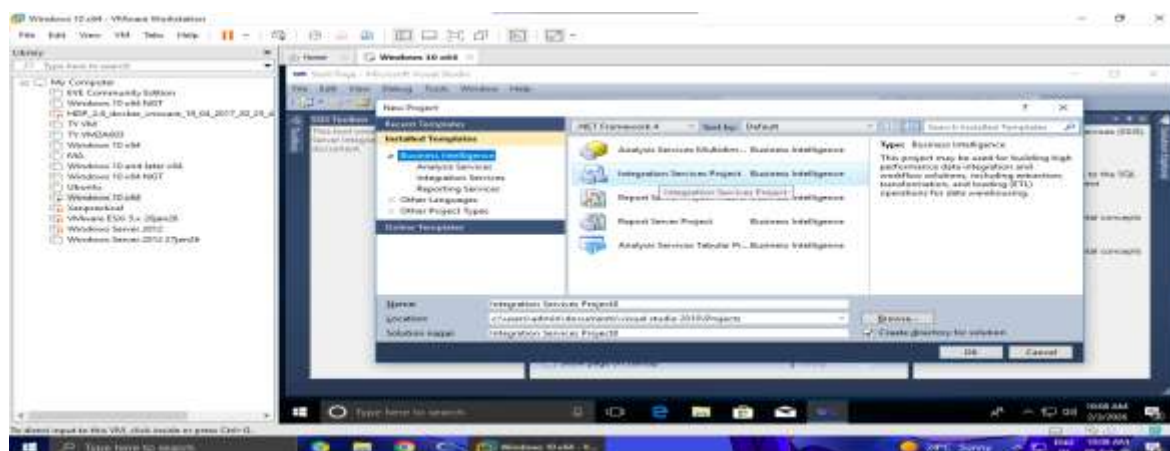
Practical 2

Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sqlserver.

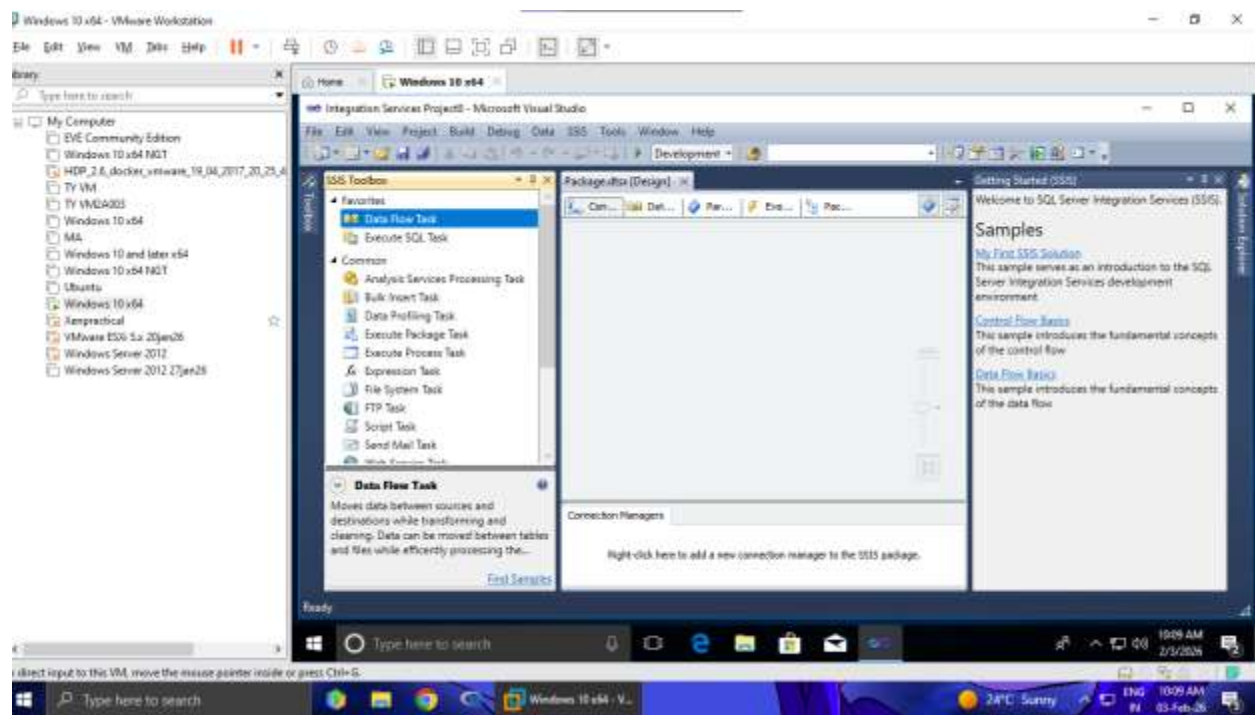
Step 1: In VMware open the windows 10x64 virtual machine> and search for sql server data tool> open > new > project.



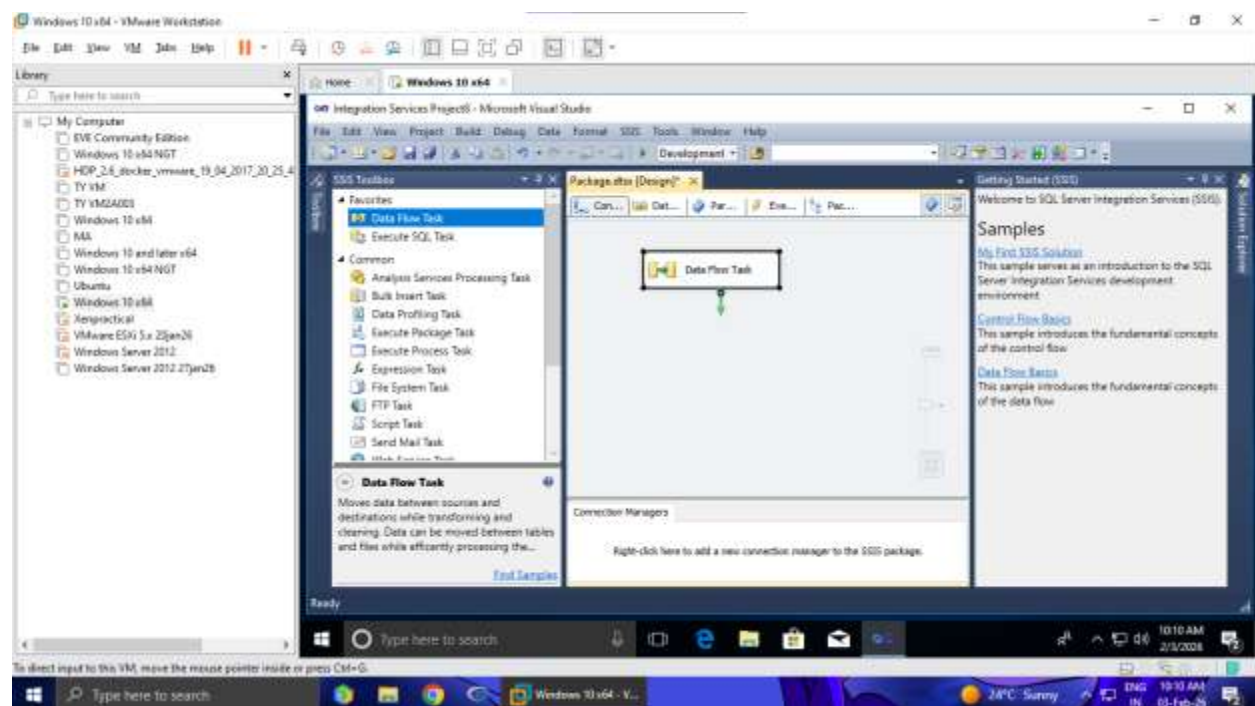
Step 2: choose “Integration service project” > Open.



Step 3: Drag and droop the “ data flow task “



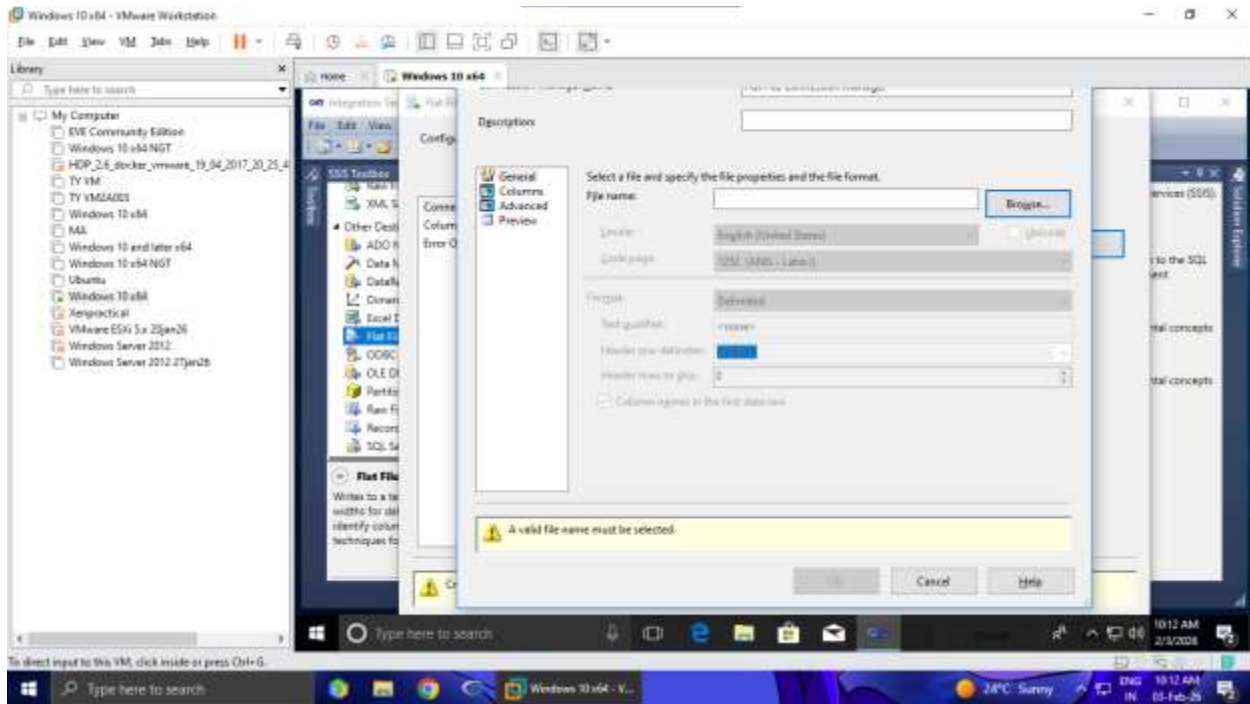
Step 4 : double click on it



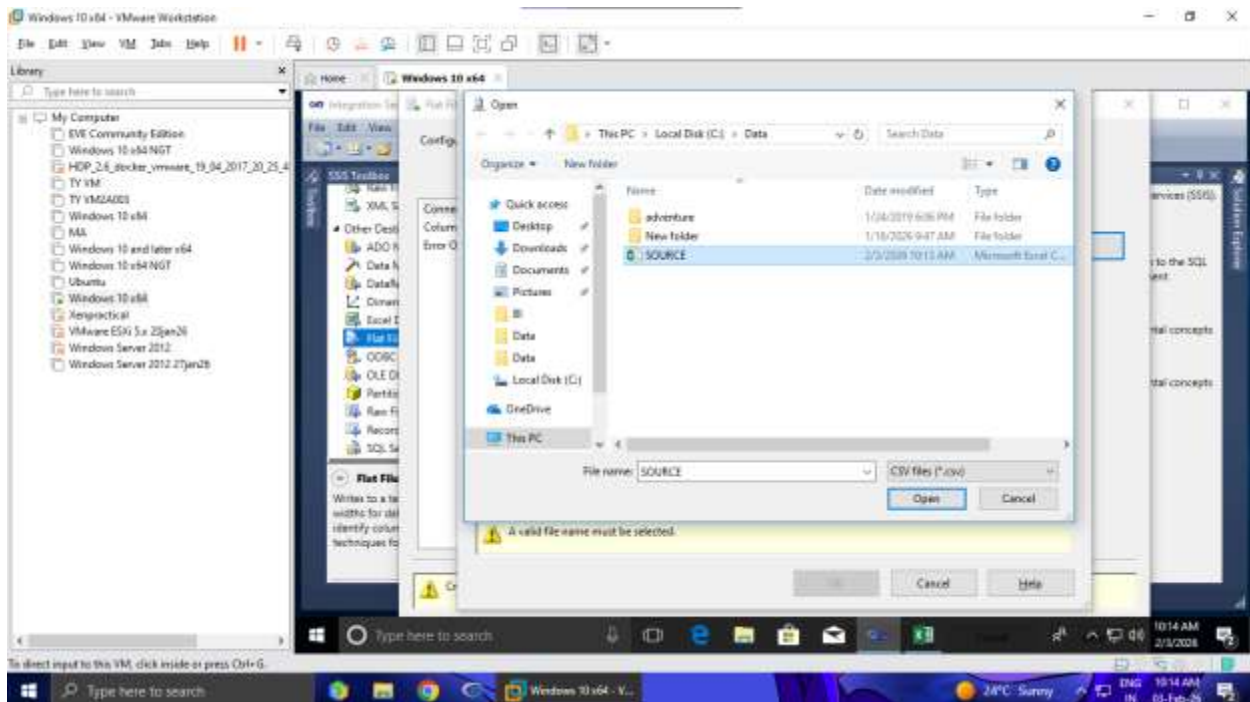
The screenshot displays the Microsoft Visual Studio environment for developing SQL Server Integration Services (SSIS) packages. The main window is titled 'Integration Services Project - Microsoft Visual Studio'. The 'SSIS Toolbox' on the left lists various data sources and destinations, with 'Flat File Destination' selected. The central 'Data Flow Task' diagram shows a 'Flat File Source' connected to a 'Flat File Destination'. The right sidebar displays 'Getting Started (SSIS)' and 'Samples' sections, including 'My First SSIS Solution' and 'Control Flow Basics'. The bottom status bar indicates the VM is running on VMware Workstation.

The screenshot shows the 'Flat File Source Editor' in SQL Server Data Tools. The 'Columns' tab is active, displaying a list of columns with their respective data types. The 'Flat File Connection Manager' is selected in the left pane. The main area shows the 'Columns' and 'Error Output' sections. A yellow banner at the bottom says 'Create a new Flat File connection manager by clicking New.'

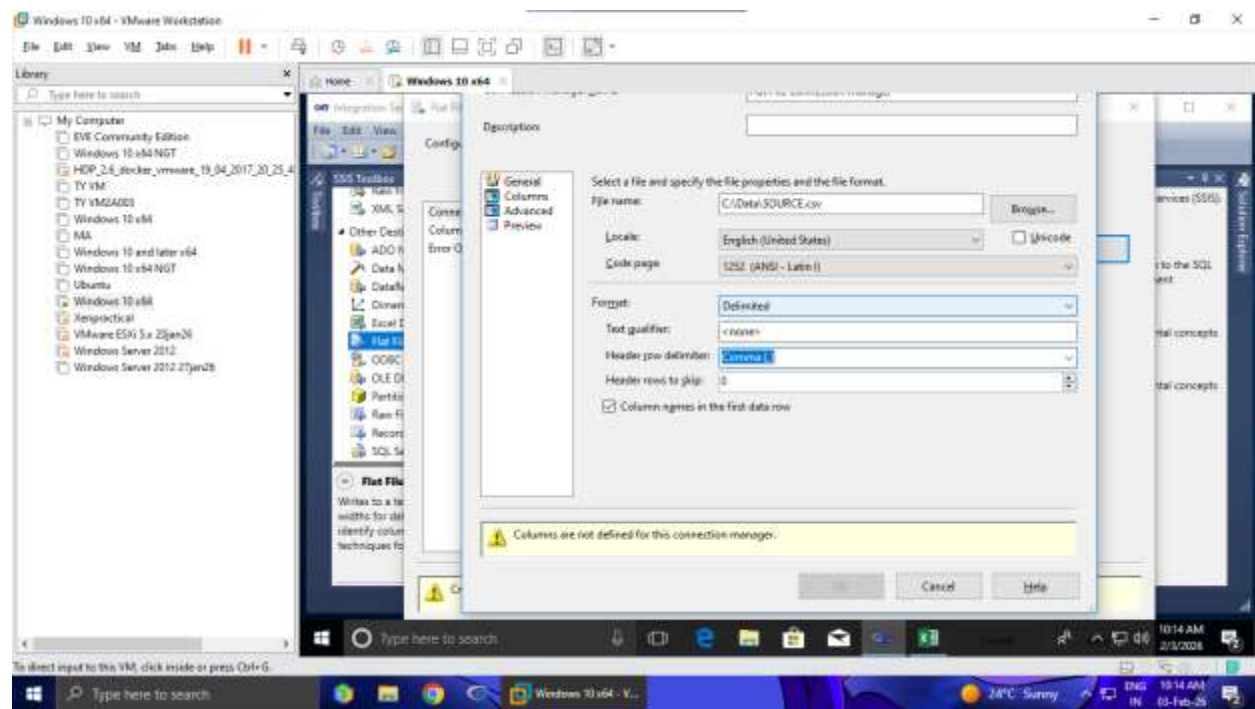
Step 7 : Browse the file If already existing > other wise go in excel and create one file containing the simple data like” name and marks column” and save in CSV file format.



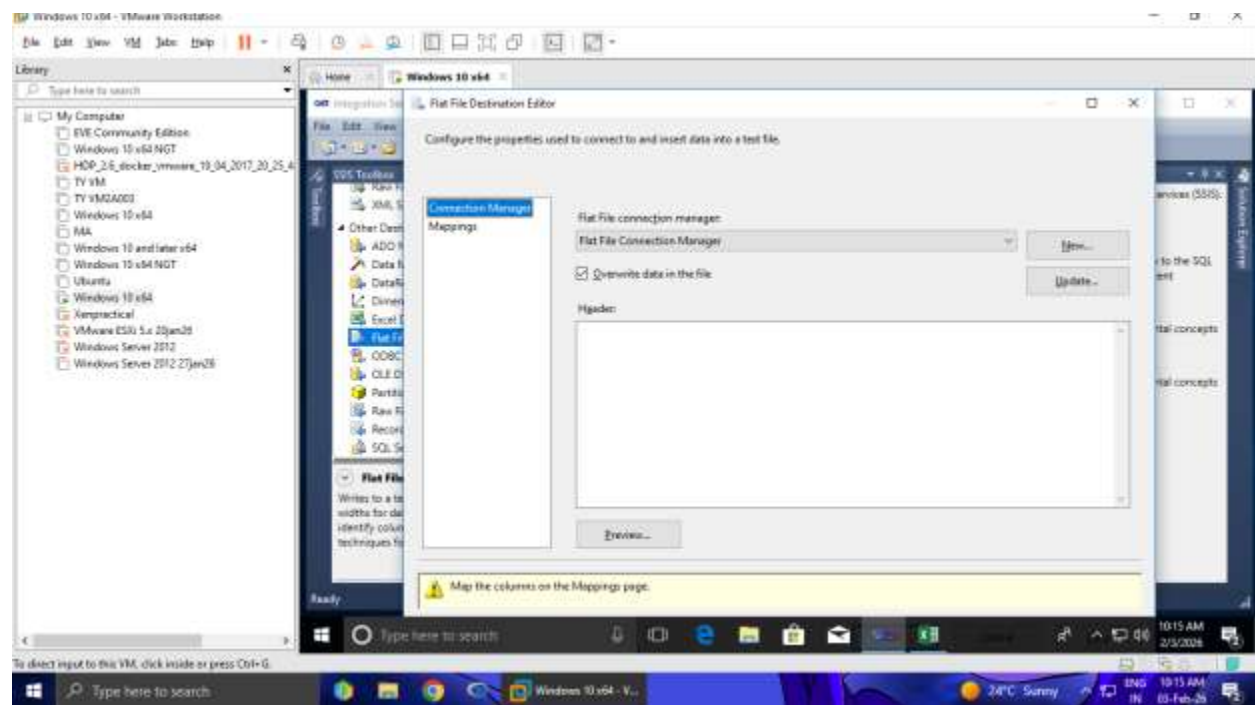
Step 8: Take the source file > open

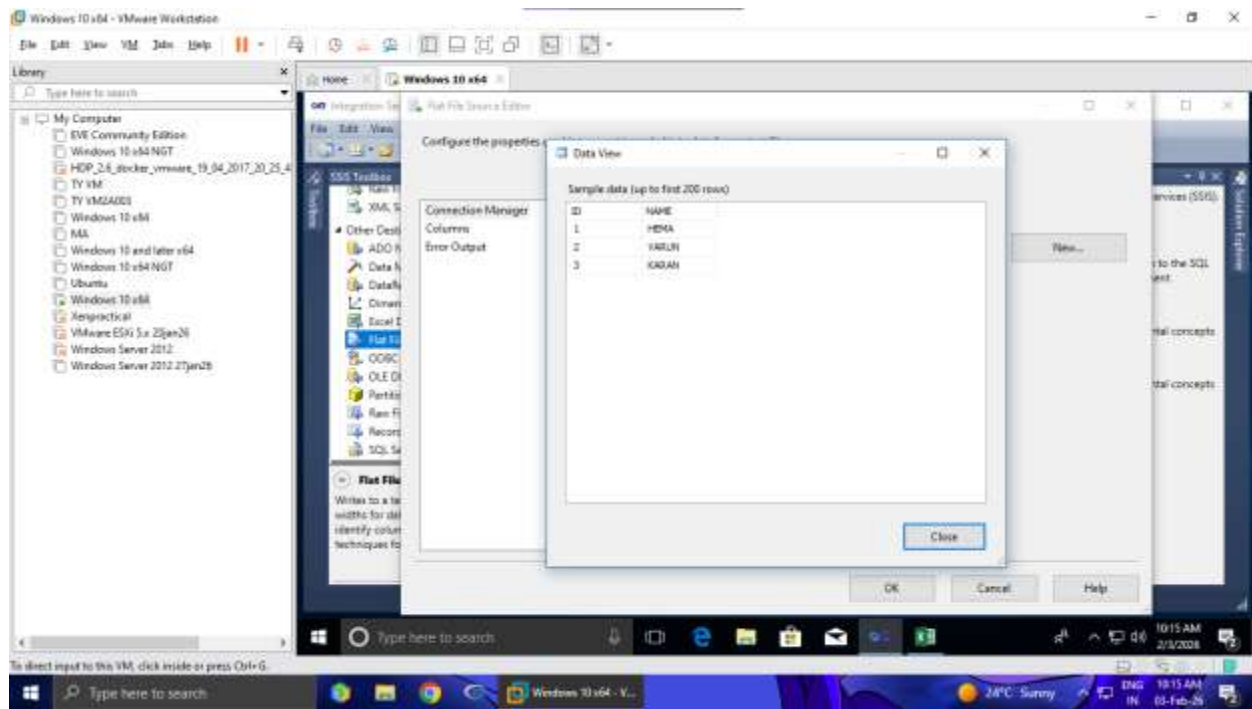


Step 9 : Select a;; the columns properly as mentioned in image

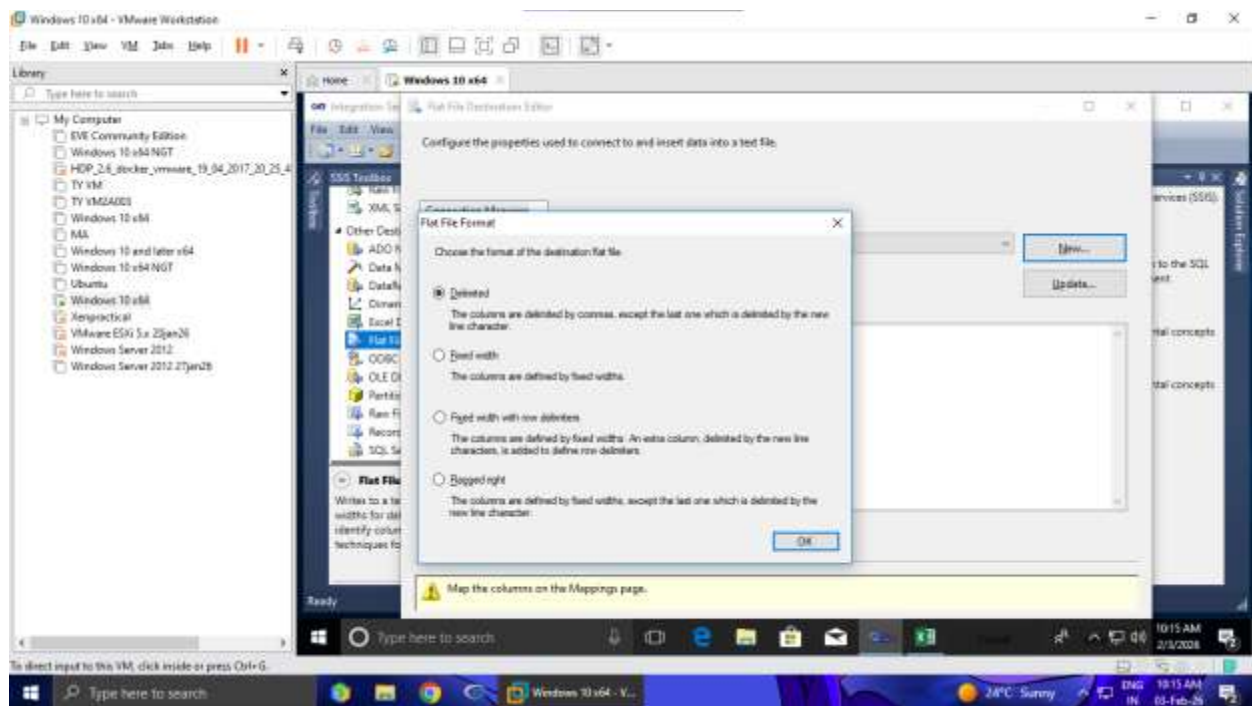


Step 10 : To check whether the data is properly came or not > go in preview

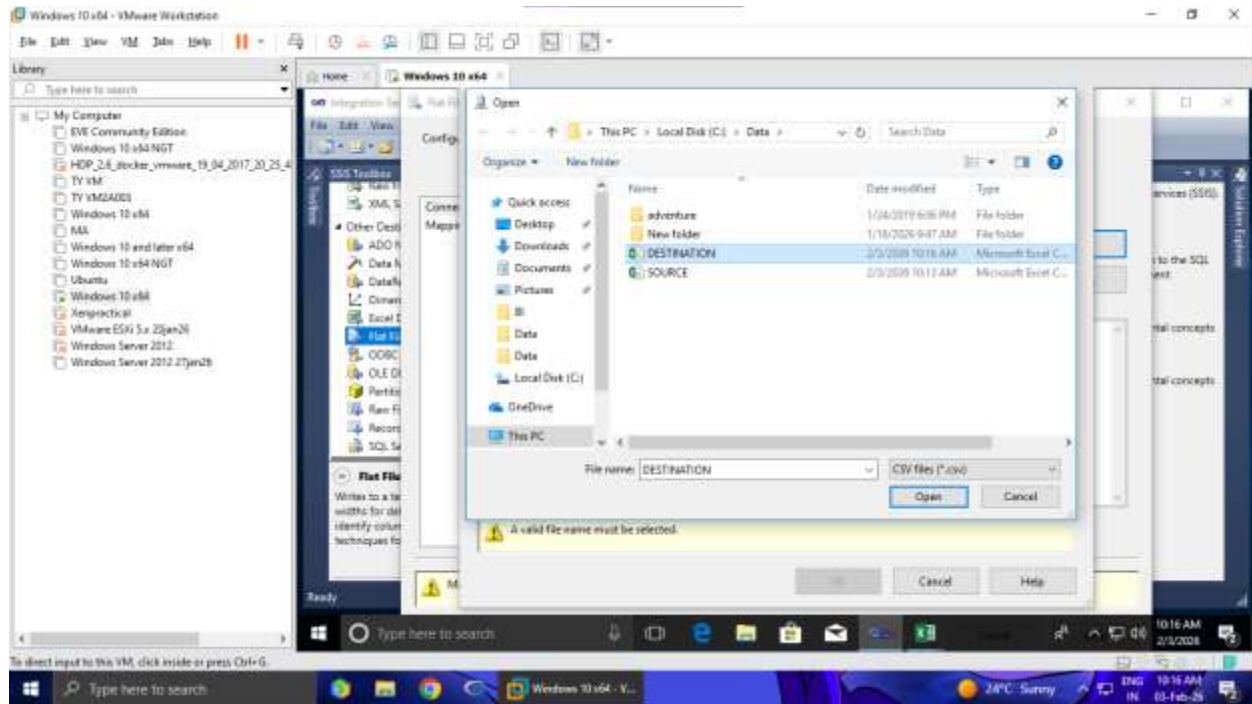




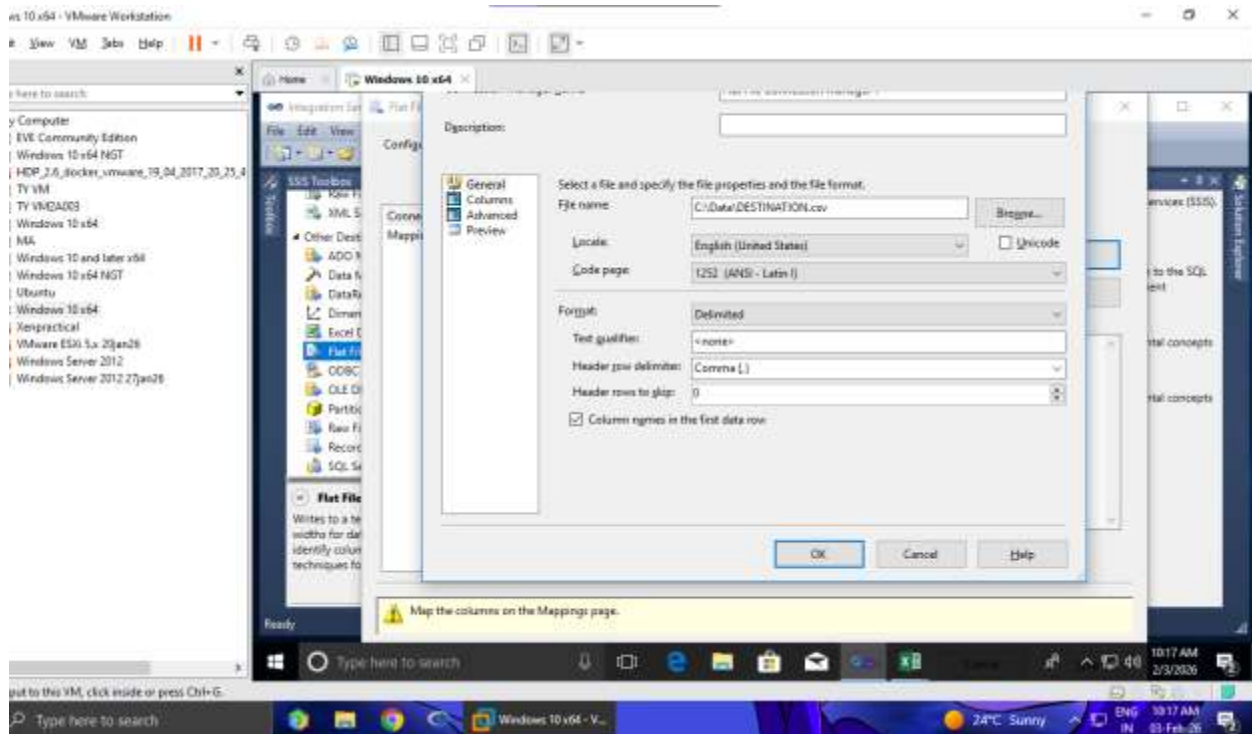
Step 11: Now go in destination flat file > new > delimited > ok



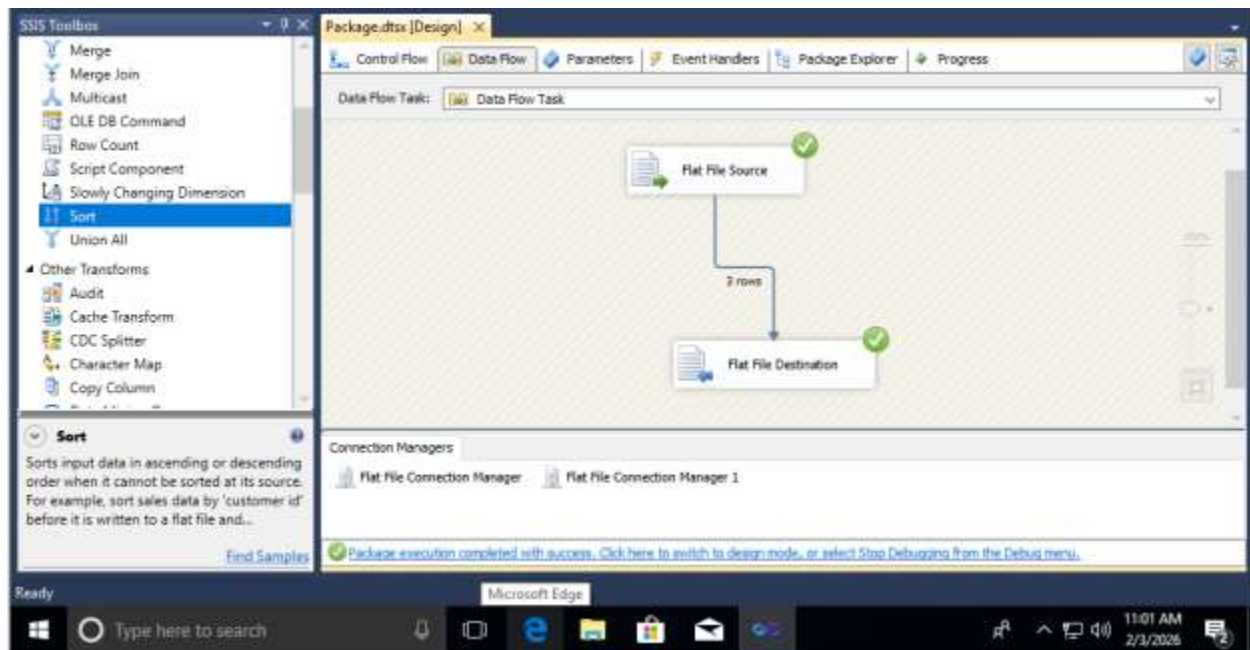
Step 12 : create a .CSV file having only the column name same as source file. >
and select it



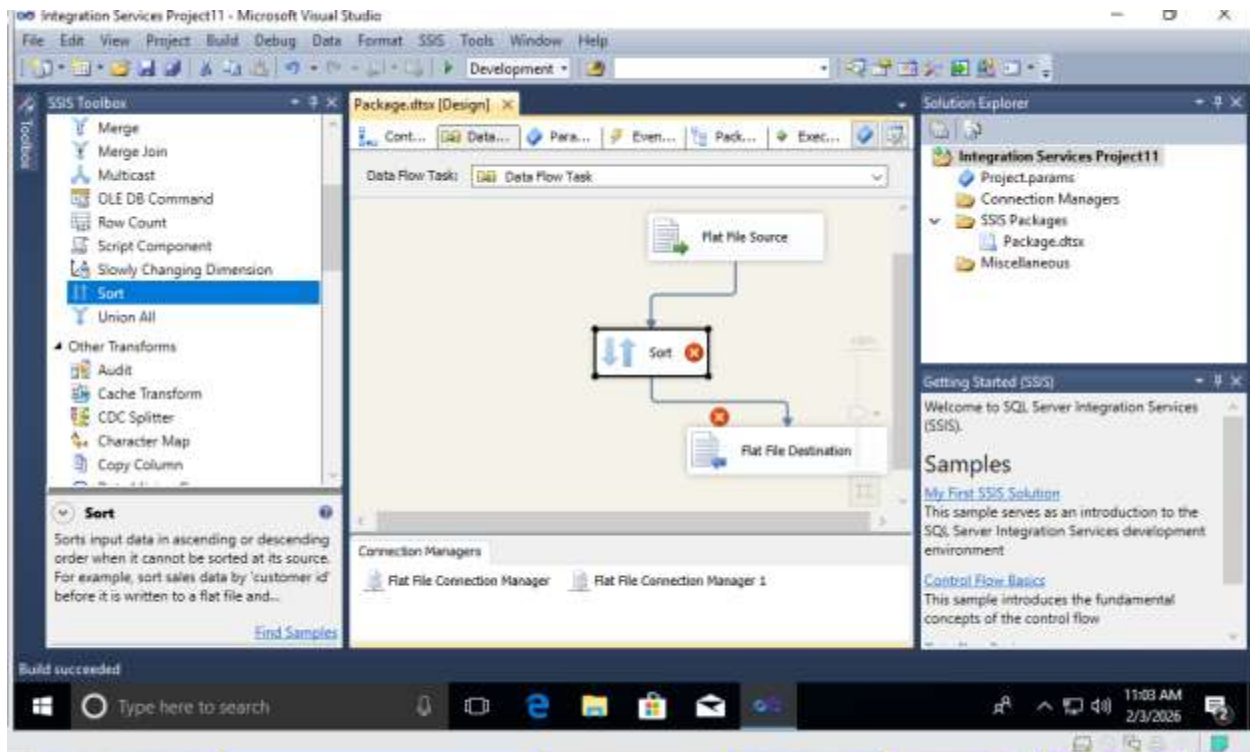
Step13: Ok



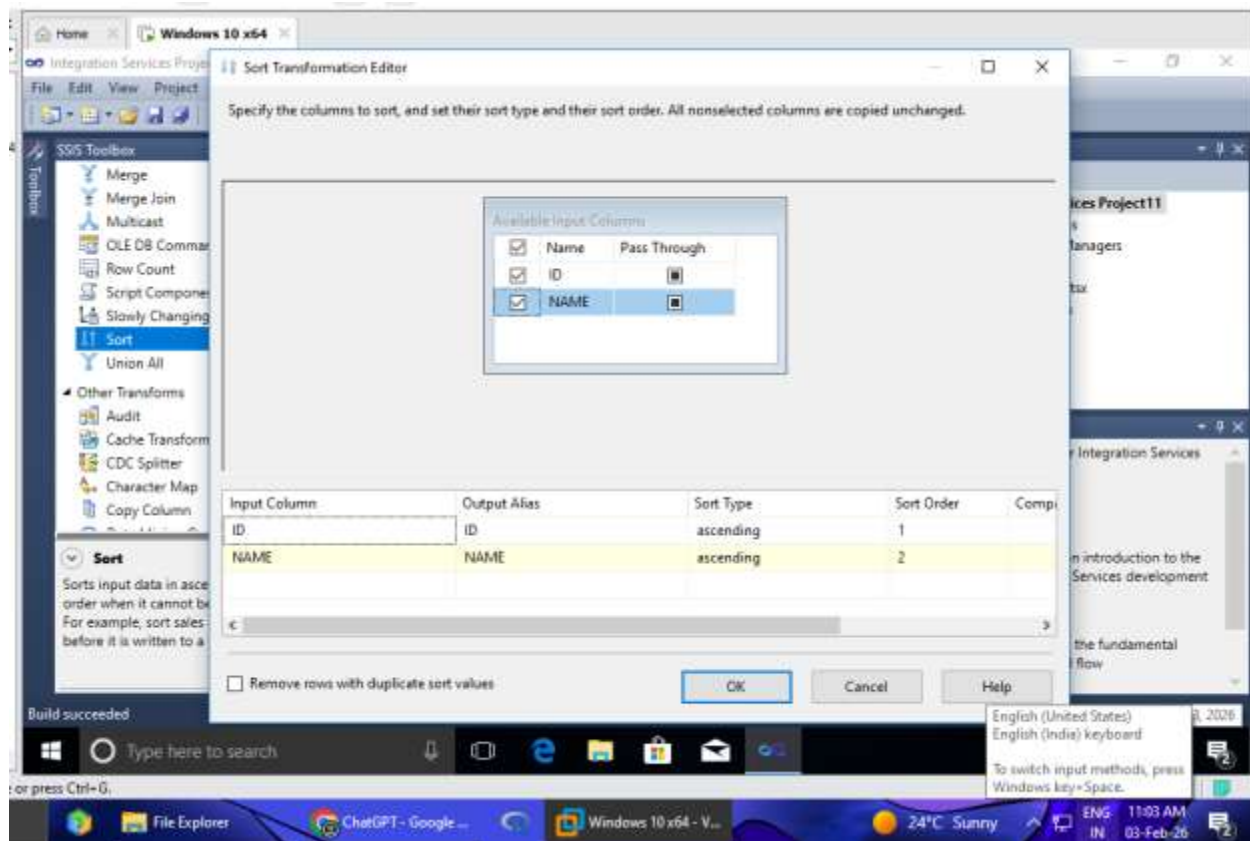
Step 14 : Successfully executed



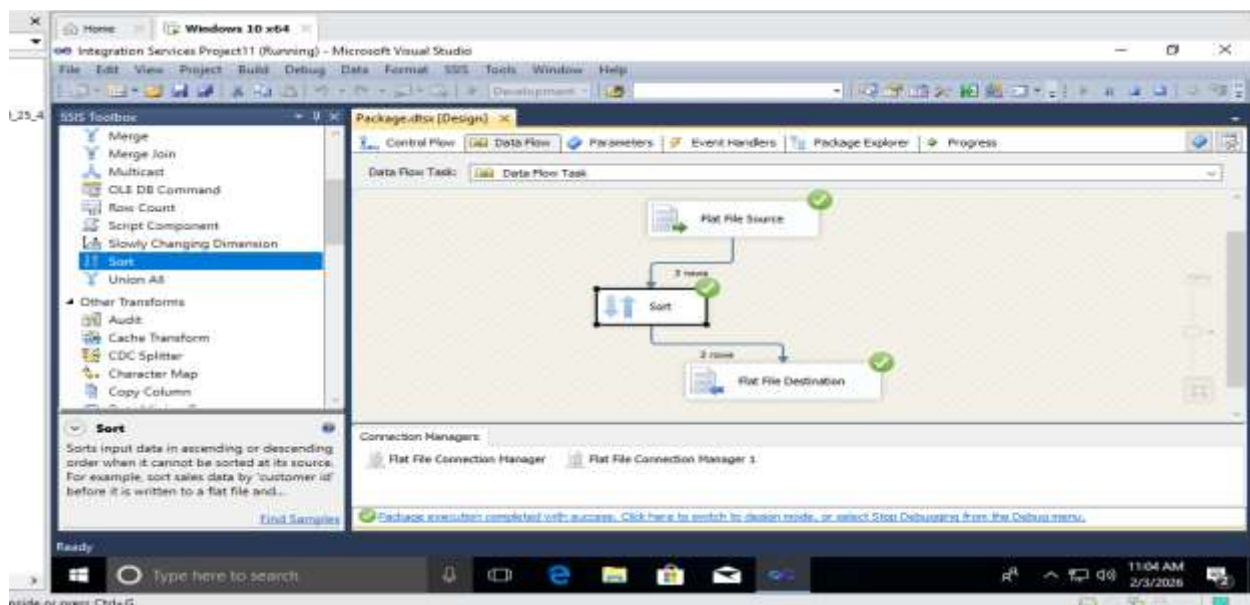
Step 15 : After that we have added one transform method i.e. sort



Step 16: go inside the sort > and choose the column that we want to sort and also the sort type > ok



Step 17: Successfully executed

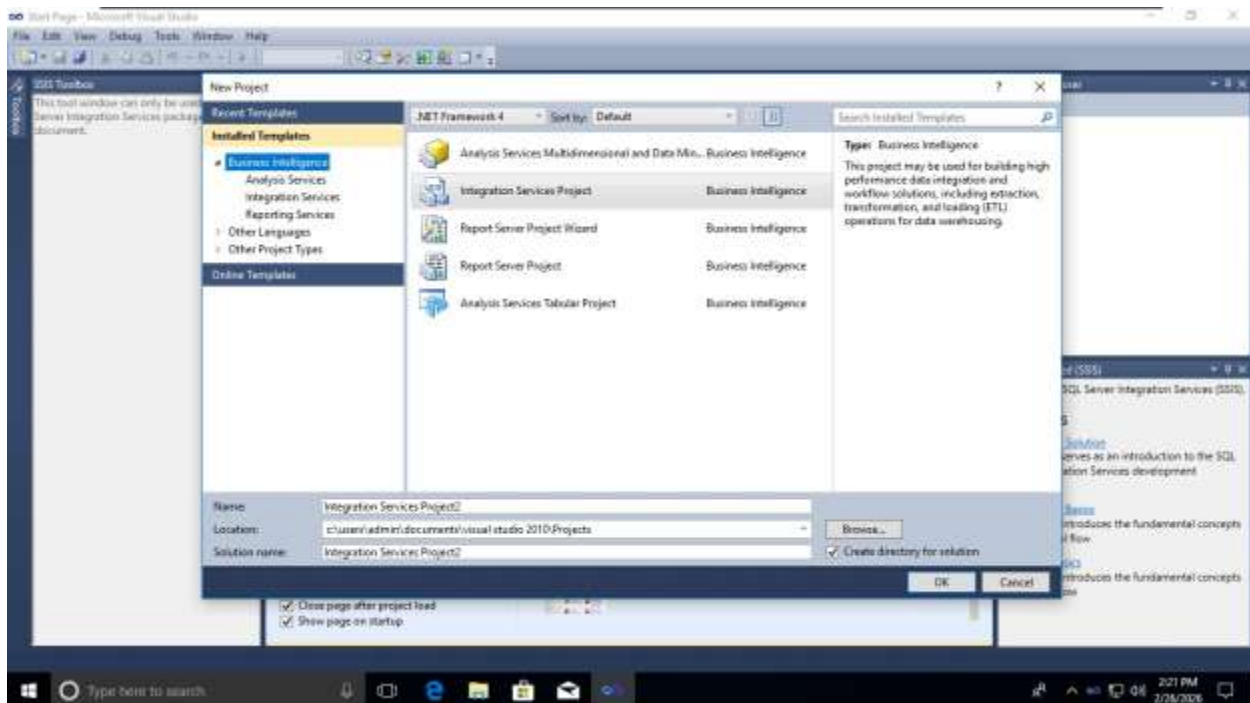


Practical 3

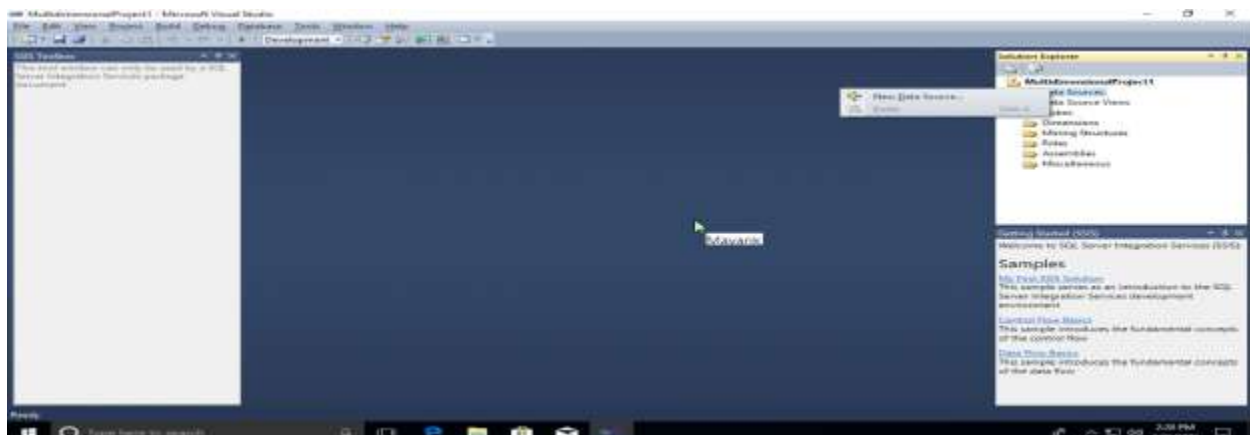
b. Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.

Step 1 : Open SQL Data Server Tool and create a new project

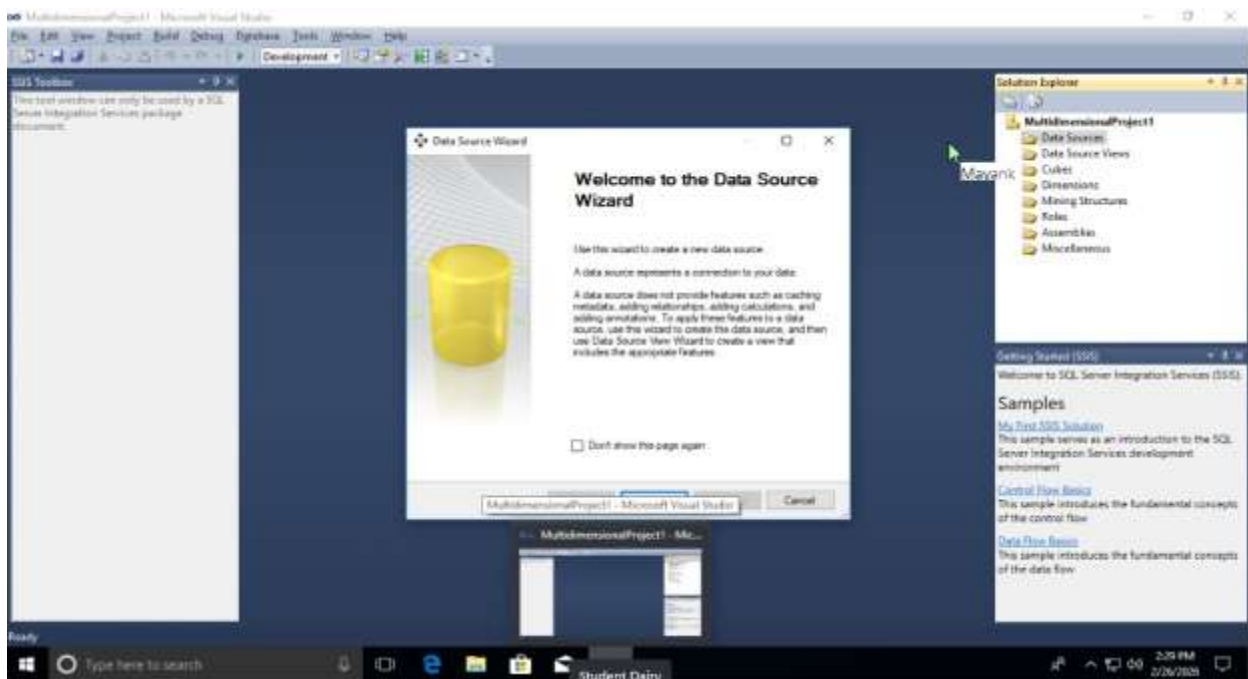
While creating select : “Analysis services Multidimensional”



Step 2 : In solution explorer Locate to data sources > Right click > New Data source

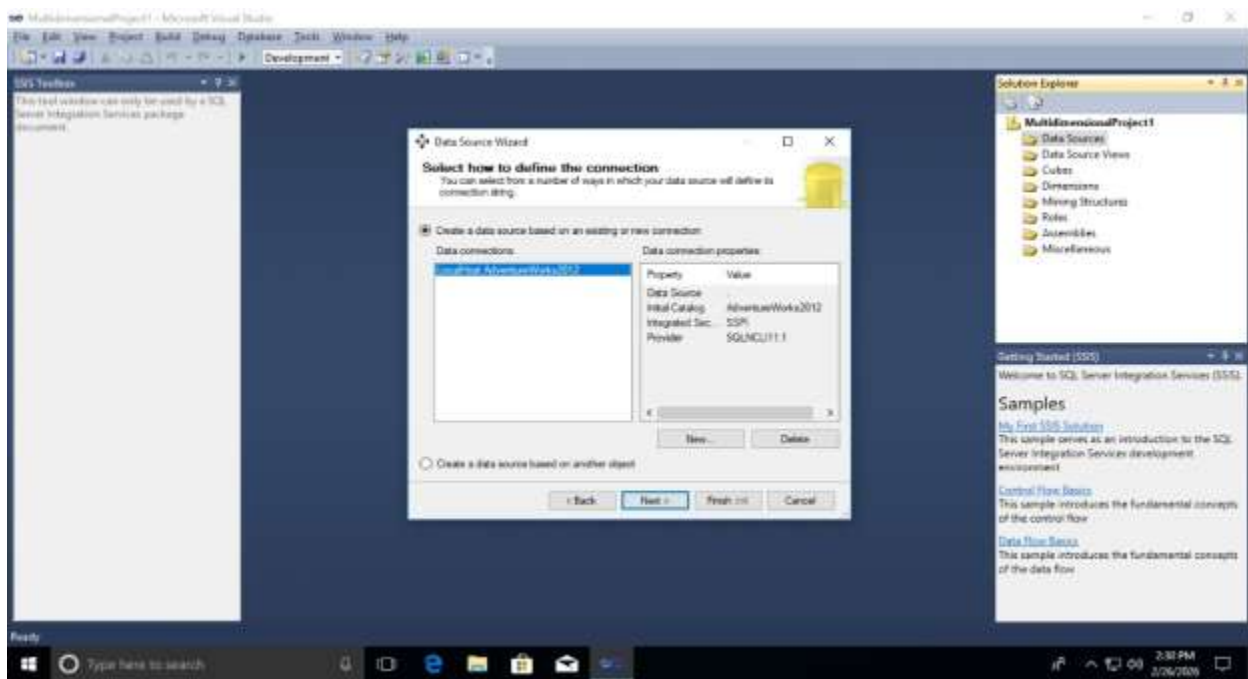


It will show a pop-up like this (Data Source Wizard)

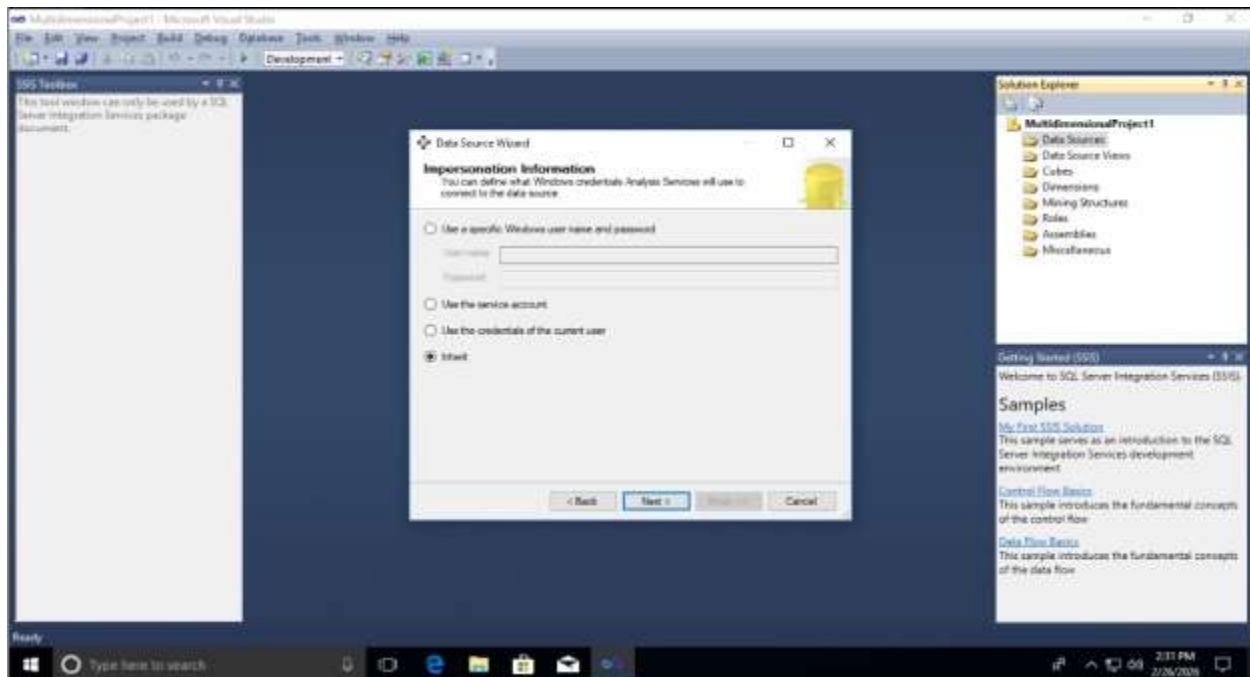


In this click next

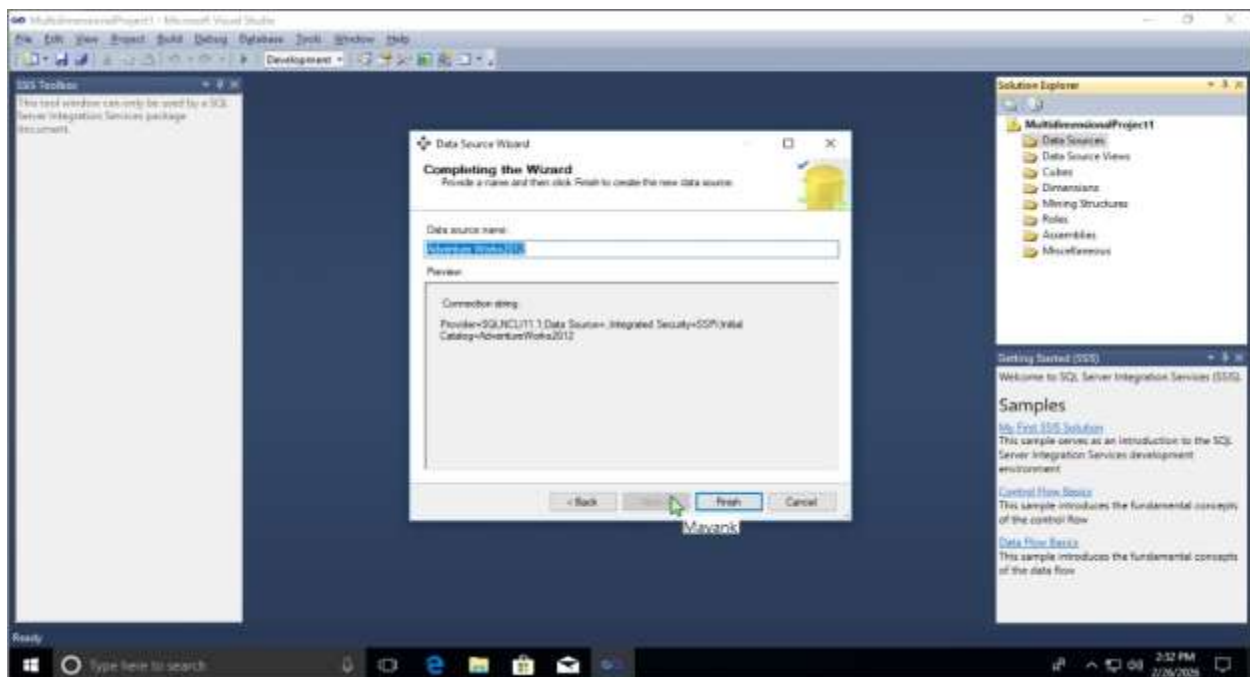
Step 3 : Select AdventureWorks2012 Database and click next



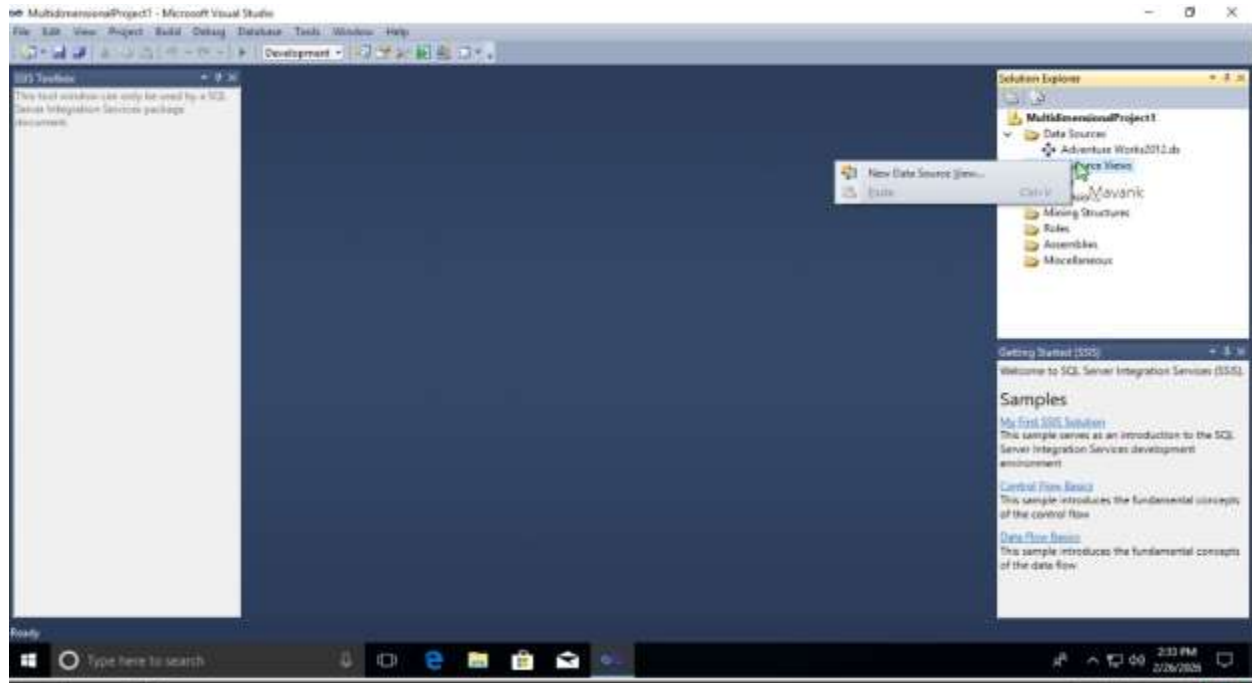
Step 4 : In Impersonation Information window Select The “Inherit” Option and click next



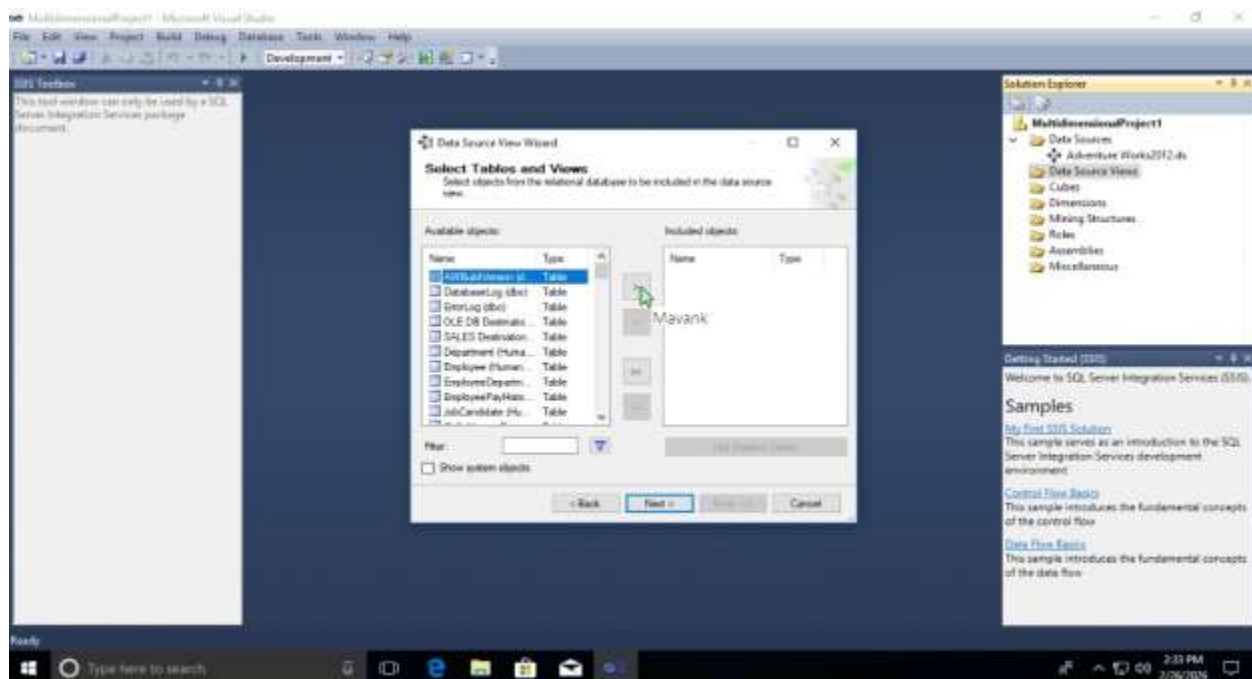
Then click on Finish



Step 5 : Locate again to solution explorer and right click on “Data source View” > click on New Data Source View

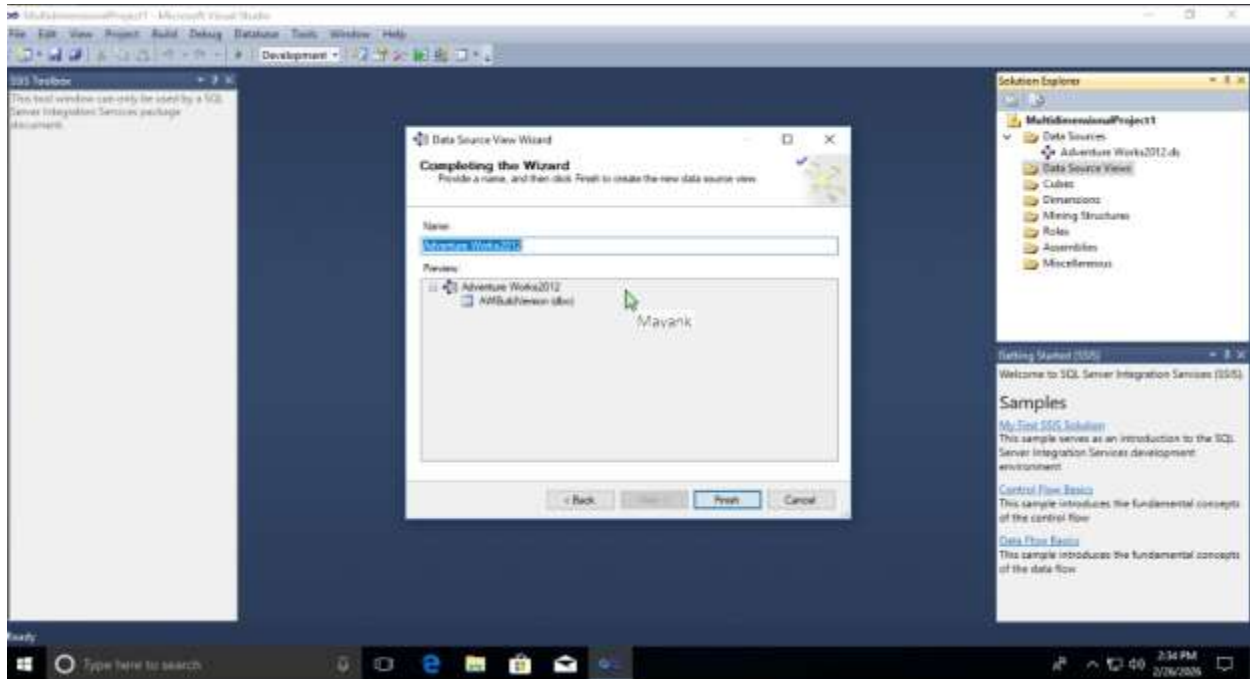


Step 6 : while selecting tables and views select any of the tables and click on “>” Button for adding that Table or view and press next

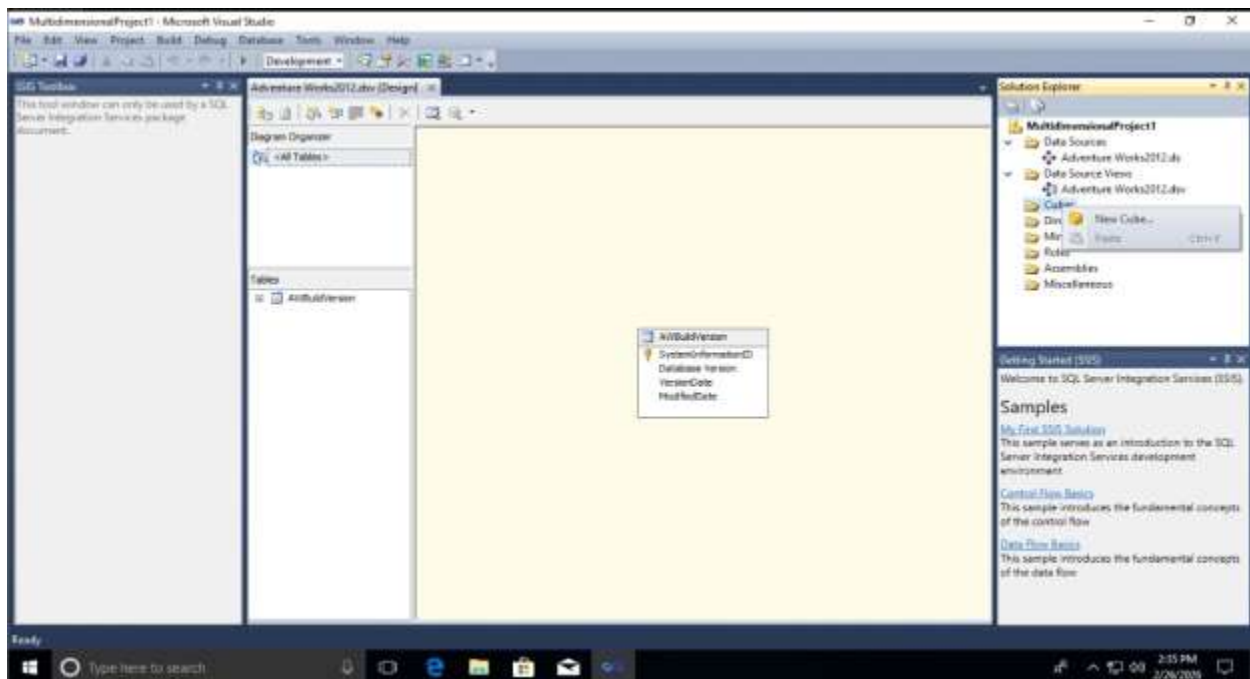


Click Next > Next

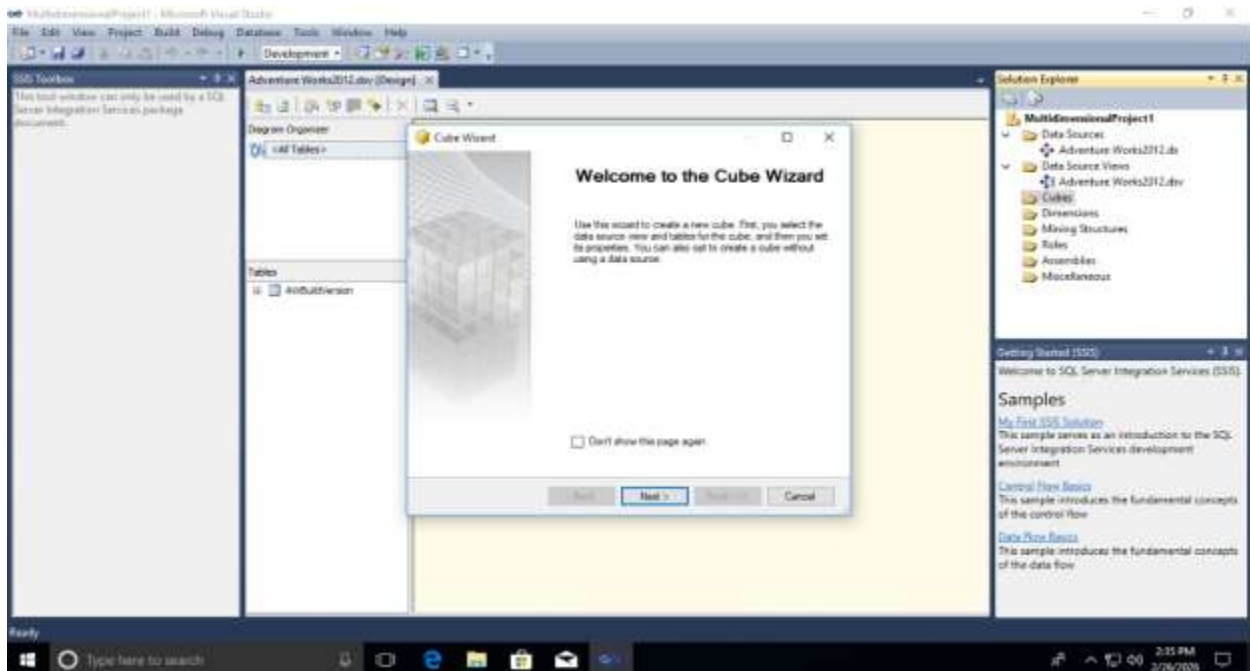
Step 6 : In completing wizard widow check for the database name and click on finish



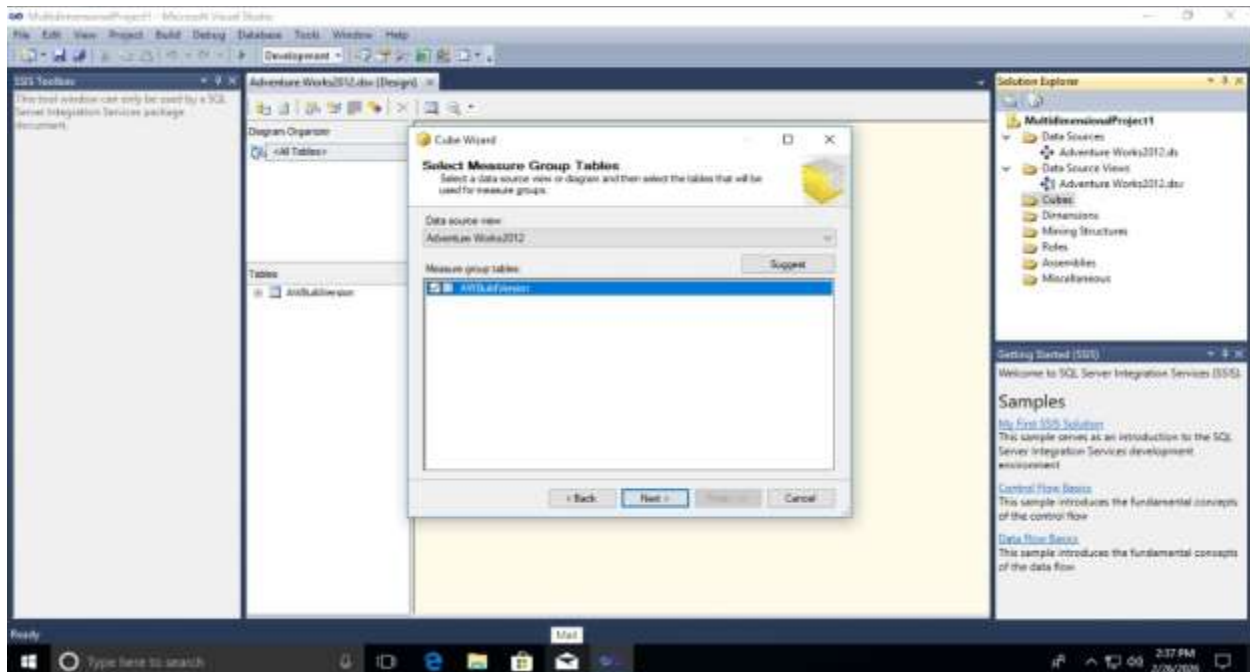
Step 7 : Go to Solution explorer > Right click on cube > New Cube



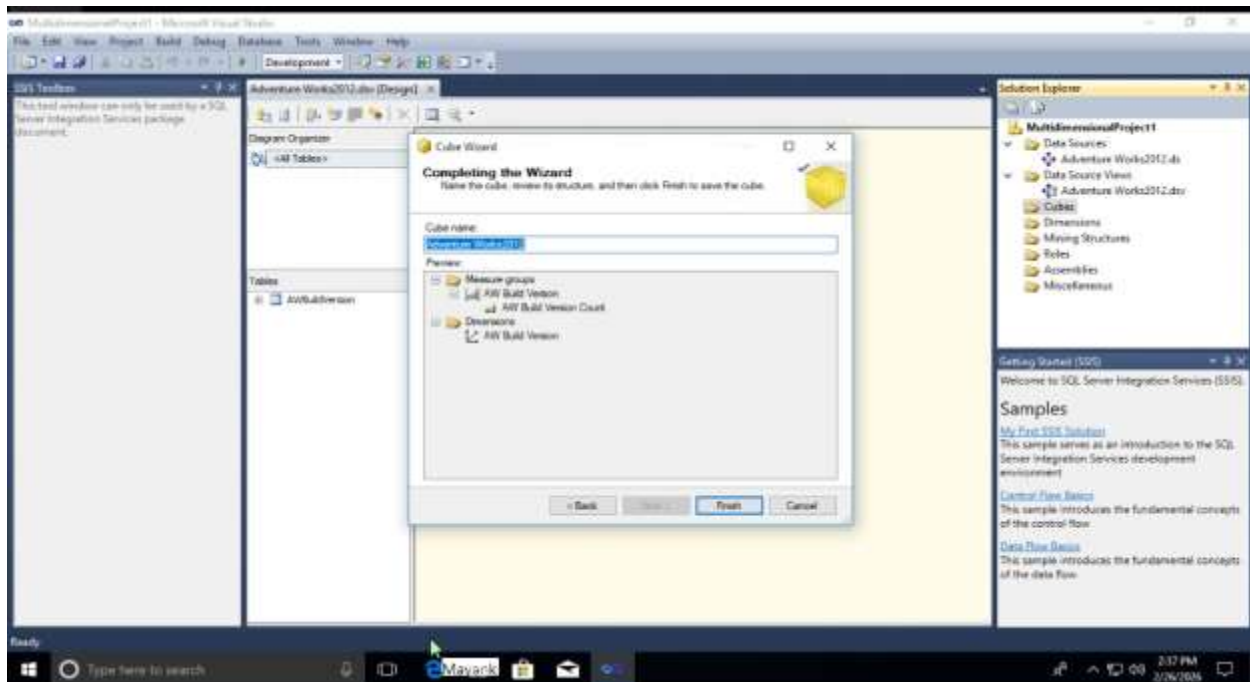
Step 8 : Inside cube wizard > click next



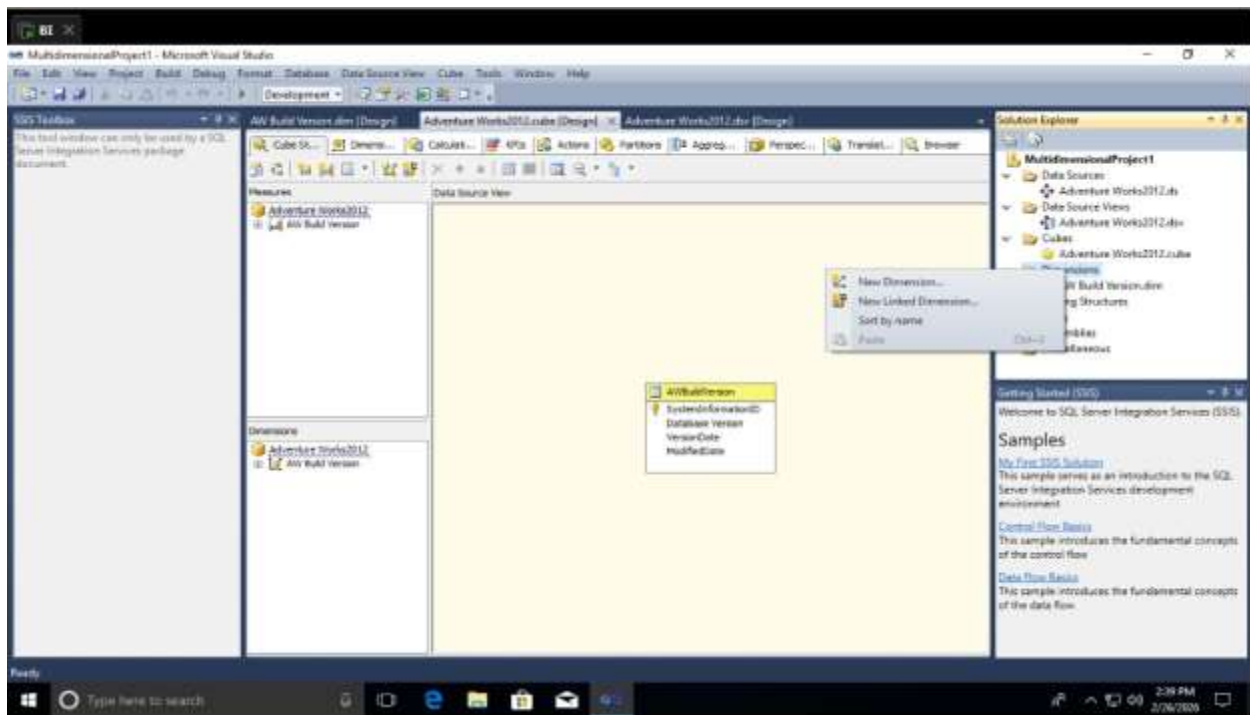
Step 9 : In selecting measure group tables > select All boxes and click on next



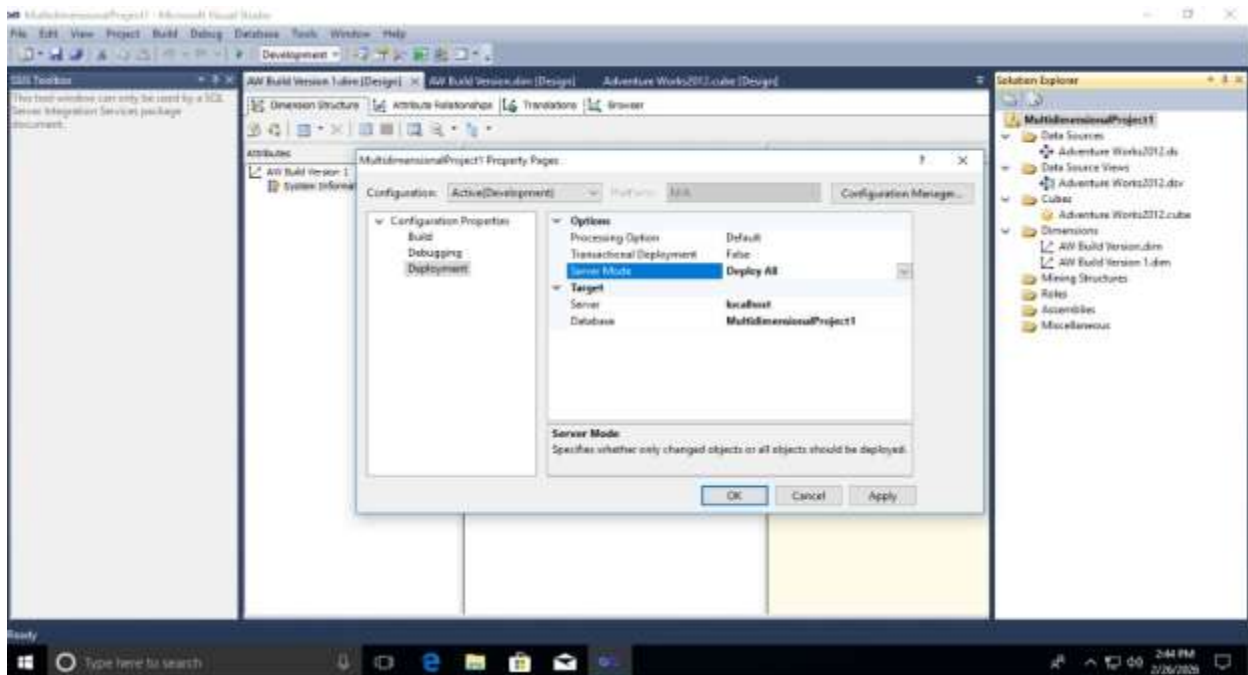
Click on finish



Step 10 : Do the same for dimensions which is located in solution explorer

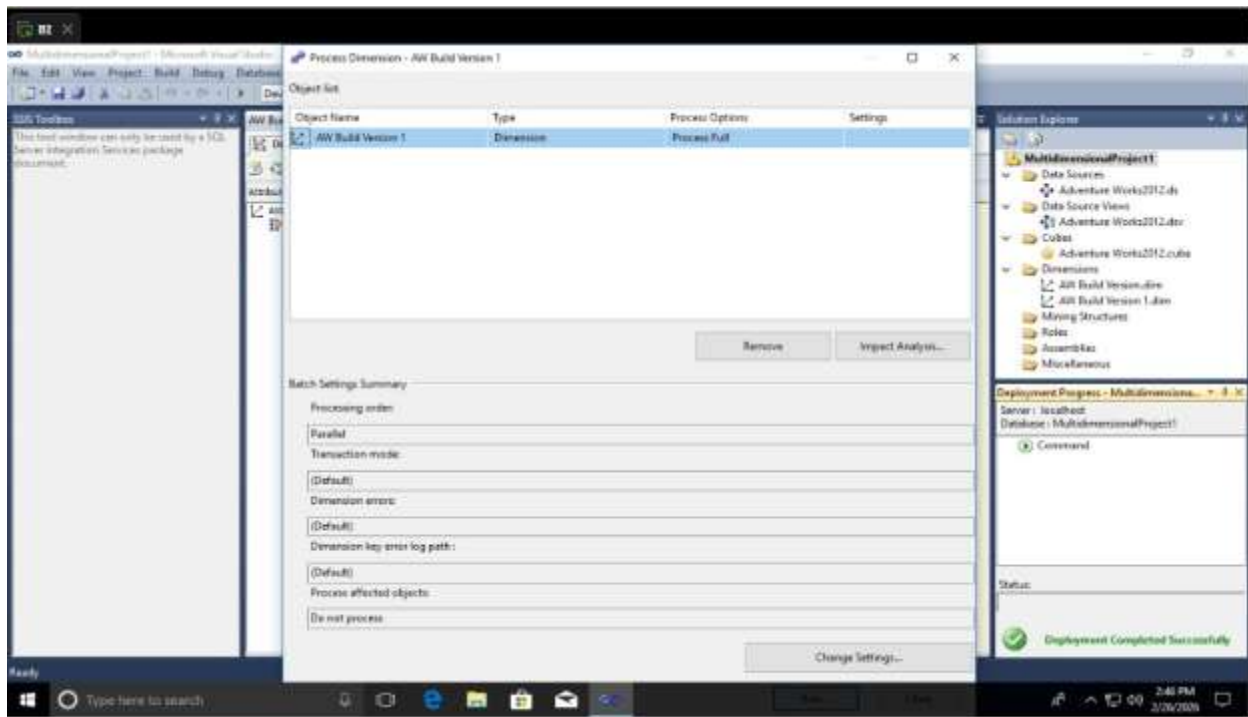


Go to dimension property page > change server mode to “Deploy All”

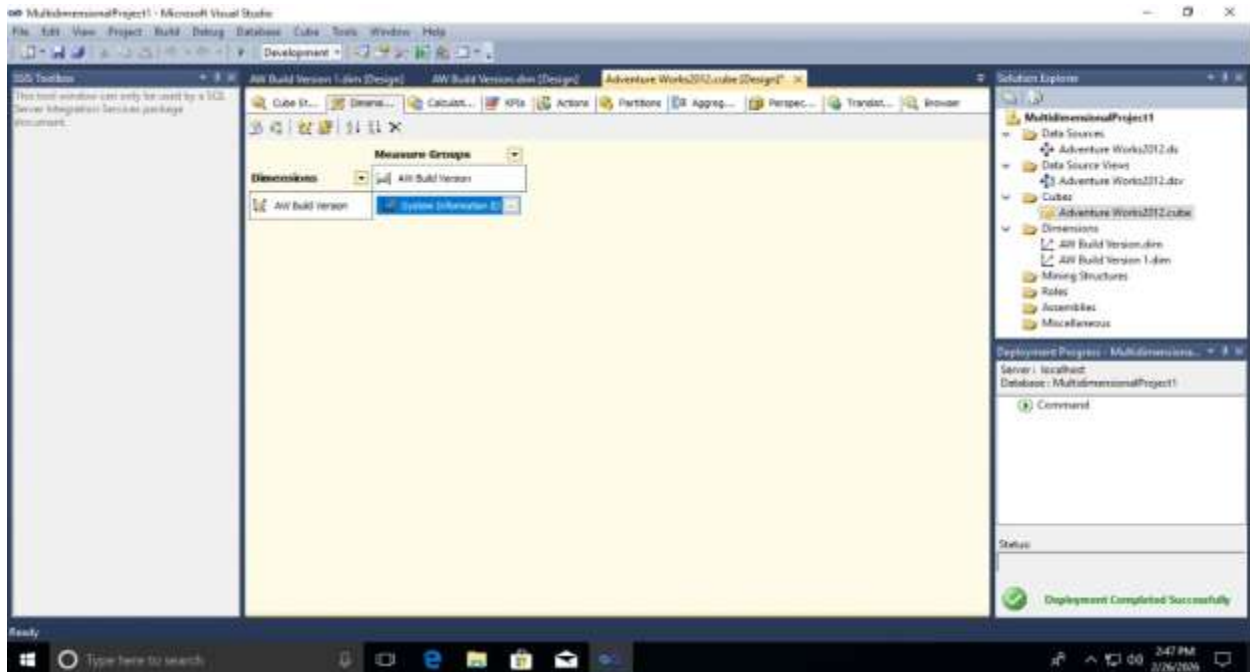


Then click on Process button which is located on the main widow

If deployment is successful then you will an output like this



Step 11 : Now go to design tab and click on Execute button and check for the output



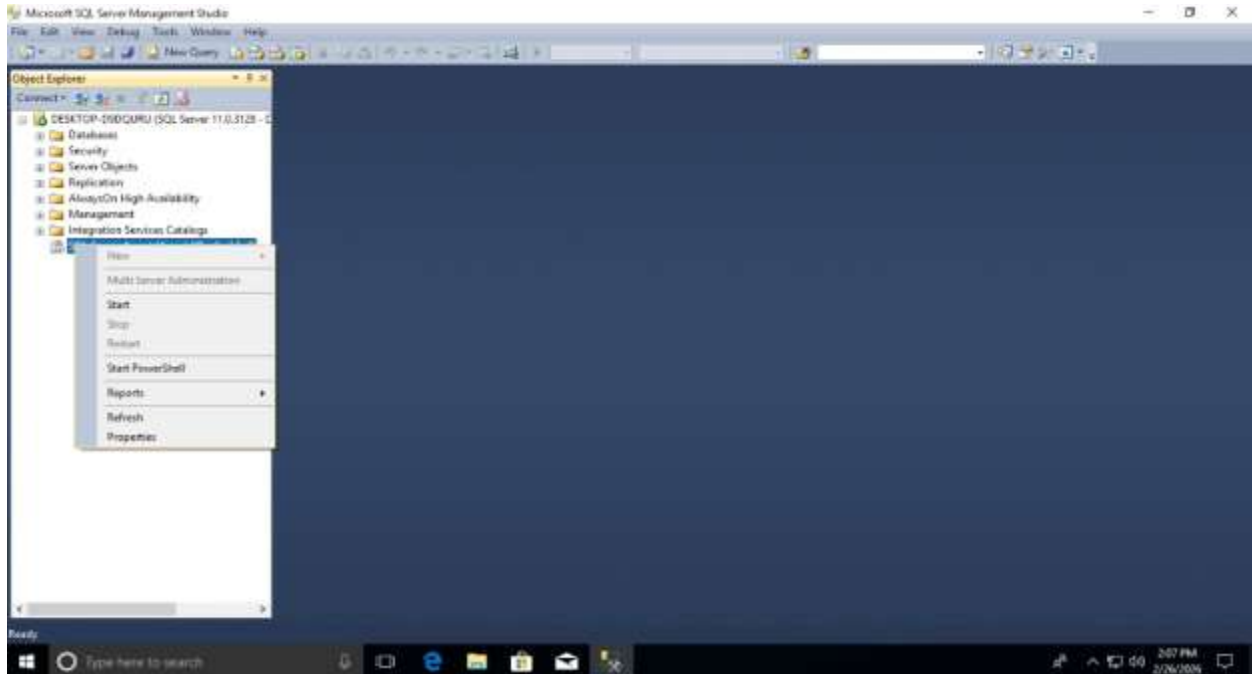
Practical 4

- a. Create the ETL map and setup the schedule for execution.

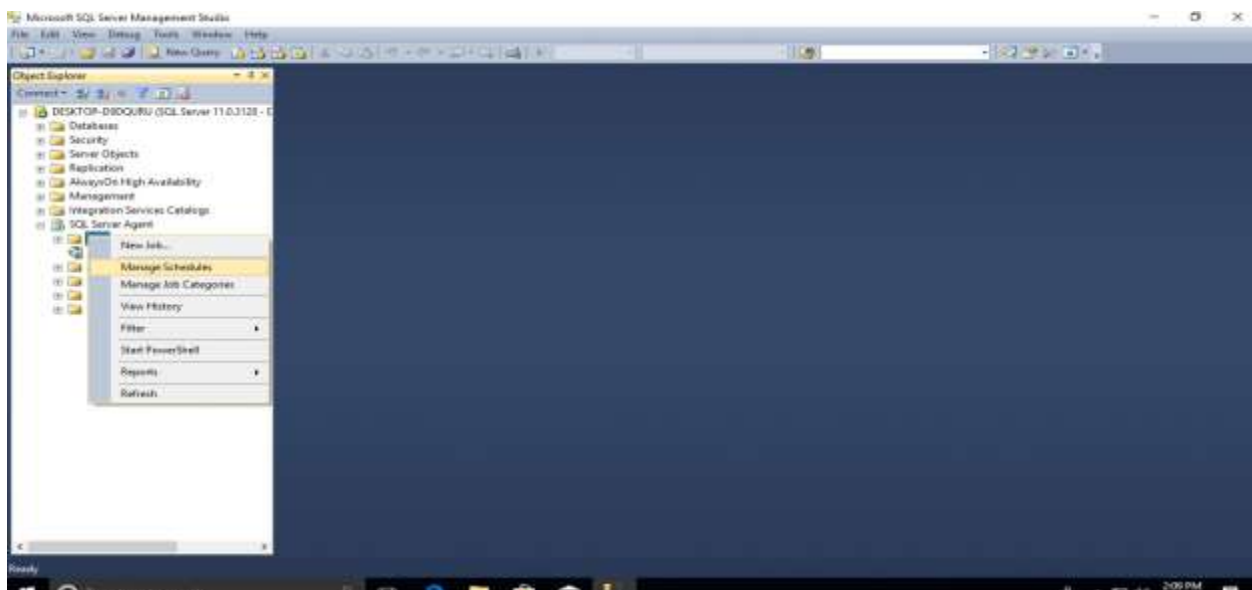
Step 1 : Open SQL server Management studio

In the Object Explorer Locate find SQL server agent

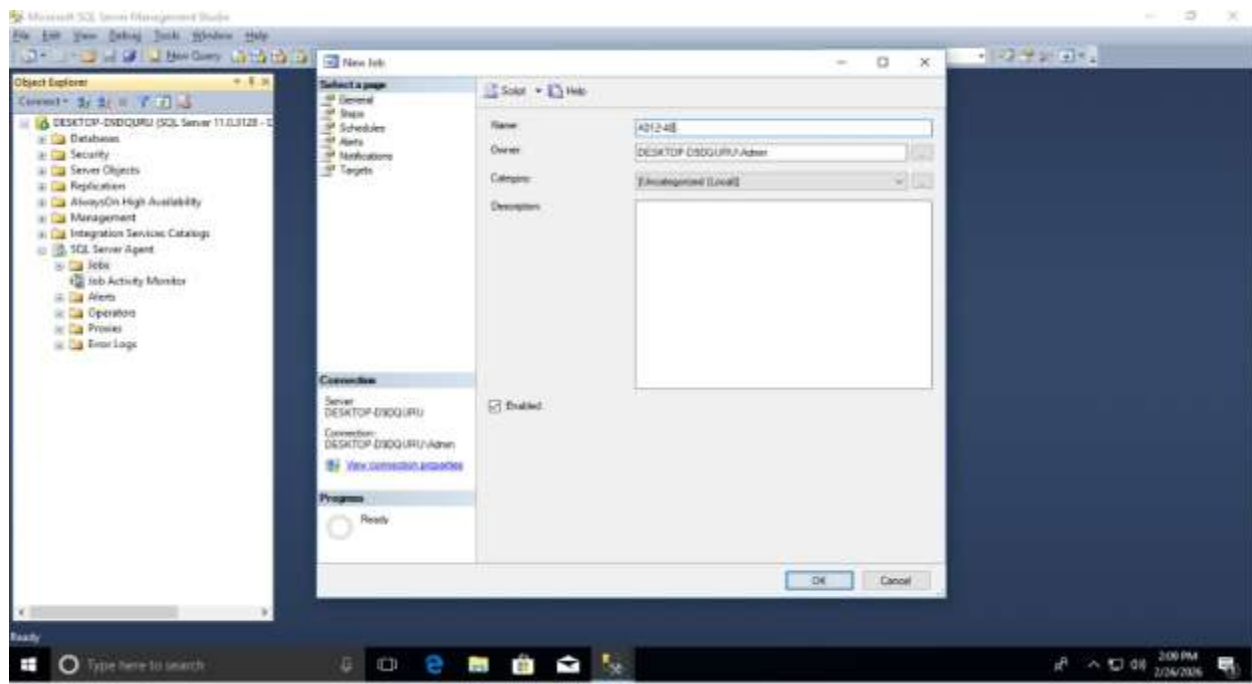
Right Click to it and click start



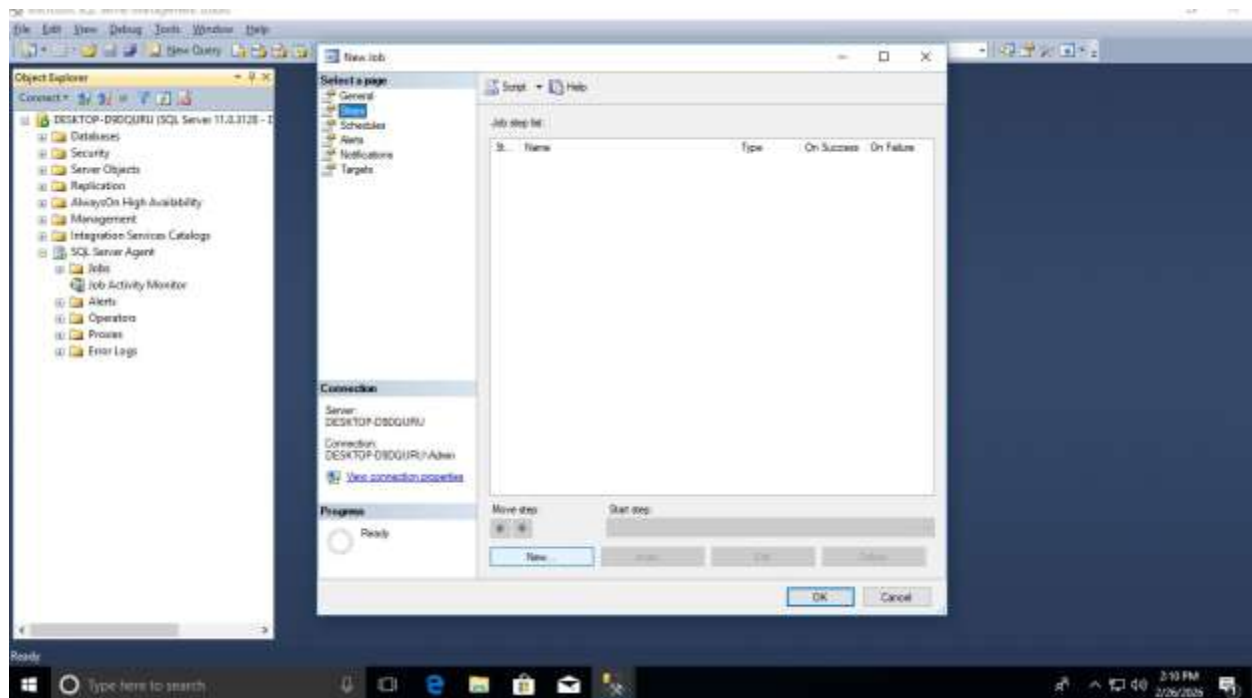
Step 2 : then right click on JOB > New Job



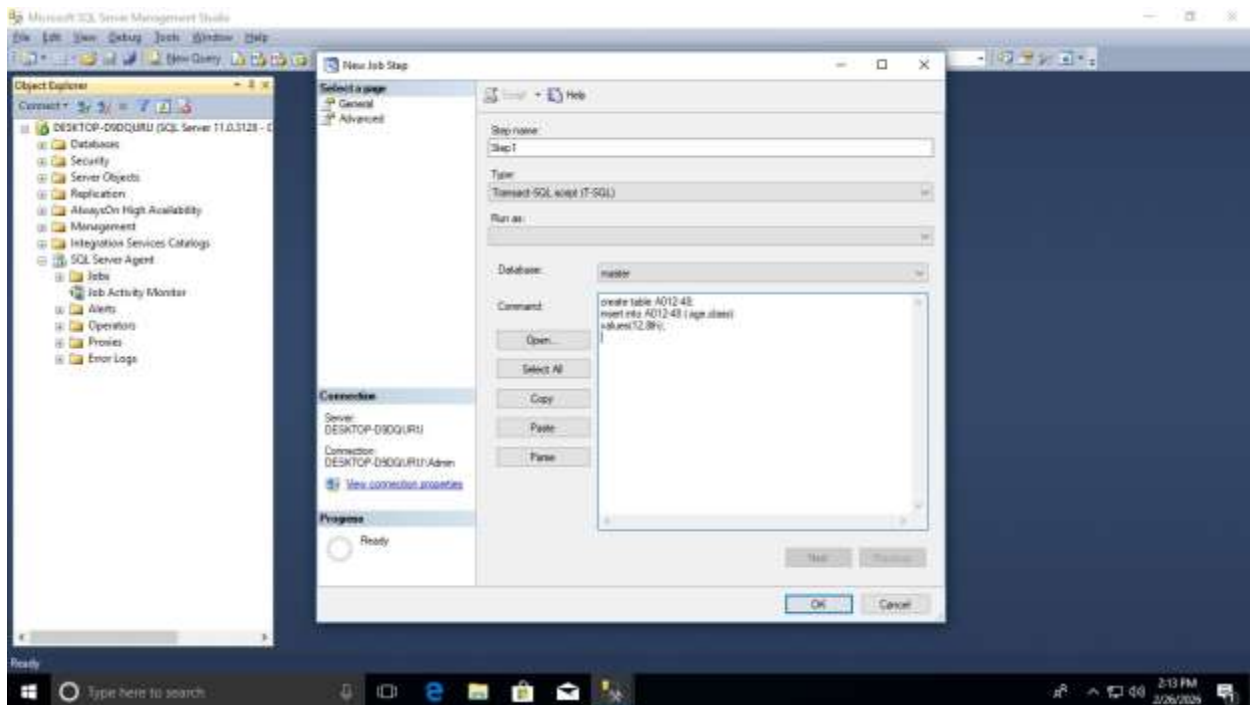
Step 3 : in general section > assign any name



Step 4 : Go to steps > Click on new

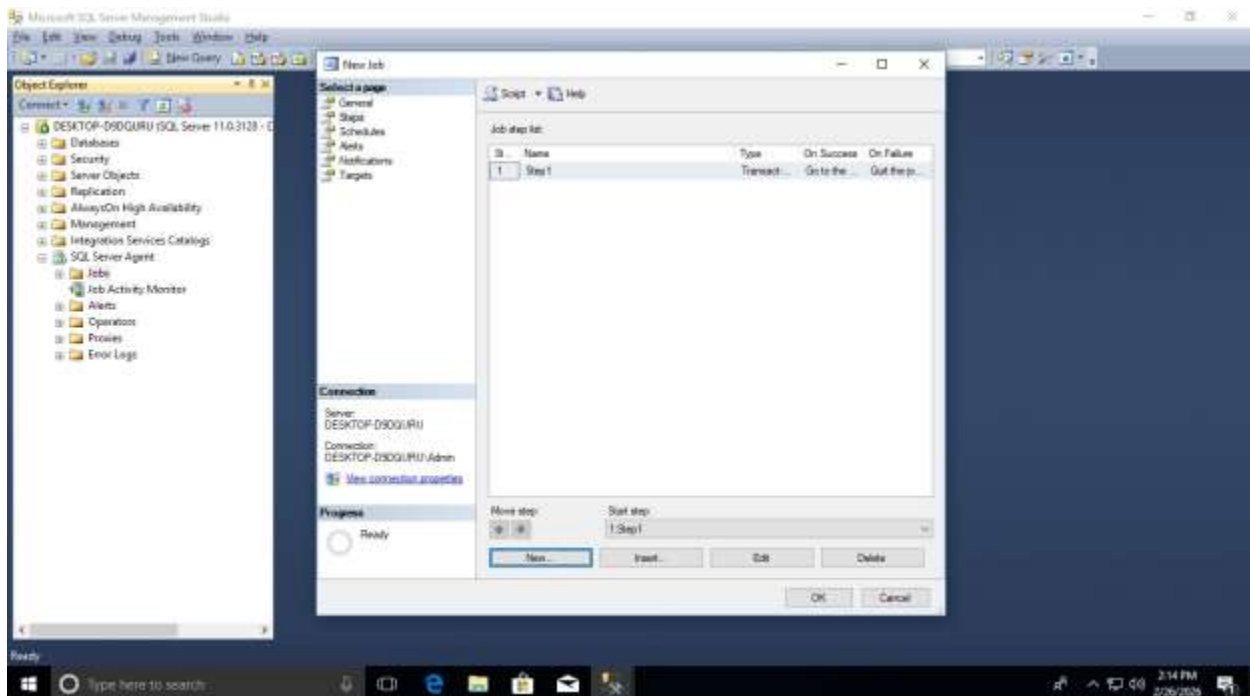


After clicking on new > assign a name to it and in command section enter the command as shown in the image

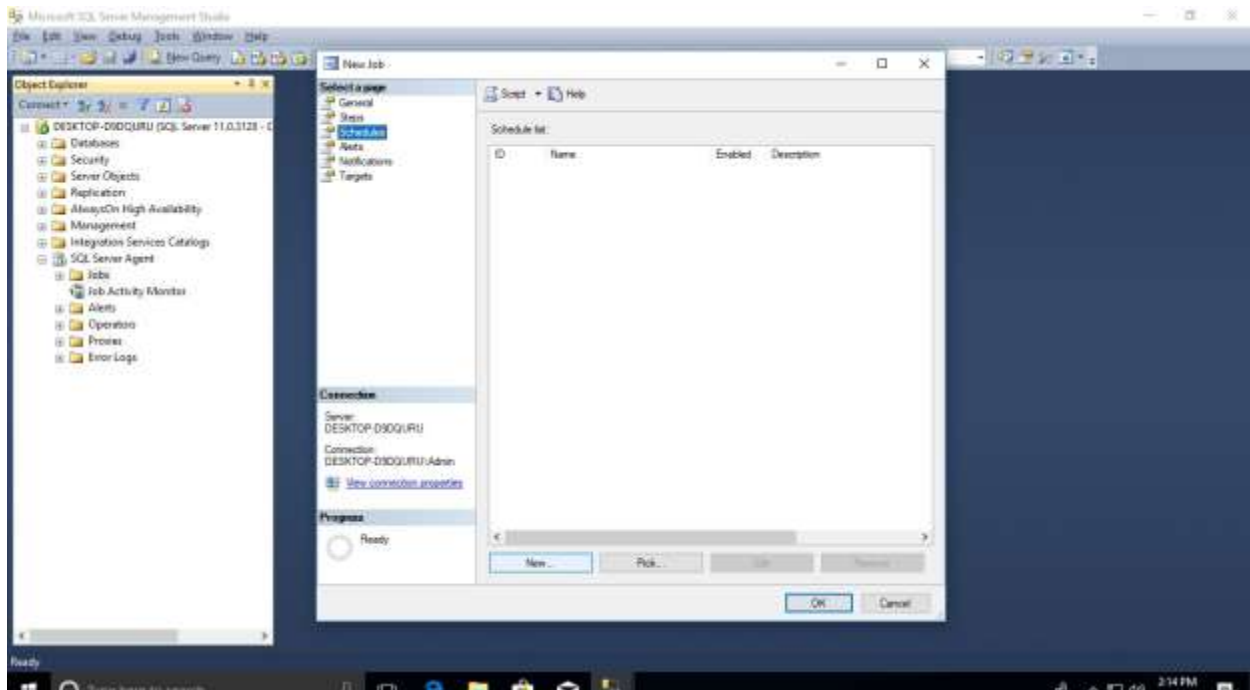


And click ok

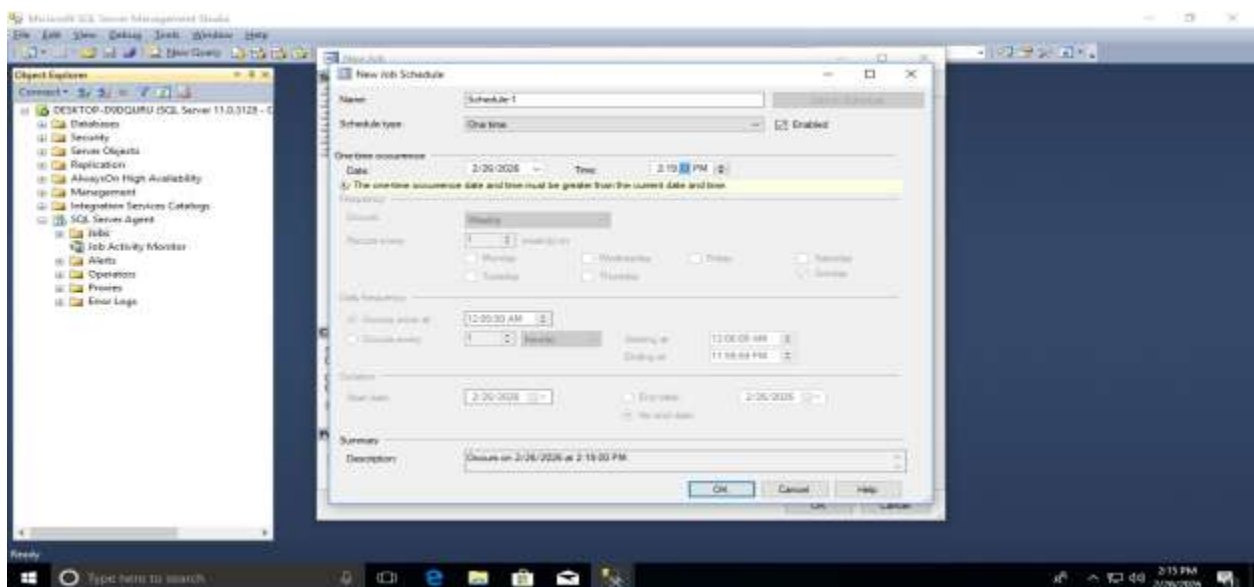
You will see a step getting added in the steps section



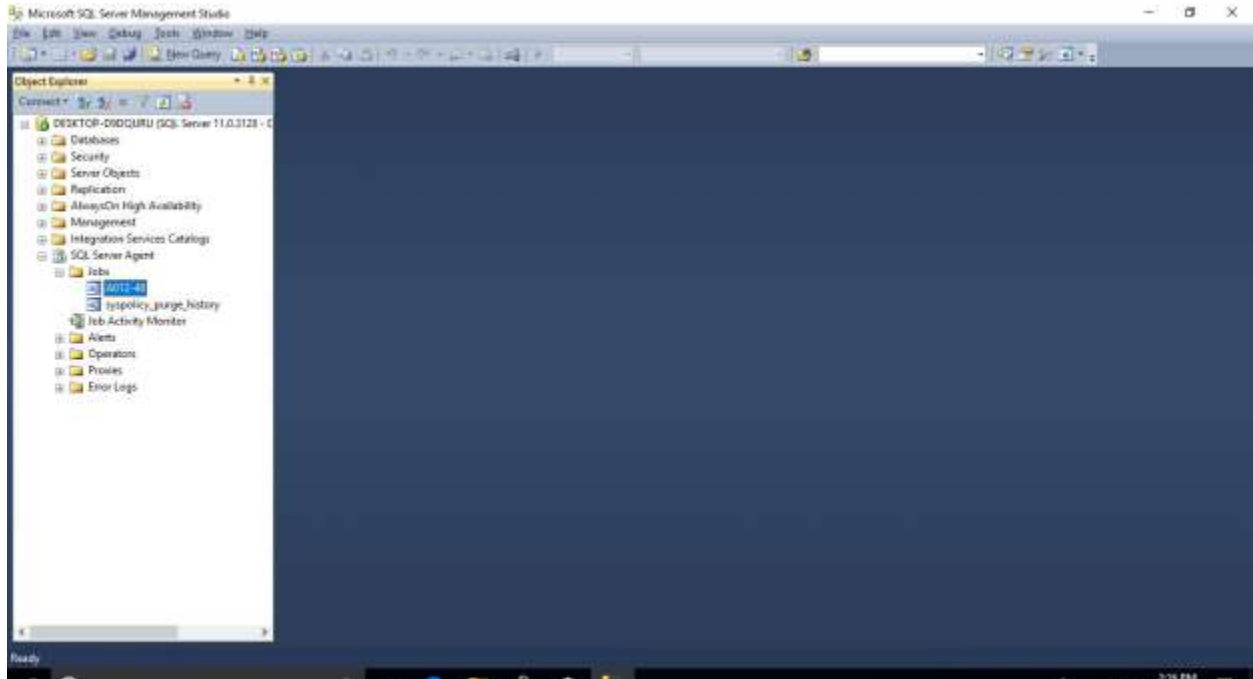
Step 5 : Go to schedules > Click on new



- Name that schedule as per your choice
- And in “Scheduling type” Select “One Time” from the dropdown
- Assign the time when you want that job to be executed and click on OK



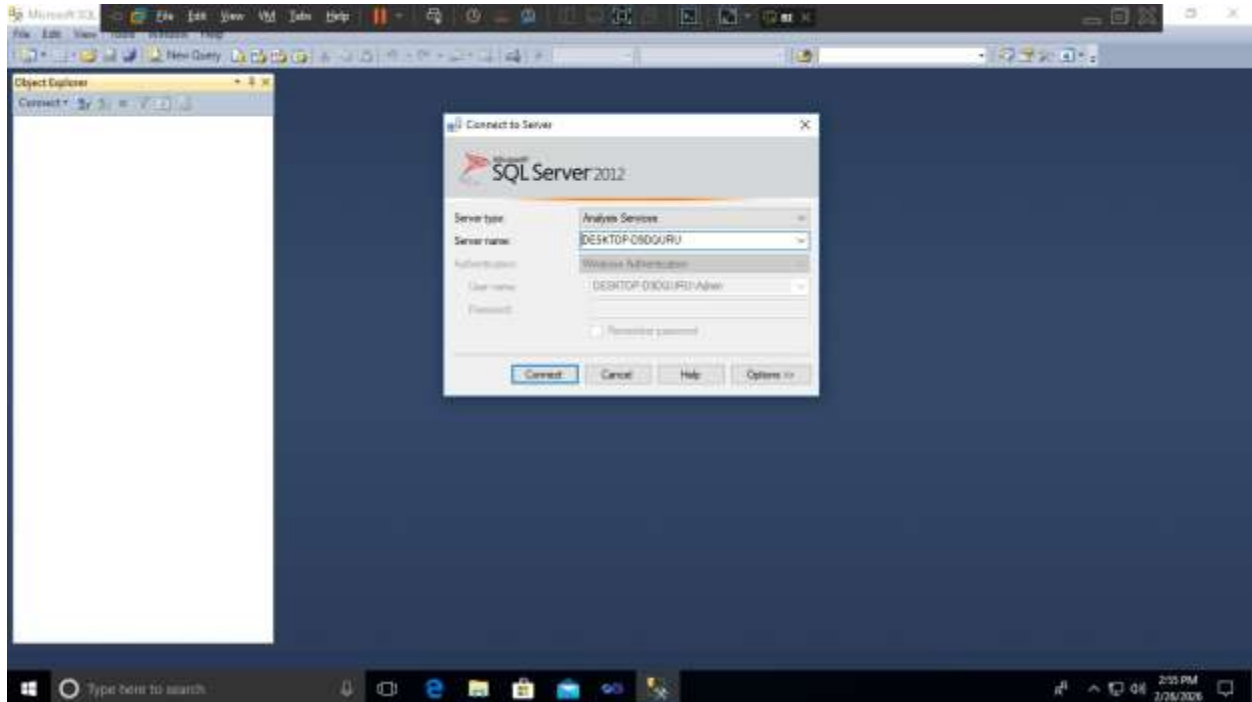
Step 6 : After that job gets executed you will see that in the jobs section your job gets added and to check it has been executed successfully or not open that jobs description



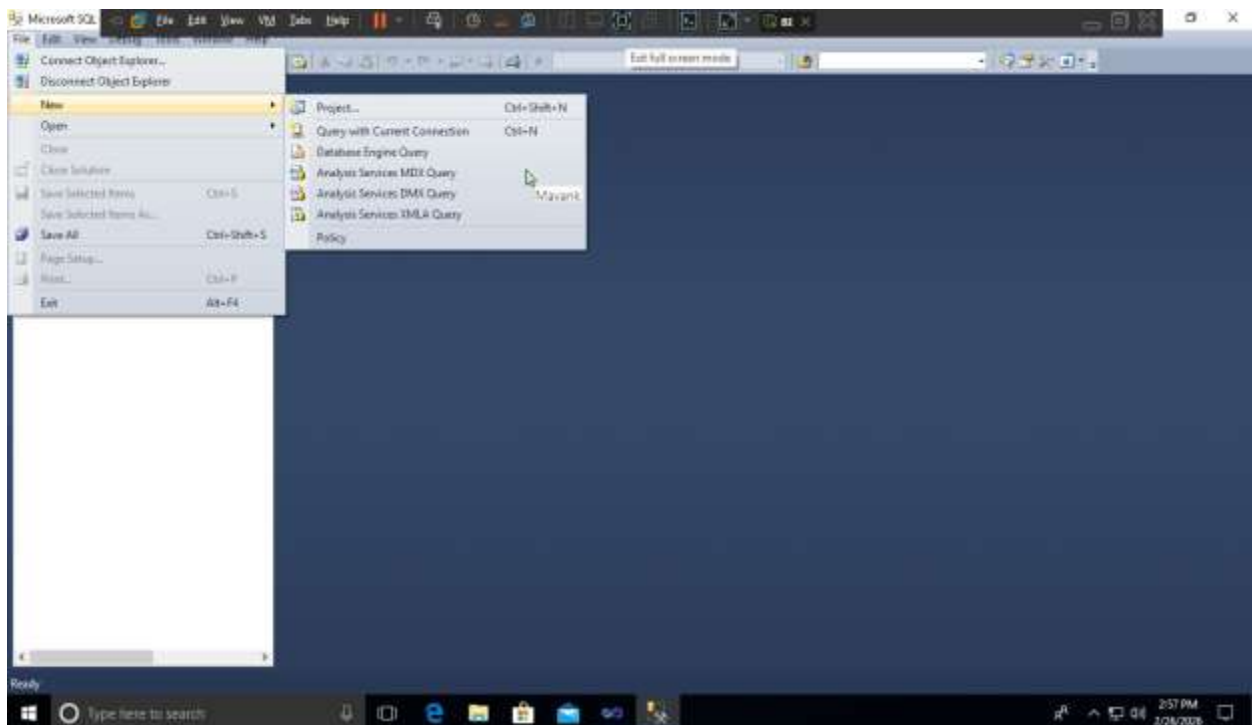
b. Execute the MDX queries to extract the data from the data warehouse.

Step 1 : Open SQL server Management studio

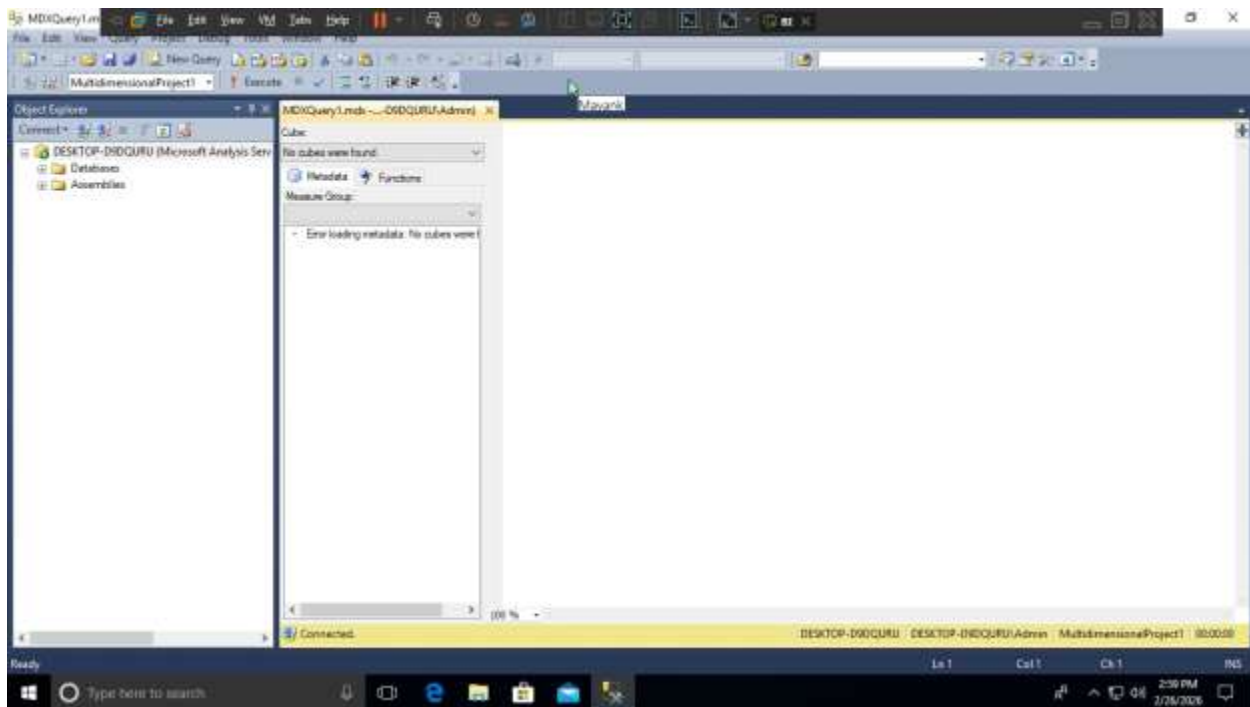
Change server type to “Analysis Services” > Click on connect



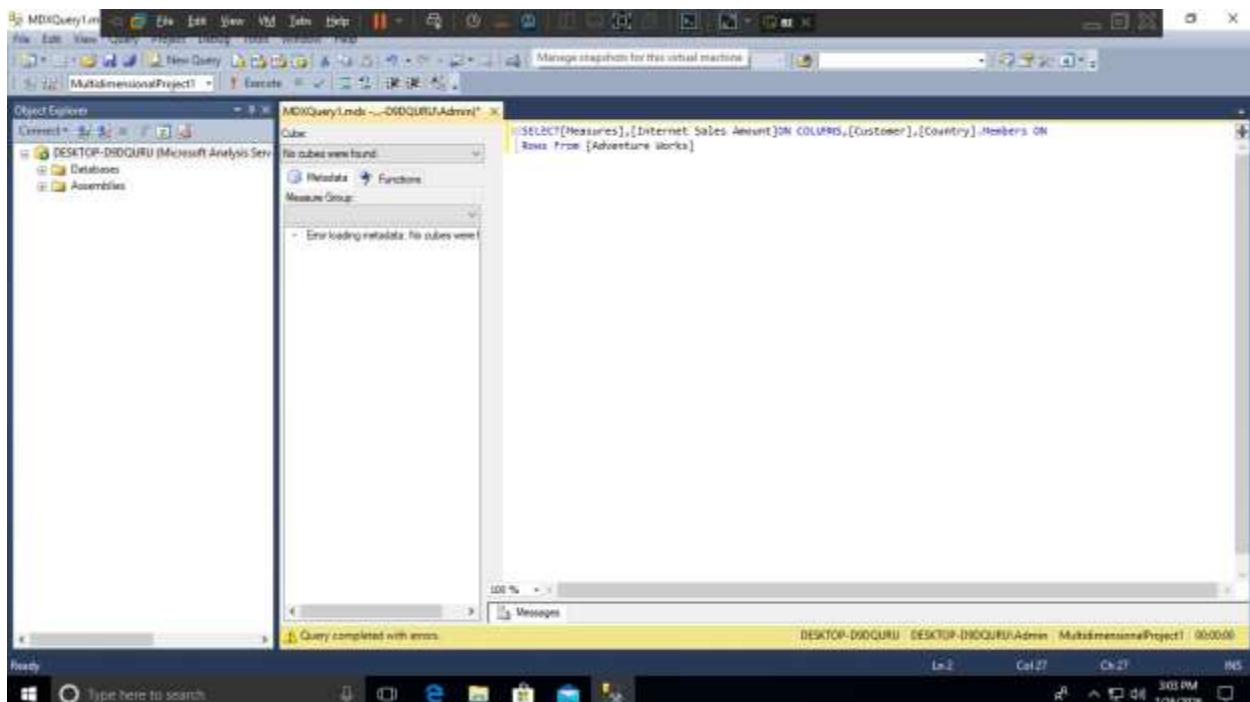
Step 2 : Go to File > New > Analysis Services MDX Query



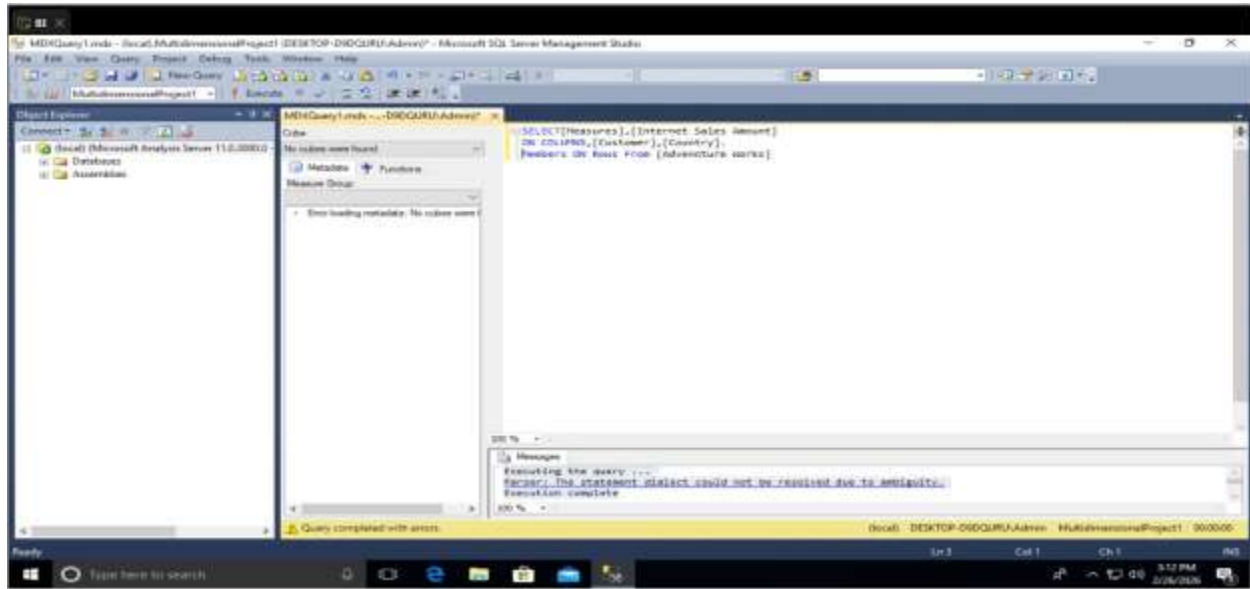
After clicking on that it will a Query Dialogue Box For executing MDX queries



In this enter the query shown in the image > and click on execute



Step 3 : If some error like this appears



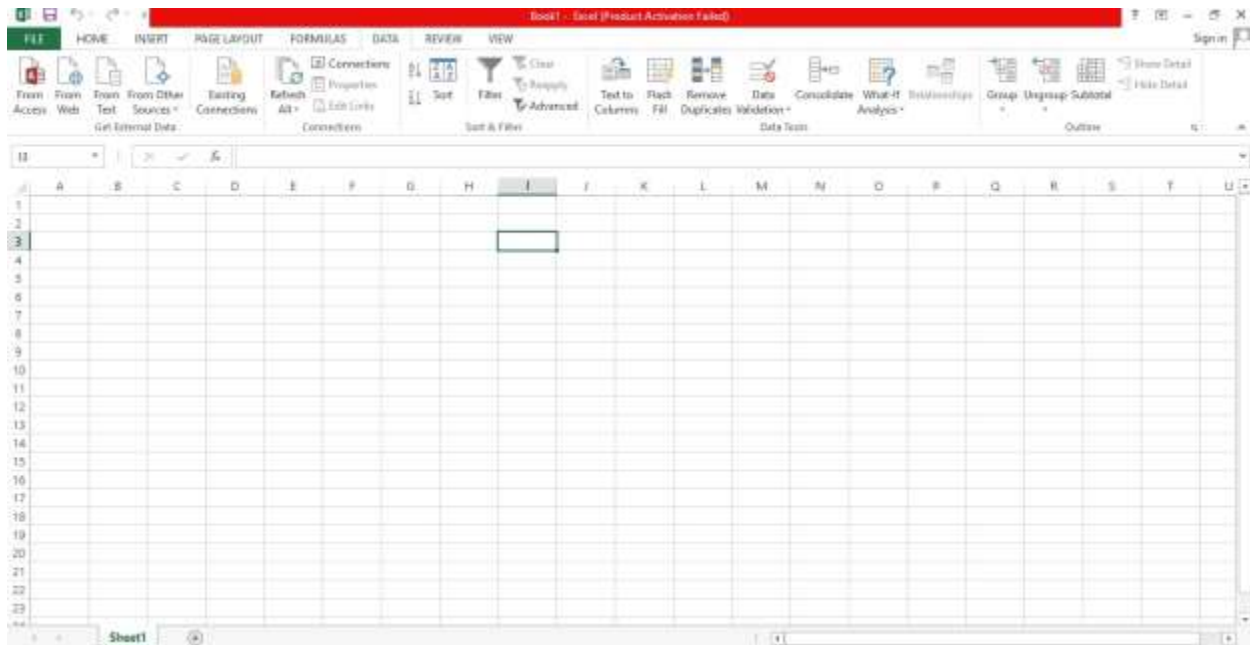
Check for the cube and execute it again.

Practical 5

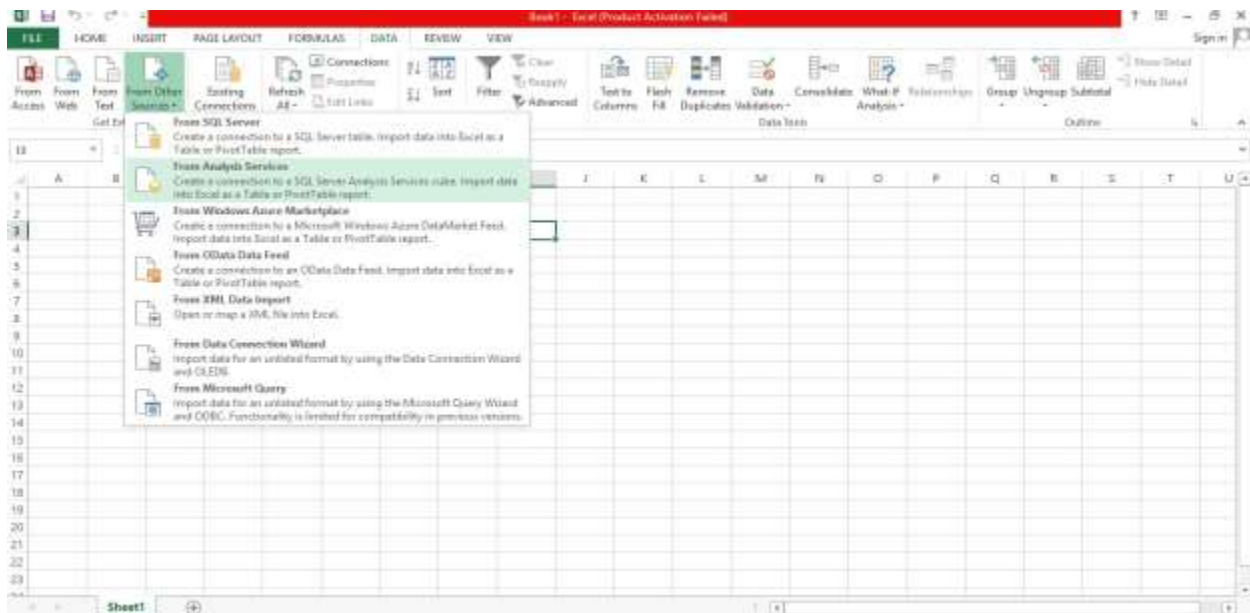
Import the data warehouse data in Microsoft Excel and create the Pivot table and Pivot Chart.

Step 1 : Open Microsoft Excel in VM

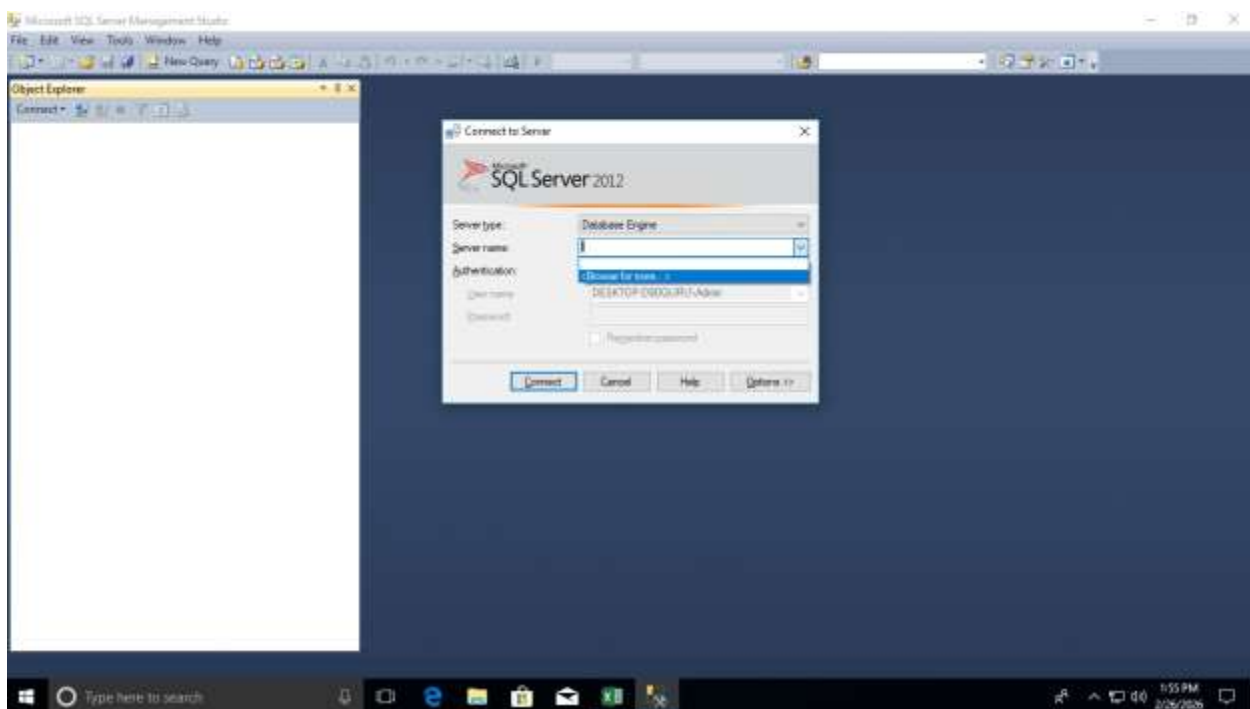
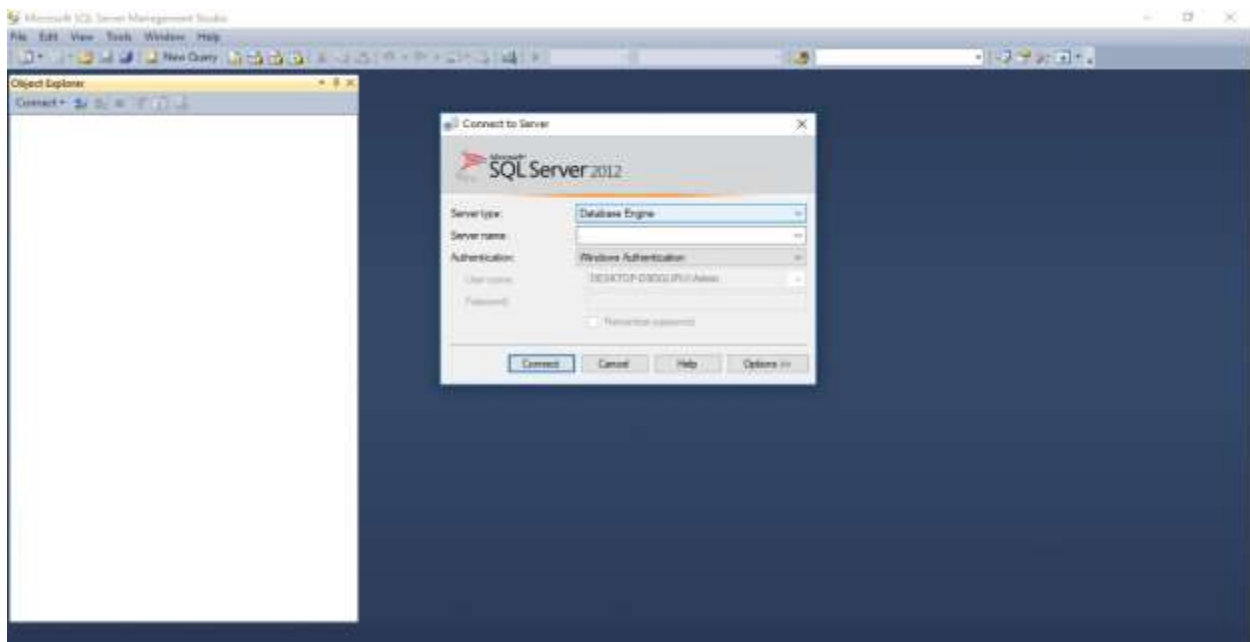
Go to Data > Select “from Other Sources”

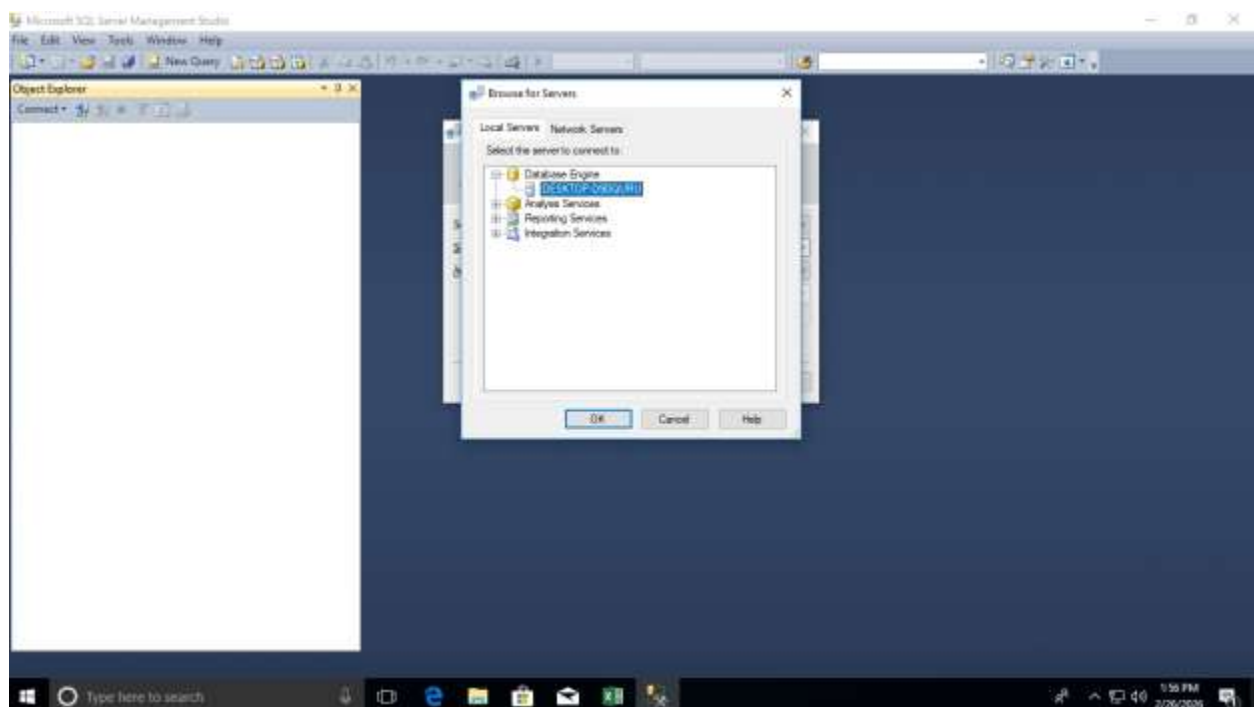


Select “Analysis Services” From Dropdown



Step 2 : Now Open SQL server Management studio and select Database engine > click on connect





Copy this DB name

And paste it in the Database wizard which was previously opened in Excel. Click on Next

Data Connection Wizard

Connect to Database Server

Enter the information required to connect to the database server.

1. Server name:

2. Log on credentials

☒ Use Windows Authentication

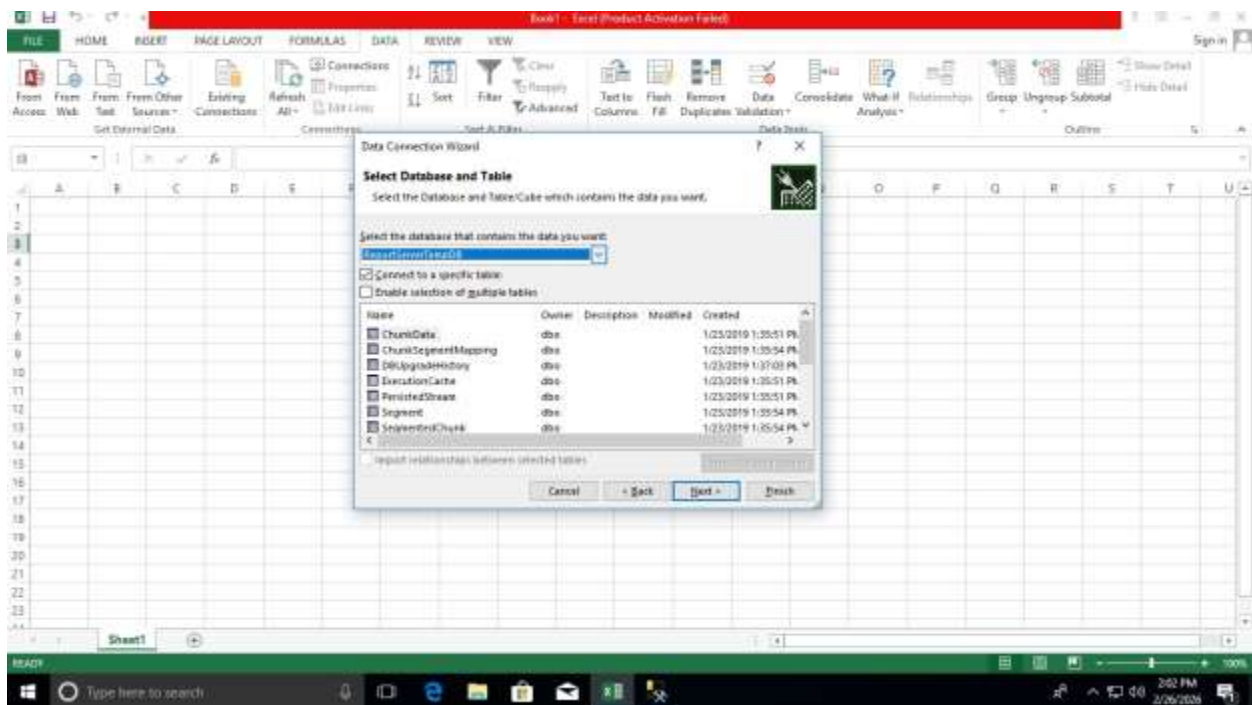
☐ Use the following User Name and Password

User Name:

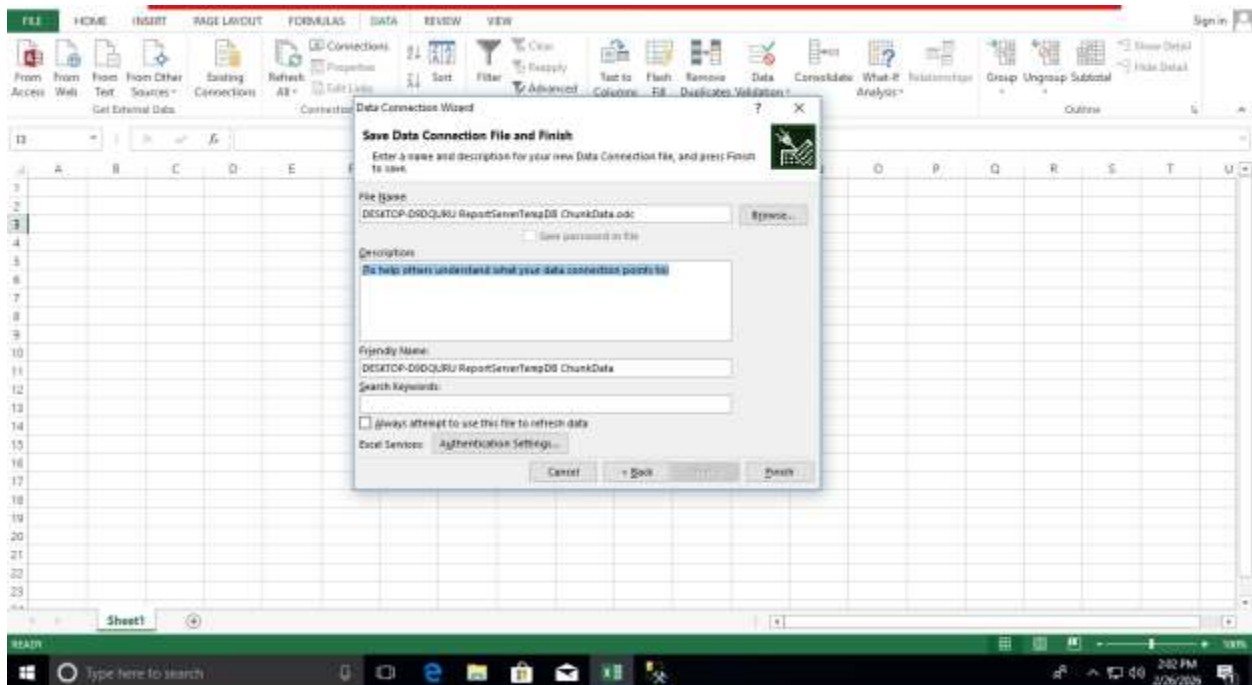
Password:

Step 3 : Then select any of on the DB which contains actual data

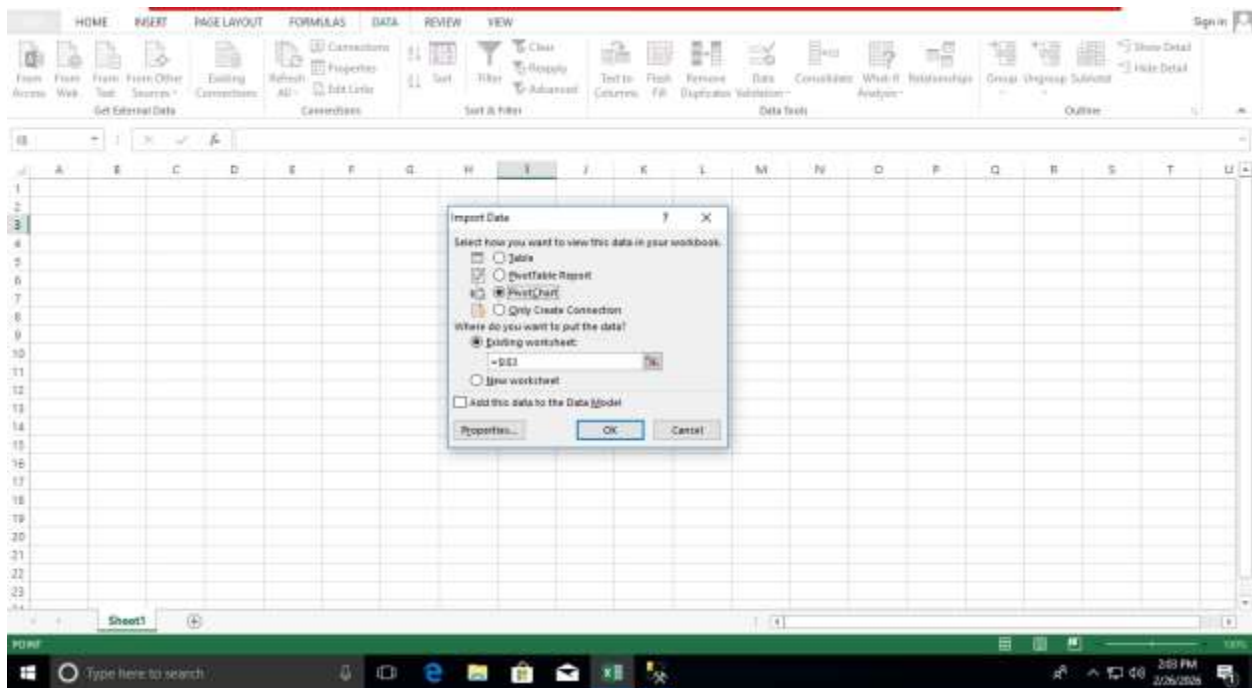
And click on next



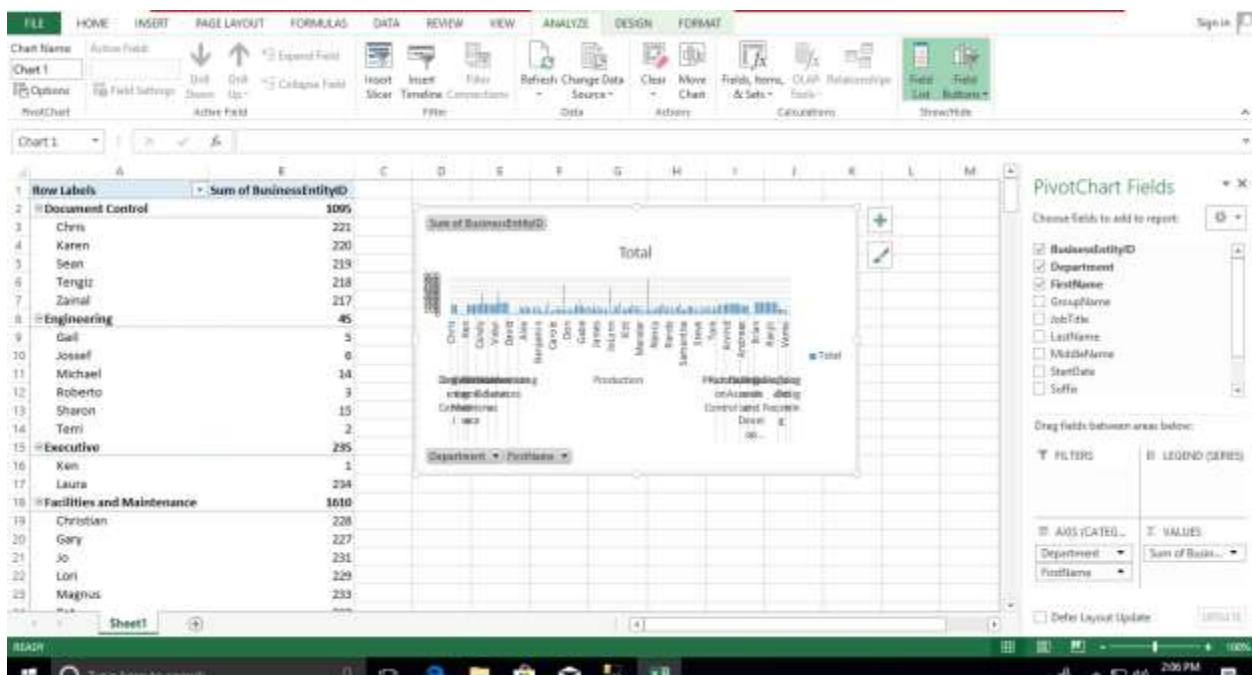
Step 4 : give the file a name for saving it and click next



Step 5 : Now in the next window select pivot chart and click OK



Step 6 Select the required fields which you want to visualize and you will see a pivot table and chart getting created automatically



PRACTICAL 6

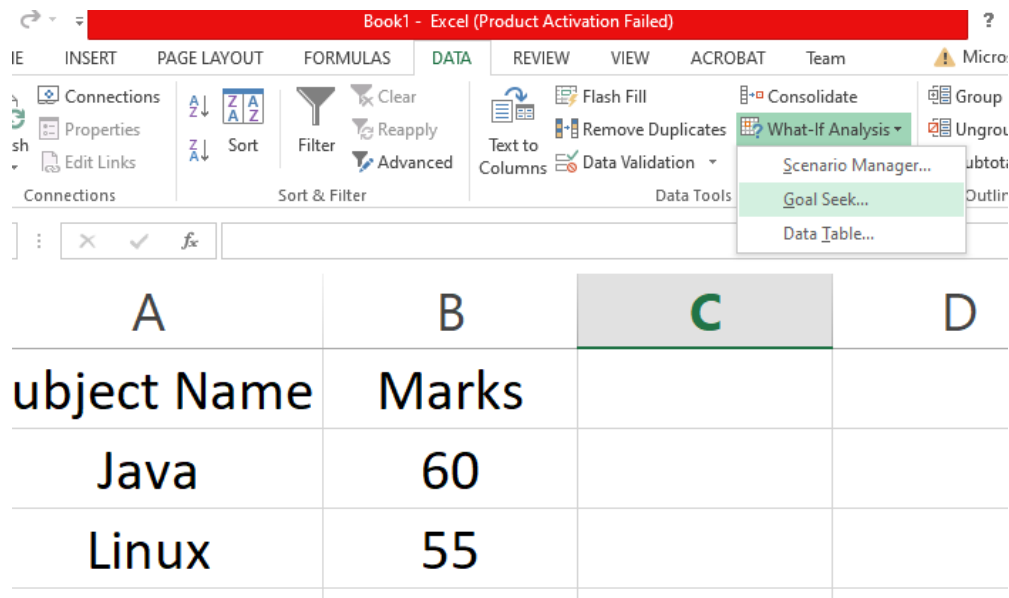
Apply the what –if Analysis for data visualization. Design and generate necessary reports based on the data warehouse data.

A) GOAL SEEK

	A	B
1	Subject Name	Marks
2	Java	60
3	Linux	55
4	AWP	50
5	SPM	
6	IOT	48
7	Total	213

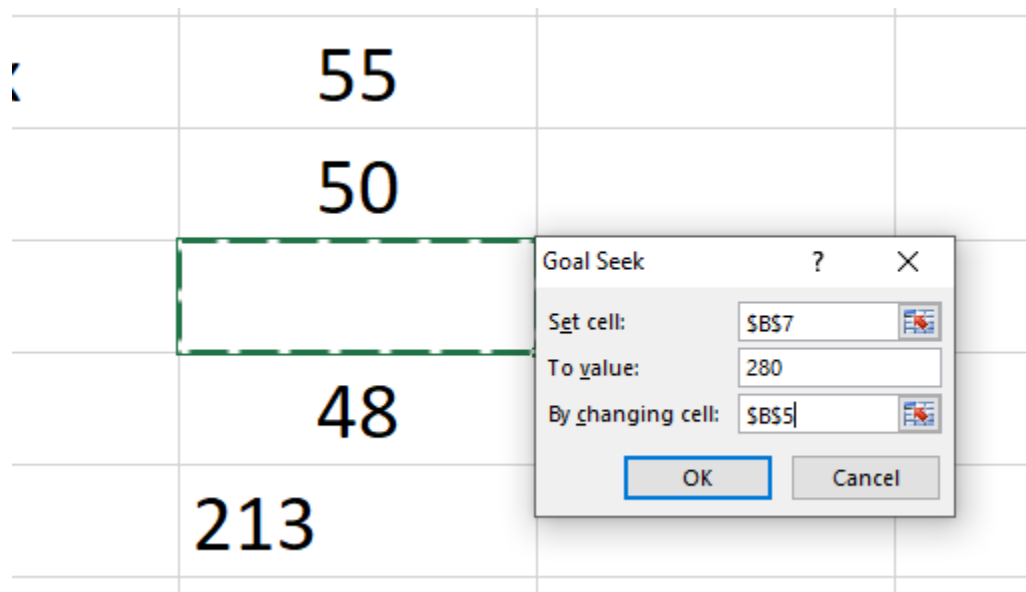
Here the question is how much marks need to get add in the SPM subject to reach the goal i.e 280 marks

Go to DATA tab > What if Analysis > Goal Seek



A	B	C	D
Subject Name	Marks		
Java	60		
Linux	55		

Goal Seek > Set Cell (Cell which value needs to get changed)
 > To Value (The target value) > By changing the cell (Cell which is blank
 were changes needed to be made to set the value)



55	
50	
48	
213	

Goal Seek

Set cell:

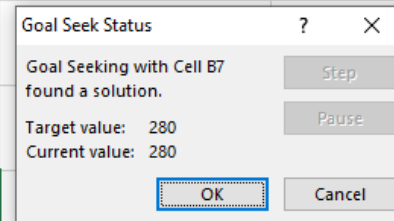
To value:

By changing cell:

OK Cancel

OUTPUT:-

	A	B	C	D
1	Subject Name	Marks		
2	Java	60		
3	Linux	55		
4	AWP	50		
5	SPM	67		
6	IOT	48		
7	Total	280		

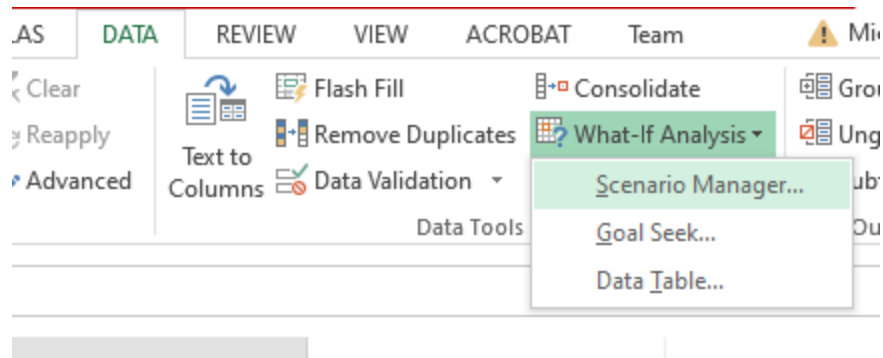


In the above output we can see that to seek the goal of total 280 marks the marks in the SPM subject must be 67.

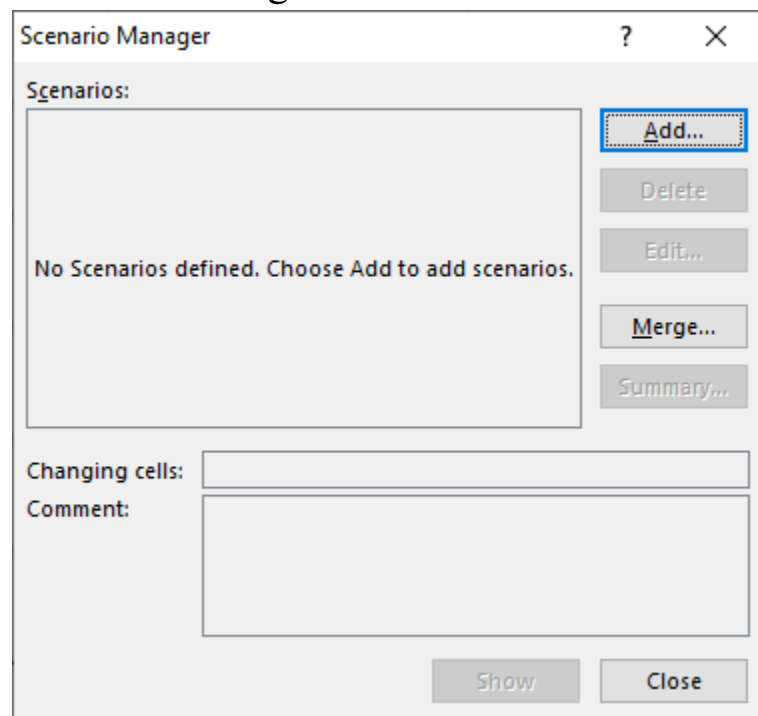
B) SCENARIO MANAGER:-

Selling Price	500
Cost	300
Quantity Sold	100
Profit	20000

Go to DATA tab > What if Analysis > Scenario Manager



Scenario Manager > Click on Add



Name the Scenario on Scenario name TAB, here it is “Low Sales”
> Select the Cell where the value need to be changed, here it is value of Quantity Sold > Click on OK after changing the value

Add Scenario ? X

Scenario name:
Low Sales

Changing cells:
B14

Ctrl+click cells to select non-adjacent changing cells.

Comment:
Created by User on 26-02-2026

Protection
☒ Prevent changes
☐ Hide

OK Cancel

Scenario Values ? X

Enter values for each of the changing cells.

1: \$B\$12 50

Add OK Cancel

Repeat the same steps and create another two scenarios here Medium Sales & High Sales

Scenario Manager ? X

Scenarios:
Low Sales
Medium Sales
High Sales

Add...
Delete
Edit...
Merge...
Summary...

Changing cells: \$B\$12

Comment:
Created by User on 26-02-2026

Show Close

Here 3 scenarios are created Low Sales with Value 50, Medium & High Sales with values 100 & 200 respectively.

OUTPUTS:-

To see the output we just need to click on the scenarios > show
this will show the outputs based on different scenarios

9			
10	Selling Price	500	
11	Cost	300	
12	Quantity Sold	50	
13	Profit	10000	
14			

9			
10	Selling Price	500	
11	Cost	300	
12	Quantity Sold	100	
13	Profit	20000	
14			

Scenario Manager

Scenarios:

- Low Sales
- Medium Sales
- High Sales

Buttons: Add..., Delete, Edit..., Merge..., Summary...

Changing cells: \$B\$12

Comment: Created by User on 26-02-2026

Buttons: Show, Close

Scenario Manager

Scenarios:

- Low Sales
- Medium Sales
- High Sales

Buttons: Add..., Delete, Edit..., Merge..., Summary...

Changing cells: \$B\$12

Comment: Created by User on 26-02-2026

Buttons: Show, Close

9			
10	Selling Price	500	
11	Cost	300	
12	Quantity Sold	200	
13	Profit	40000	
14			

Scenario Manager

Scenarios:

- Low Sales
- Medium Sales
- High Sales

Buttons: Add..., Delete, Edit..., Merge..., Summary...

Changing cells: \$B\$12

Comment: Created by User on 26-02-2026

Buttons: Show, Close

PRACTICAL 7:-

Perform the data classification using classification algorithm.

```

rainfall <-c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)
rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)
print(rainfall.timeseries)

png(file = "rainfall.png")

plot(rainfall.timeseries)

dev.off()

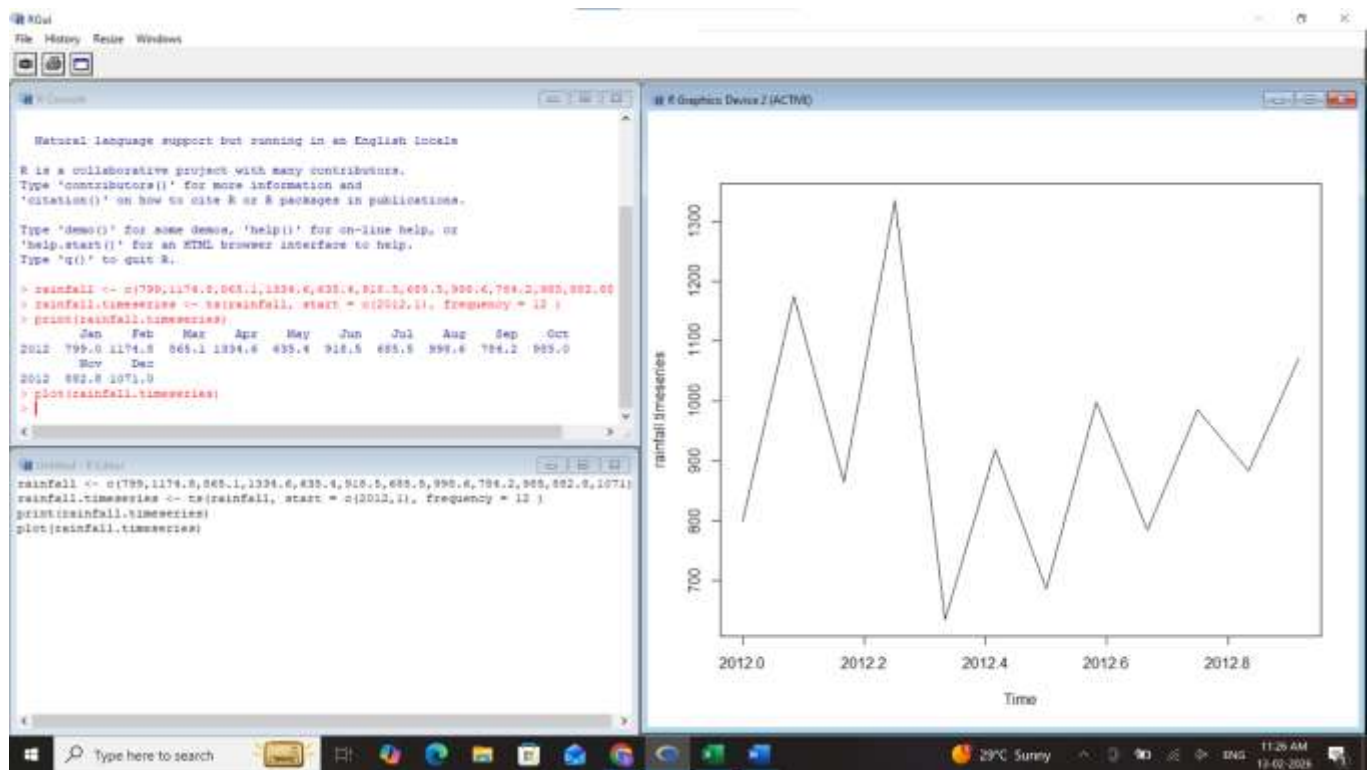
```

OUTPUTS:-

```

> rainfall <-c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)
> rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)
> 
> rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)
> print(rainfall.timeseries)
      Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct
2012  799.0 1174.8  865.1 1334.6  635.4  918.5  685.5  998.6  784.2  985.0
      Nov   Dec
2012  882.8 1071.0
> rainfall <-c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)
> rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)
> print(rainfall.timeseries)
      Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct
2012  799.0 1174.8  865.1 1334.6  635.4  918.5  685.5  998.6  784.2  985.0
      Nov   Dec
2012  882.8 1071.0
> 
> png(file = "rainfall.png")
> plot(rainfall.timeseries)
> 
> dev.off()
null device
      1
> |

```



PRACTICAL 8

Perform the data clustering using clustering algorithm.



```
newiris <- iris
newiris$Species <- NULL
(kc <- kmeans(newiris,3))

table(iris$Species,kc$cluster)

plot(newiris[c("Sepal.Length","Sepal.Width")],col=kc$cluster)
points(kc$centers[,c("Sepal.Length","Sepal.Width")],col=1:3,pch=8,cex=2)
dev.off()

plot(newiris[c("Sepal.Length","Sepal.Width")],col=kc$cluster)
```

OUTPUTS:-

[illegible]

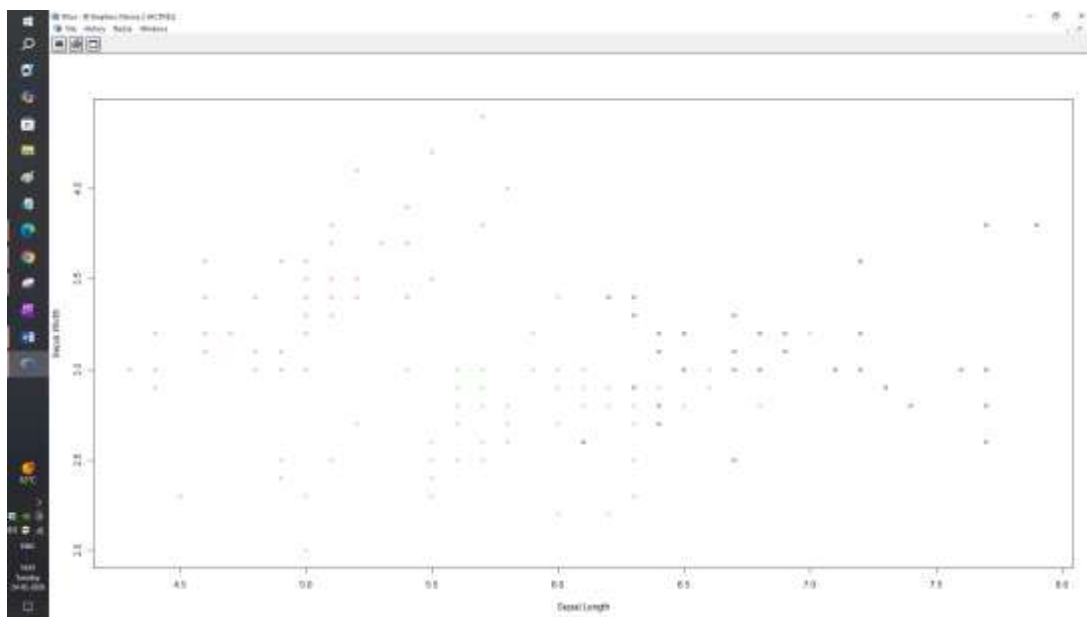
```

table(iris$Species,kc$cluster)

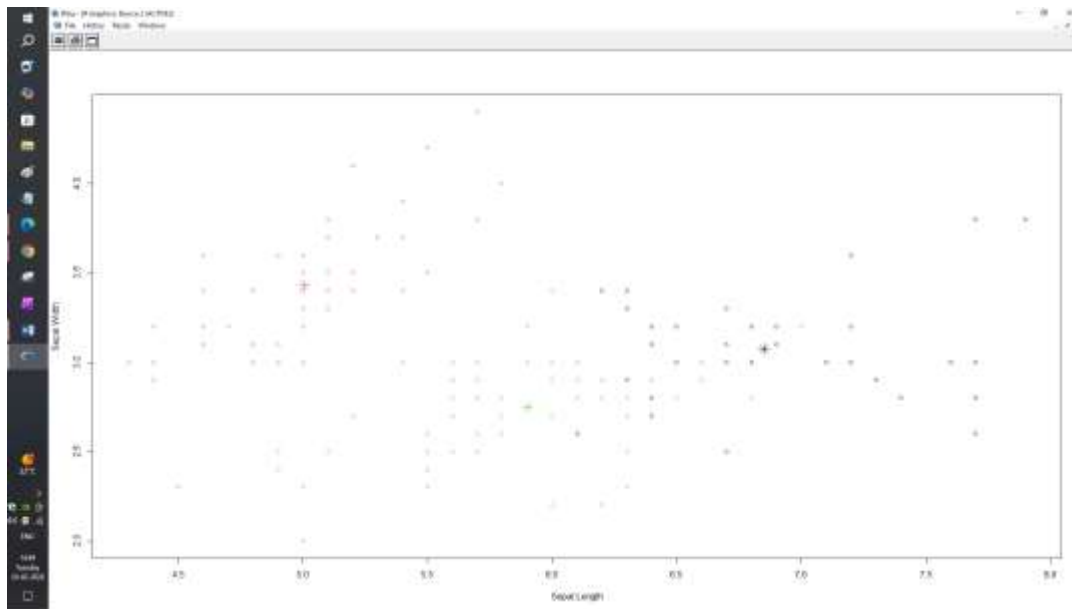
      1  2  3
setosa   0 50  0
versicolor 2  0 48
virginica 36  0 14

```

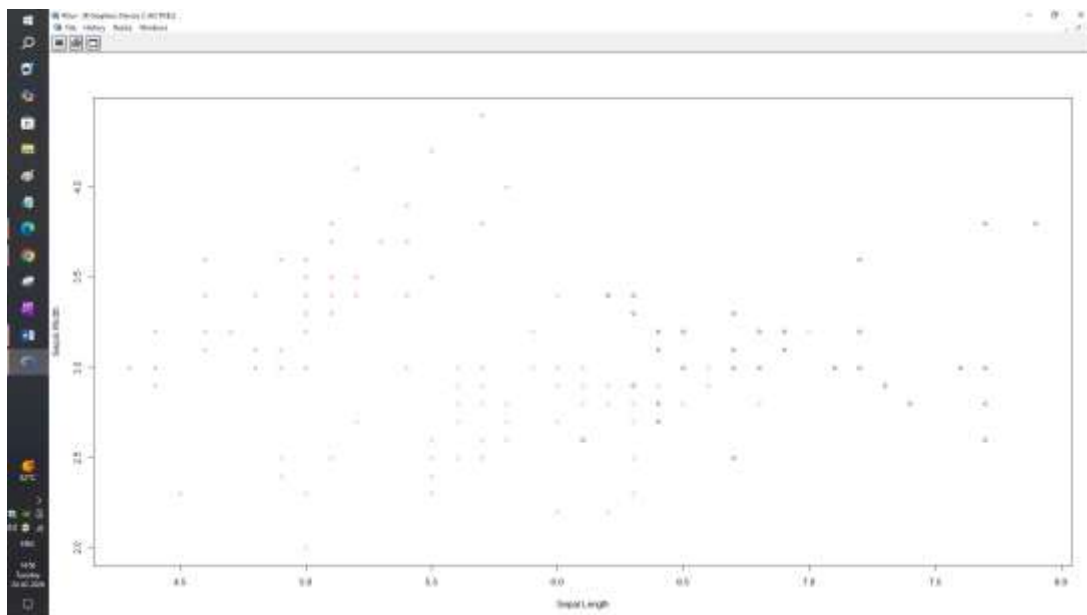
```
plot(newiris[c("Sepal.Length","Sepal.Width"),col=kc$cluster)
```



```
points(kc$centers[,c("Sepal.Length","Sepal.Width"),col=1:3,pch=8,cex=2)
```

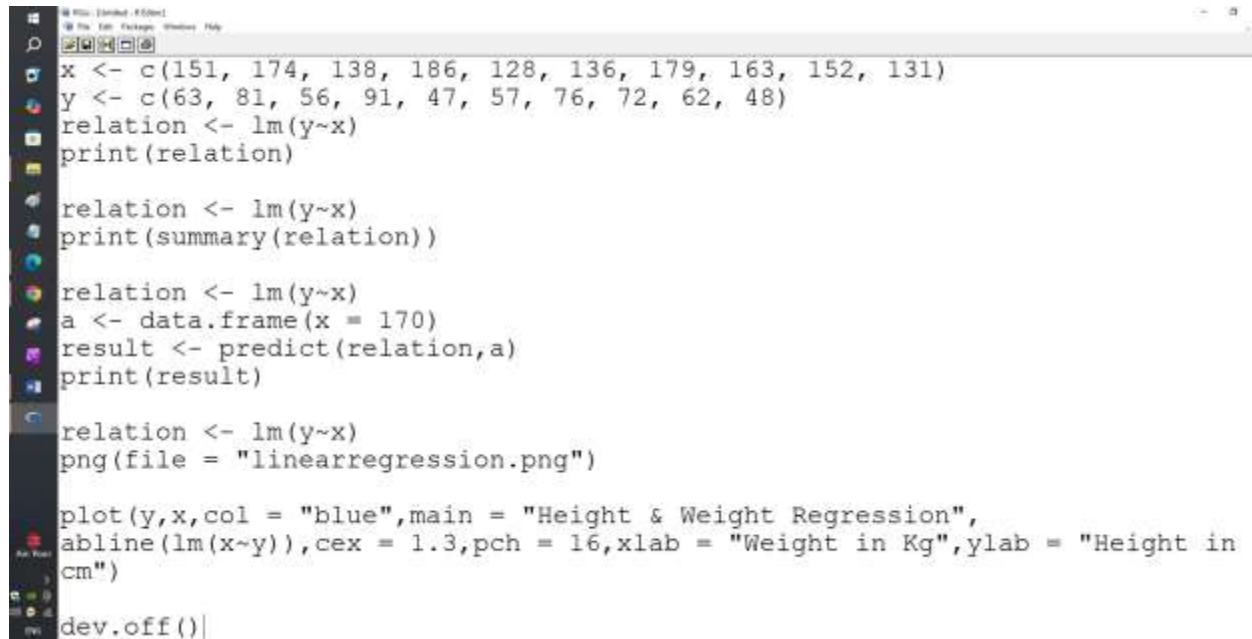


```
plot(newiris[c("Sepal.Length","Sepal.Width"),col=kc$cluster)
```



Practical 9

Perform the Linear regression on the given data warehouse data.



```
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)
relation <- lm(y~x)
print(relation)

relation <- lm(y~x)
print(summary(relation))

relation <- lm(y~x)
a <- data.frame(x = 170)
result <- predict(relation,a)
print(result)

relation <- lm(y~x)
png(file = "linearregression.png")

plot(y,x,col = "blue",main = "Height & Weight Regression",
abline(lm(x~y)),cex = 1.3,pch = 16,xlab = "Weight in Kg",ylab = "Height in
cm")
dev.off()
```

OUTPUTS:-

```

> plot(newiris[c("Sepal.Length","Sepal.Width")],col=kc$cluster)
> x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
> y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)
> relation <- lm(y~x)
> print(relation)

Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)          x
    -38.4551      0.6746

>
> relation <- lm(y~x)
> print(summary(relation))

Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-6.3002 -1.6629  0.0412  1.8944  3.9775

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -38.45509      8.04901  -4.778  0.00139 **
x              0.67461      0.05191  12.997 1.16e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.253 on 8 degrees of freedom
Multiple R-squared:  0.9548,    Adjusted R-squared:  0.9491
F-statistic: 168.9 on 1 and 8 DF,  p-value: 1.164e-06

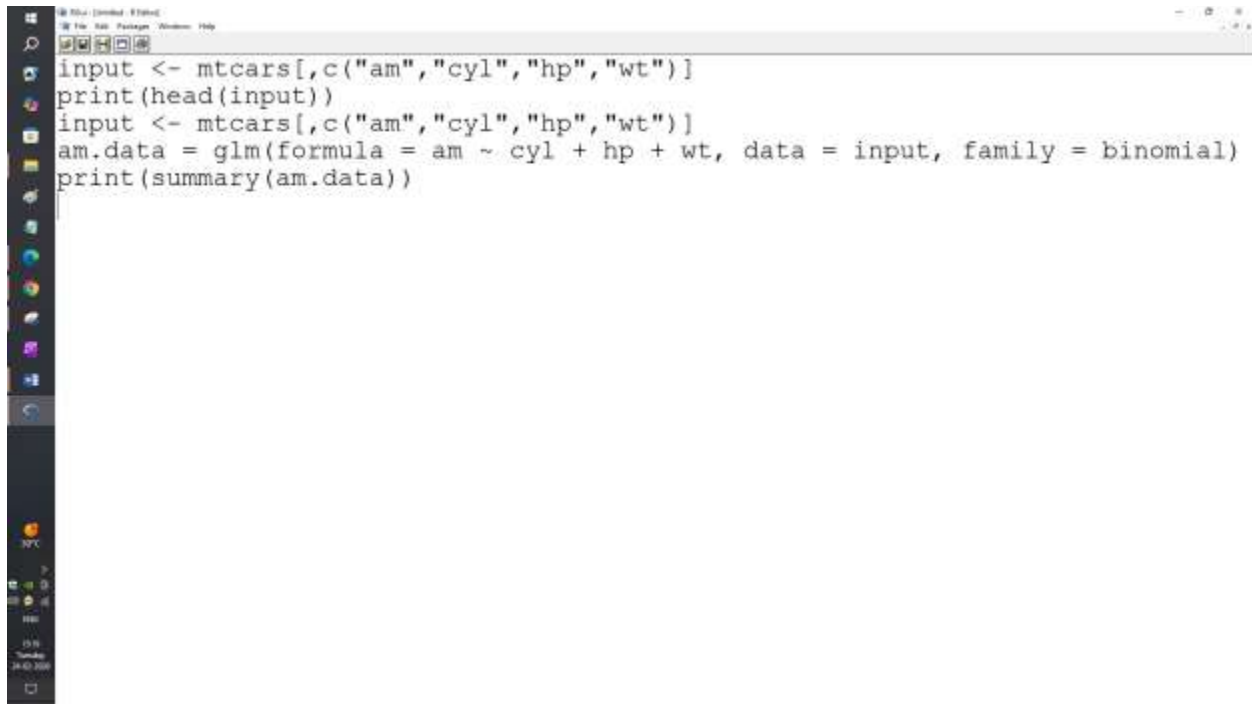
>
> relation <- lm(y~x)
> a <- data.frame(x = 170)
> result <- predict(relation,a)
> print(result)
      1
76.22869

>
> relation <- lm(y~x)
> png(file = "linearregression.png")
> plot(y,x,col = "blue",main = "Height & Weight Regression",
+ abline(lm(x~y)),cex = 1.3,pch = 16,xlab = "Weight in Kg",ylab = "Height in
+ cm")
>
> dev.off()
windows
      2
> |

```

Practical 10:-

Perform the logistic regression on the given data warehouse data.

A screenshot of an RStudio console window. The window has a title bar that says "RStudio [Untitled1] RStudio". Below the title bar is a menu bar with "File", "Edit", "Packages", "Session", and "Help". The main area of the window is a text editor showing R code. The code is as follows:

```
input <- mtcars[,c("am", "cyl", "hp", "wt")]
print(head(input))
input <- mtcars[,c("am", "cyl", "hp", "wt")]
am.data = glm(formula = am ~ cyl + hp + wt, data = input, family = binomial)
print(summary(am.data))
```

The console window has a dark theme. On the left side, there is a vertical toolbar with various icons for file operations, editing, and running code. At the bottom of the window, there is a status bar showing the current file name "D:\R", the date "Tuesday", and the time "14:00:00".

Output

```

> input <- mtcars[,c("am", "cyl", "hp", "wt")]
> print(head(input))
      am cyl  hp  wt
Mazda RX4      1   6 110 2.620
Mazda RX4 Wag  1   6 110 2.875
Datsun 710      1   4  93 2.320
Hornet 4 Drive  0   6 110 3.215
Hornet Sportabout 0   8 175 3.440
Valiant        0   6 105 3.460

> input <- mtcars[,c("am", "cyl", "hp", "wt")]
> am.data = glm(formula = am ~ cyl + hp + wt, data = input, family = binomial)
> print(summary(am.data))

Call:
glm(formula = am ~ cyl + hp + wt, family = binomial, data = input)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  19.70288    8.11637   2.428  0.0152 *
cyl           0.48760    1.07162   0.455  0.6491
hp            0.03259    0.01886   1.728  0.0840 .
wt           -9.14947    4.15332  -2.203  0.0276 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 43.2297 on 31 degrees of freedom
Residual deviance: 9.8415 on 28 degrees of freedom
AIC: 17.841

Number of Fisher Scoring iterations: 6

> |

```